

Peer Effects in Social Networks: Evidence from an Entrepreneurship Experiment [†]

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Abstract

We implement a randomized entrepreneurship program with women in rural Nepal that varies both the assigned training peer and whether participants are encouraged to share their network contacts. We find that the most effective peer depends on the time horizon. Pairing with a socially close peer increases short-term aspirations, with gains that appear to persist a year later. By contrast, pairing with a more connected peer significantly reduces long-term steps toward starting a business. The average long-term peer effect is close to zero, masking substantial heterogeneity: the effect of a peer is positive for those matched with the least connected peers, declining by 0.07σ per connection. Finally, encouraging peers to exchange contacts does induce sharing, but generates no additional gains. Sharing remains concentrated within social groups, suggesting that bridging can be difficult to engineer under high homophily.

JEL Codes: D85, C93, L26, L14, O12, Z13.

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1 Introduction

Development policies often leverage social ties to improve economic outcomes: peers can raise aspirations (Macours & Vakis 2014, Bernard et al. 2026), encourage entrepreneurship (Field et al. 2016, Lafortune et al. 2018, Brooks et al. 2018), and accelerate technology adoption (Banerjee et al. 2013, Beaman et al. 2021).¹ However, the features that make some peers more valuable than others remain less well understood. Social ties vary along multiple dimensions and can perform different functions. A popular peer may serve as an effective role model, while a socially close peer may be a more trusted collaborator. Moreover, some peers may be equipped to provide direct support, whereas others act as bridges to disconnected parts of the network (Granovetter 1973).² The value of a specific peer may therefore depend both on their individual characteristics as well as the desired outcome of interest.

In this paper, we provide causal evidence on how heterogeneous peers affect individual outcomes, how these effects evolve over time, and whether peers can be leveraged as bridges to broader social networks. To do this, we implement a randomized entrepreneurship program in rural Nepal that induces peer interactions among women, experimentally varying both the identity of the assigned peer and whether peers share their network contacts to help each other set up a business. We find that while social proximity generates both short-term and long-term benefits, peer popularity is detrimental in the long-term. We also show that bridging can be difficult to engineer in settings with high homophily (i.e., preference for interacting with similar people) as contact sharing is concentrated among existing social groups. By distinguishing the *relative* advantages of peers as collaborators, motivators, and connectors, we show that peer composition can potentially be a low-cost design choice that can affect program effectiveness.

Causal evidence on how peer identity affects outcomes is scarce, in part because development programs often pair peers endogenously, and in part because detailed network data are rarely available. To proceed, we first mapped the social networks of 2,840 women across 31 villages in Nepal and then randomized a subset to attend an intensive entrepreneurship training program.³ While individuals in pure control villages did not attend the program, individuals in treated villages were allocated into one of three groups. They either completed the program alone, attended with another woman assigned by the experimenter, or did not attend at all. Among those attending in pairs, we additionally randomized the implementation of a ‘connections’ treatment in which trainees were encouraged to share their social contacts with each other and think of ways these can help them set up a business.

¹Breza (2016), Breza et al. (2019) provide a review of the literature on social networks and development.

²This distinction relates to Putnam’s (2000) characterization of “bonding” and “bridging” social capital.

³Our main network measure captures all women in the village with whom an individual reports interacting in any capacity, including visiting homes, giving or taking advice, going to a temple together, or contacting during a health emergency.

Our setting is well-suited for studying peer interactions, as social networks are sparse⁴ and exhibit a strong community structure with high homophily, suggesting that networking opportunities may be limited.⁵ We first exploit the variation in whether women attend the program alone versus with another woman in their village to assess whether social support can improve outcomes on average.⁶ We then leverage whether individuals attended the connections treatment to compare the direct value of being trained with peers to the indirect value of accessing each other’s social contacts. This allows us to assess whether increasing the perceived benefit of linking to new contacts and reducing the cost of reaching them can improve outcomes. Finally, we leverage the variation in the identity of the peer to study whether the treatment effect differs depending on their network position. In particular, we exploit variation in social distance (i.e., the number of network steps between individuals) and peer centrality (i.e., the number of connections an individual has within the village network) across two survey waves to provide reduced-form evidence of heterogeneity and to estimate heterogeneous peer effects using a linear-in-means model (Manski 1993) that allows the weight on a peer to vary with their network position. Importantly, our analysis plan pre-registered heterogeneity along social distance and peer centrality, together with a framework motivating these as the two key dimensions of interest.⁷

We first examine short-term outcomes immediately after the training program. Training substantially improves business knowledge, increasing scores by 0.70 to 0.80 standard deviations. Training alone additionally raises an index of self-reported intentions to start a business by 0.25 standard deviations and leads to a higher willingness to take preparatory actions (e.g., opening a savings account or taking a loan) than training with a peer. Being paired does not deliver additional gains over training alone on average. However, this null effect masks substantial heterogeneity by peer identity. We find that pairing with a socially closer peer significantly improves business aspirations: each additional unit of social distance to the peer lowers aspirations by 0.08σ ($p < 0.01$, $q < 0.05$) and the effects are highest among those paired with socially close and more connected peers. The motivational benefit of a more connected peer depends on closeness: being matched with a peer who is close and more central raises the aspirations index significantly more than being matched with a more central yet socially distant peer ($p = 0.008$, $q = 0.040$), consistent with an aspirations window (Ray 2006, Genicot & Ray 2017).⁸ Moreover, each additional peer connection marginally

⁴These networks have an average density of 5.5%, meaning that very few of the possible links are present. This contrasts with the average density of 11.9% in villages in Karnataka in Banerjee et al. (2024).

⁵The inbreeding homophily index (Currarini et al. 2009) is 0.6 in our sample, indicating strong in-group linking. Links with individuals of the same social group, i.e., caste, are also 21.5 percentage points higher, on average, than the proportion of same-caste members in the village population, indicating high levels of homophily.

⁶Peers can provide support through skill complementarities, motivation, and risk-sharing, especially since education, aspirations, and risk aversion are correlated with opening a business in our baseline data.

⁷Details on how we depart from our pre-analysis plan are provided in Section 3.

⁸We define a person as socially close when their network distance is less than or equal to 2, i.e., strictly lower than the median social distance equal to 3.

increases take-up of resources (+0.03 per connection, $p < 0.10$). Disentangling mechanisms, we find that socially close pairs collaborate better during training (performing significantly better in a joint business plan exercise) and are more likely to report wanting to meet and start a business together. On the other hand, respondents matched with a more central peer are significantly more likely to report encouragement as the main reason pairing was useful.

In the connections treatment, we find that individuals share 6.5 village-level contacts on average with their matched peer (about 70% of the combined network size of both peers) to help each other set up a business. Network structure also makes bridging valuable: 73% of pairs in this treatment arm do not share any friends in common. Because peers were assigned rather than self-selected, we can ask how the characteristics of the pair affect sharing. We find that those with greater social distance or a larger gap in network connections share significantly more contacts. Sharing more contacts is also associated with a significantly greater willingness to start a business together with the peer ($p < 0.01$). However, contact-sharing remains strongly constrained by social identity. Same-caste pairs share about 1.5 more contacts on average ($p < 0.01$), making caste similarity the strongest predictor of contact-sharing, outweighing other factors such as social distance, differences in connectedness, and education. This tendency to concentrate sharing within caste groups may help explain why the connections treatment does not translate into significant improvements in short-term outcomes. Taken together, these results indicate that even when structural conditions make weak ties valuable, homophily can limit bridging across social boundaries where it is most beneficial.

We conducted a follow-up survey with a subset of women one year after the intervention. Those trained alone are significantly more likely to have taken steps to open a business, but those matched with a peer do not show a significant average effect. Decomposing this by peer identity as before, we find that those matched with a socially close peer have a marginally higher treatment effect on average outcomes compared to those matched with a socially distant peer. However, unlike the short-term results, those matched with more central peers are significantly *less* likely to have taken steps to open a business compared to those matched with less central peers ($p < 0.01$, $q < 0.05$).⁹ In terms of network interactions, one year later, 31% of paired trainees report contacting their matched peer for advice, money, or business discussions, but those matched with socially close but less central peers are significantly more likely to discuss business with others in the village, and pairs across social boundaries are significantly less likely to have remained in contact. This suggests that any short-term advantage from being matched with a more central peer does not translate into sustained interactions, which instead occur mainly among proximate and same-group peers.

Next, we use the field data to estimate peer effects by exploiting the experiment to address endogenous sorting into pairs and by using data from two survey waves to mitigate the

⁹We complement these findings with machine learning techniques that show that peer network characteristics are consistently ranked among the top three predictors of outcomes, compared to peer demographics.

“reflection problem” (Manski 1993). We find that the average peer effect is close to zero, but this masks substantial heterogeneity: the estimated effect of a 1 standard deviation increase in a peer’s outcome on long-term outcomes ranges from 0.29 standard deviations for the least connected peers to a negative effect for the most connected, declining by 0.07 standard deviations for each additional peer connection, consistent with our reduced form findings. This heterogeneity persists even after controlling for individual and peer characteristics. These patterns are consistent with the mechanisms documented above: the short-term gains from being matched with a more central peer are concentrated in take-up, while the costs of sustained interaction with busier peers accumulate over time.

While pairings based on social distance can be easy to implement, the feasibility of pairing based on centrality is uncertain, since network data can be hard to collect and there may be a limit to how many centrality-based matches we can construct within a fixed network. We find that the majority of our sample guesses their peers’ centrality correctly within one standard deviation, implying that additional data collection may not be necessary. Moreover, as a policy implementation exercise, we simulate 10,000 counterfactual reassignments of peer pairings within each village in our dataset and plot the distributions of the average centrality gap by village. The resulting distributions show wide dispersion, indicating that strategic pairings based on network centrality are feasible. The returns to strategic pairing, however, depend on the network dimension that is targeted: pairing on social *closeness* raises outcomes for both members of a pair and can increase aggregate outcomes (it is Pareto-improving), whereas pairing on *centrality* is largely redistributive, reallocating who takes entrepreneurial steps across individuals rather than generating aggregate gains (Section 6).

Related Literature: Our paper contributes to the existing literature in three ways.

First, we contribute to the large literature on peer effects by showing that their magnitude and sign vary systematically with peer identity and change over time. While peer effects have been measured in various contexts, including the adoption of new technology (e.g. Beaman et al. (2021)), financial products (e.g. Banerjee et al. (2013), Cai et al. (2015)), education (e.g. Calvó-Armengol et al. (2009), Duflo et al. (2011)), and entrepreneurship (e.g. Lerner & Malmendier (2013), Field et al. (2016)), the majority of literature treats peers as homogeneous.^{10,11} We show that peer effects are heterogeneous and provide evidence on the channels through which peers influence outcomes: closeness drives aspirations and sustained interaction over both time horizons, while peer centrality does not translate into longer-term entrepreneurial activity. By doing so, we demonstrate that peer assignment can be a low-cost

¹⁰The limitation of the linear-in-means assumption has also been noted in the context of peer effects in education (Sacerdote 2011) and more generally in Boucher et al. (2024), who show that taking an average can be misleading. Instead, we focus on the weights that individuals place on different types of peers.

¹¹Exceptions include a growing literature that leverages variation in the peer’s network position to show how it can heterogeneously affect different types of behavior, ranging from contract enforcement (Chandrasekhar et al. 2018), to savings (Breza & Chandrasekhar 2019), and the development of social skills (Zárate 2023).

lever that can substantially affect the effectiveness of development policies. Relatedly, we contribute to the literature on role models (Bernard et al. 2026, Riley 2024, Macours & Vakis 2014, Beaman et al. 2012): central peers are more likely to be described as encouraging by their pair partners, but the short-term aspirations gain operates primarily through social proximity rather than peer centrality on its own.

Second, our experimental design allows us to distinguish between two contrasting mechanisms through which social capital can be beneficial, mechanisms that have not previously been disentangled experimentally. Peers can directly influence outcomes, or they can serve as an indirect gateway to a broader social network, potentially generating additional peer effects. We disentangle these direct and indirect channels using a novel ‘connections’ treatment and highlight the extent to which peer effects can be engineered. A key methodological advantage is that because peers are not self-selected, we can estimate the causal effects of dyad-level characteristics that affect bridging, such as social distance within the pair, differences in centrality, caste similarity, and overlap in existing contacts. By doing so, our design contributes to the large literature on social capital (e.g., Granovetter (1973), Glaeser et al. (2002), Durlauf & Fafchamps (2005)) and in particular, to the literature on “bonding” and “bridging” social capital introduced by Putnam (2000). We show that contact-sharing responds to network structure as theory would predict. However, it remains concentrated within caste and does not affect downstream outcomes, suggesting that the binding constraint is the willingness to cross social boundaries.

Third, we contribute to the literature on entrepreneurship and the role of peers in improving business outcomes (Field et al. 2010, De Mel et al. 2014, Field et al. 2016, Cai & Szeidl 2018, Vasilaky & Leonard 2018, Fafchamps & Quinn 2018, Brooks et al. 2018, Carranza et al. 2018, Campos et al. 2019, Asiedu et al. 2023, Vega-Redondo et al. 2023). Unlike most existing business training RCTs, which focus on existing entrepreneurs or individuals who expressed interest in starting a business (see McKenzie et al. (2021) for a review), we mainly focus on non-business-owning women in our villages regardless of entrepreneurial intent. This allows us to examine whether training and peer effects operate for a broader, less selected population. We also show that peer effects in entrepreneurship mask significant network-based heterogeneity in that there are optimal peer matches that can be leveraged to improve outcomes. Related to our paper, Vega-Redondo et al. (2023) document the role of peer heterogeneity by showing the effects of diverse and non-diverse peer groups on entrepreneurship and how this varies based on interaction format (virtual or in-person). In contrast, we hold the interaction format fixed and vary peer identity within the same social network, allowing us to measure how network position affects entrepreneurial outcomes over time.

The paper is organized as follows. Section 2 describes the context and experiment design. We discuss the reduced form and peer effects estimation strategy in Section 3. Section 4

presents the short-term results of the intervention and discusses potential mechanisms, while Section 5 presents the long-term results and mechanisms. This is followed by Section 6, where we estimate peer effects, and discuss the feasibility of strategic pairing. Section 7 concludes.

2 Data

We first conducted a detailed survey and network elicitation with 2,840 women across 31 villages in rural Nepal in September-October 2021. We collected data on demographic characteristics (age, marital status, caste, education), willingness to start a business, reasons for not doing so, risk aversion, and aspirations related to agricultural investments, non-agricultural investments, and income.¹²

In addition, we administered a network survey to elicit information about social networks, as in Banerjee et al. (2013). These questions measure with whom individuals report interacting in any capacity, including visiting homes, giving or taking advice, going to a temple, or contacting during a health emergency. Individuals were asked to list as many names as they liked. For the majority of the analysis, we treat links as undirected, i.e., individual i is assumed to be a friend of j if either mentions the other’s name for any of the above interactions. We successfully elicited social networks for roughly 80% of village populations on average.^{13,14}

2.1 Baseline

Baseline summary statistics are presented in Table A.1. The average age of women in our sample is 38, and 92% of them are married. About 45% of our sample has no formal schooling: 32.6% report no education and a further 12.6% report only informal education. Roughly 22% of women report having started businesses already, and 42% report a willingness to open a non-agricultural business. We find that 89% aspire to earn an income higher than their current income, while 23% aspire to spend more on non-agricultural business expenditures than their current investment. We also elicited risk preferences using a standard choice experiment involving a series of lotteries and a fixed payment. On average, women are risk-averse with a risk-aversion level of around 4.6, where 6 stands for very risk-averse and 1 stands for risk-loving.

¹²Data on aspirations were collected as per the procedure in Bernard & Seyoum Taffesse (2014).

¹³The village population is defined as the set of women who were either surveyed at baseline or listed as a network contact by a surveyed woman.

¹⁴Our main results are robust to correcting the centrality measure to account for our sampling strategy. These results are discussed in the supplementary appendix (Section SA14).

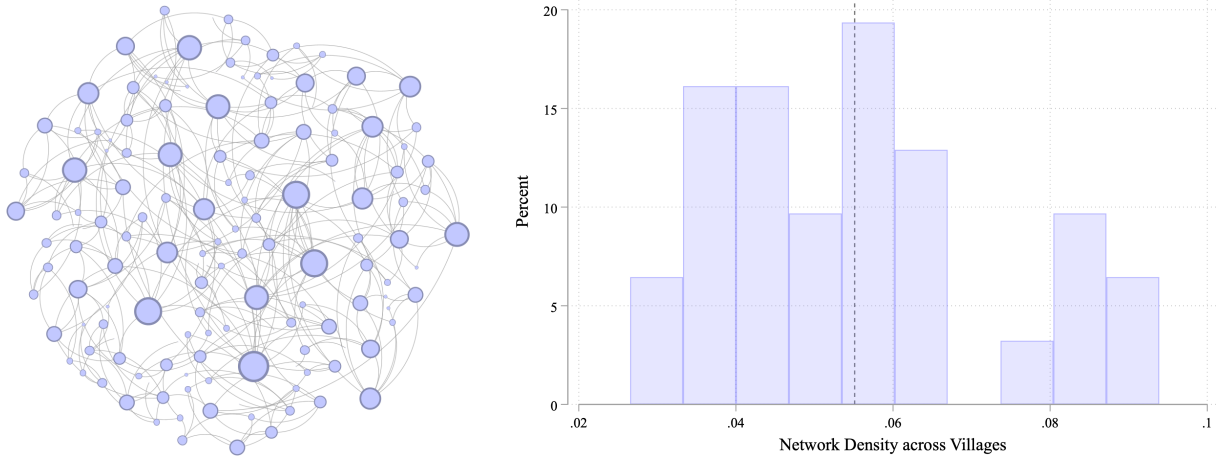


Figure 1: The figure on the left shows the social network in a village from our data. Larger nodes represent women with more connections. The figure on the right plots the network density for each village in our sample. Density is defined as the number of realized edges in the social network as a proportion of total possible edges. The number of total possible edges is given by $\frac{n(n-1)}{2}$ where n is the number of individuals in the network.

2.1.1 Network Statistics

Degree centrality is defined as the total number of undirected links an individual has in the village social network, i.e., the count of people who either named or were named by the individual as a friend. The average degree centrality in our sample is about 4.6.¹⁵ The left panel of Figure 1 plots the social network of one of the villages in the sample, showing the heterogeneity of the number of connections reported. Table SA.1 shows that the number of connections is correlated with other demographic variables of interest, including education, caste, and marital status, with significantly fewer links among unmarried women, women with no education, or women of a lower caste.

Networks in these settings have two key features. First, networks are sparse, i.e., each person is connected to a small share of others in their village and most potential links are not present. We plot the density of networks, i.e., the number of observed links divided by the total possible links, computed for each village (Figure 1). The average network density across all villages is 5.5%, i.e., only 5.5% of links exist out of the total possible number of links given the number of individuals observed in each network. The sparsity persists even after we correct each individual’s degree centrality (i.e., number of links) to account for our sampling strategy. Average density is 7% in that case. In comparison, the villages in rural Karnataka in Banerjee et al. (2024) have an average density of 11.9%.

¹⁵Degree centrality is measured at baseline for 95.3% of respondents. For the remaining 4.7%, who were not part of the baseline sample, degree is computed using their endline nominations, including any baseline nominations they received from others.

Second, we find that these networks exhibit a community structure and a high level of homophily (i.e., in-group linking bias): intra-caste links are more common than links across castes, even after accounting for the number of same-caste members present in the village. On average, the proportion of same-caste links exceeds the proportion of same-caste village members by 21.5 percentage points. We measure caste homophily using the inbreeding homophily index (Currarini et al. 2009):

$$\text{Homophily} = \frac{\text{share of same-caste links} - \text{share of own caste in population}}{1 - \text{share of own caste in population}}.$$

Intuitively, the index measures what fraction of an individual’s maximum possible in-group bias they exhibit, accounting for the fact that members of larger groups could appear biased even if they link randomly. The index ranges from -1 (maximum cross-type preference) through 0 (no excess same-type preference) to 1 (maximum same-type preference). The average inbreeding homophily in our sample is 0.6 , implying a high in-group linking bias.

Together, these findings suggest that the experiment induces social interactions that may not occur naturally.

2.1.2 What are the barriers that prevent women from opening businesses?

We find that women who already own a business have a 50% higher median income than those with no business ownership. Correlating whether or not they have opened businesses with their baseline characteristics in Table SA.2, we find that those who own a business are younger, more educated, less risk-averse, and have higher aspirations for investment in non-agricultural business. This is also shown in Figure SA.1 that employs a Lasso regression (at various penalty levels) to show which variables correlate with whether or not a business has been opened already out of a list of demographic and network characteristics. We find that being educated, and especially having obtained a university education, matters the most in explaining the decision to have already opened a business.

When asked to give reasons for not having opened a business, 77% say that they do not have the skills or do not feel capable, while 24% say that they lack the financial ability. We correlate their willingness to open businesses with baseline demographics, number of network connections, aspirations, and other variables such as risk aversion. These results are shown in Table SA.3. We find that those who are older and more risk-averse are correlated with being less willing to open businesses. This is also reinforced by Figure SA.2 where we implement a Lasso regression, which selects age, risk-aversion, and education as the relevant correlates at high values of the penalty parameter. This suggests that risk-sharing and skill complementarities with peers could be helpful in setting up a business. Finally, those who have higher aspirations are more willing to open businesses, suggesting that peers can potentially be used to motivate and boost aspirations.

2.2 Experiment Design

We conducted the experiment in September 2022. The experiment consisted of a three-day entrepreneurship training program motivated by the SIYB module developed by the International Labour Organization. The training typically lasted three hours per day, and individuals were given NPR 100/day (i.e., \$0.7) for participation. The training focused on various topics, including defining entrepreneurship, listing the characteristics of a successful entrepreneur, learning how to build a business plan, learning about market scoping and market access, setting savings goals, and boosting aspirations using a video highlighting a successful female entrepreneur. The training also included interactive activities such as developing a business plan and playing an investment game. Participants would complete these activities alone or with their assigned peer depending on their treatment status.

Figure B.1 represents our two-step randomization design. First, we randomly allocated villages to Pure Control and Treatment. Following this, women in treatment villages were randomly allocated into one of the treatment groups as discussed below.¹⁶

Treatment 1: Trained Alone: Women in this group attended the training alone, i.e., the enumerator seated the participant alone without pairing them with another individual. Women in this treatment arm completed all interactive activities of the training alone.

Treatment 2: Trained with a Peer: Women in this group attended the training paired with another woman from their village. They were allocated a peer by the enumerator at the training center, ensuring that this peer is not self-selected. These pairs completed all interactive activities of the training with each other.

Treatment 3: Trained with a Peer + Connections Module: As in Treatment 2, women in this group also attended the training with another woman from their village. However, in addition, the pair received a novel “connections module” at the end of the last day of training that encouraged them to share their network contacts with each other and think about how network ties can be helpful in opening a business.

To guide participants, we first highlighted three main ways in which networks can be helpful: i) Information-Sharing, ii) Complementarity in skills, and iii) Risk-Sharing, and then asked them to think about how they could help each other. The module activity then proceeded in two steps. Because the two women were paired at random and may not have known each other beforehand, we first asked them to discuss how they could help one another directly. Following this, the pair were asked to collectively write down and share their network contacts

¹⁶Women in treatment villages were also randomized into a within-village control group. However, this group is contaminated by two-sided non-compliance, as discussed in Section 2.3. Our main treatment effect estimates compare treated individuals to pure control villages and treated individuals across various treatment arms. However, we also discuss robustness to within-village specifications.

that could potentially help them start a business together, and think of ways each of these contacts could be helpful. Appendix Figure B.2 shows the example of a contact-sharing list that would be filled by the pair. Pairs were encouraged to share contacts but they could decide whether to share any contacts and if so, how many.

Importantly, participants attended training sessions in different classrooms depending on their treatment status. Women assigned to training alone were placed in classrooms with others receiving training alone, while women assigned to training with a peer attended sessions with others receiving training with a peer.

Conceptual and Empirical Motivation for the Connections Module:

The connections module allows us to measure the indirect benefits of peers as bridges to the wider social network. Conceptually, the module aims to test whether increasing the perceived benefits of linking and reducing the cost of reaching new contacts can induce useful interactions. Suppose individual i decides whether to interact with individual k with expected utility $E[U_{ik}] = \mu_{ik} - c_{ik}$, where μ_{ik} denotes the perceived benefit from interacting and c_{ik} is the cost of approaching k , including learning about them in the first place. In sparse, homophilous networks, μ_{ik} may be low and c_{ik} may be high. The module is designed to reveal who can be reached through the matched peer j and how these contacts might be useful, increasing μ_{ik} and lowering c_{ik} at the same time.

The scope for bridging depends on how much the pair’s networks already overlap. When overlap is low, the peer provides access to genuinely new contacts; when overlap is high, the shared contacts are largely redundant. Given the sparsity of our networks (average degree ≈ 4.6 and average density of 5.5%), most experimentally assigned T3 pairs have non-overlapping neighborhoods: the average overlap of friends among T3 pairs is just 4%, with 73% of pairs sharing zero common contacts. This implies that the potential for bridging is high in our setting, but realizing this potential requires that individuals are willing to share contacts with others in the first place.

2.3 Sampling and Implementation

We describe sampling and balance for the training effect in the remainder of this section, and discuss identification of peer effects in Section 2.3.1.

Table B.1 in the appendix summarizes the full set of implementation details by treatment group, including invitation, arrival location, peer assignment, training logistics, and survey location. Figure B.3 shows the sample flow across various stages of the design.

Sample Construction and Take-up: Out of 2,840 women surveyed at baseline, we excluded those already operating businesses and four villages where training could not be

implemented ex ante. The remaining 1,970 women were randomized either into the pure control group (if their village had been randomly assigned to pure control) or, if in a treated village, into the various arms outlined above. Nearly half of the randomized sample is part of our experiment, and 44% of those invited in treated villages attended the training venue. These attendance rates are consistent with the business-training literature, where take-up rates of 39–51 percent are common even among samples screened for initial interest, and average take-up is around 65 percent when training is free (McKenzie & Woodruff 2014).

Compliance: We define compliance as a binary indicator equal to one if an individual’s realized treatment status matched their assignment. Overall compliance in the endline sample is 68%, or 58% when excluding pure control villages, where compliance is mechanical. Non-compliance is two-sided. On one side, among those assigned to training who arrived at the venue, 82% were actually trained. On the other, some within-village controls were themselves trained: because they came to the training center for their endline survey, some were reassigned to fill vacant training slots, contaminating that comparison group. We do not find meaningful differences in baseline characteristics between compliers and non-compliers in treated villages (Table B.3).

Given this two-sided non-compliance, our main ITT specification compares treated individuals to women in pure control villages rather than within-village controls. For robustness, we also discuss results from analogous within-village specifications, reported in the online appendix. Moreover, we implement LATE specifications that instrument actual treatment with treatment assignment; first-stage F-statistics are large in all specifications. Further details on dropouts and non-compliance are provided in Appendix Section B.1.¹⁷

Balance and Attrition (Treated vs. Pure Control): Surveys in pure control villages took place in respondent homes while surveys in treatment villages took place at the training venue. Participation rates are therefore lower in treated villages: treated women had to travel to the venue for the survey, whereas pure control women were surveyed at home. However, dropouts in treated villages do not differ from pure control participants in terms of their willingness to open a business, implying that those who come to the training venue are not selected on their willingness to open a business, the key dimension relevant to our outcomes.¹⁸ We also check for balance between pure control and treated individuals on several baseline variables, including demographic, network, and business characteristics such as income, sources of income, age, education, caste, network connections (i.e., degree centrality), and aspirations, for all individuals in the endline and follow-up samples. Tables B.8 and B.9 show these results for the endline and follow-up samples, respectively, taking pure control

¹⁷Some women assigned to train with a peer could not be matched and remained unmatched. Appendix Section B.1 compares the baseline characteristics of compliers and non-compliers across arms and of matched and unmatched individuals within the peer arms, and discusses what these differences imply for external validity.

¹⁸This is also confirmed when we compare dropouts with non-dropouts in the entire sample.

villages as the comparison group. Comparing the various arms (114 pairwise comparisons across 19 variables), only 1 has a p-value less than 0.05 and 6 have a p-value less than 0.1. Comparing only across the trained sample, none are significant at the 5% level. We find similar results when we compare within the follow-up sample. These findings are discussed in more detail in Section B.2 in the appendix.

The single significant comparison at the 5% level is a difference in degree centrality between the pure control and a treated arm. To further check this, we use Kolmogorov-Smirnov tests to measure differences in the distribution of degree centrality among pure control and treated individuals (Figure SA.3)¹⁹. We find that there are no significant differences in the distribution of network connections of individuals across the various arms. We also later show that our key results are robust to any observed cases of imbalance by employing post-double selection Lasso wherever necessary (Belloni et al. 2014) as well as correction for selection by using probability weights.

We also conducted a phone-based follow-up survey one year later with a subset of 580 women from the endline sample whom we could reach on phone. Table B.7 shows that attrition between endline and follow-up is not correlated with assignment to training. Non-response was primarily due to changes in phone numbers as resource constraints prevented an in-person follow-up.

Within-Village Comparisons (Secondary): Although the contaminated within-village control group is not our main benchmark, we report results using it for completeness. Participation rates do not differ between those assigned to treatment and those in the within-village control group (Table B.6). Using within-village controls as the reference group in balance checks, we find that 7 of 57 comparisons are significant at the 5% level (Table SA.8), specifically including degree centrality, but Kolmogorov-Smirnov tests do not detect any differences in distribution.

2.3.1 Peer Assignment and Randomization Inference

Importantly, our main results on heterogeneous peer effects do not rely on the treated-versus-pure-control comparison at all, as they restrict the sample to treated individuals and take those trained alone as the reference category. In particular, dropout rates are similar for those trained alone and those trained with a peer, which is our main comparison of interest. The two groups are also balanced across most baseline characteristics in both the endline and follow-up samples (Tables B.8 and B.9).

Crucially, peer assignment occurred after individuals arrived at the training center. This means that initial refusals of invitations to participate prior to arrival at the center cannot be correlated with peer type, since non-attendees were never assigned a peer. Peers

¹⁹Tables and figures numbered with the prefix “SA” appear in the Supplementary Appendix, available online.

were assigned on the spot by the research implementation team, with the goal of avoiding self-selection by ‘randomly’ pairing women with another attendee from the same village. For instance, women who arrived together would not be paired with each other. We assess whether the implemented assignments are as-good-as-random by conducting randomization inference on the entire sample of individuals who were actually paired for the training. Specifically, we simulate multiple alternative peer assignments within each village and compute, for each assignment, a statistic capturing the total difference in characteristics between assigned pairs. We find that the observed value of the statistic from the actual pairing lies well within the distribution generated from these simulations, suggesting that the implemented peer assignments are statistically indistinguishable from random. These results are discussed in Section B.5.

Balance by Peer Type: We additionally check whether baseline characteristics of individuals differ by the type of peer assigned within the paired treatment arms. Tables SA.4–SA.7 compare individuals assigned to close versus distant peers and more versus less central peers in the endline and follow-up samples. Characteristics are well-balanced: 14 of 17 show no significant differences by social closeness, and 15 of 17 show no significant differences by peer centrality in the endline. The follow-up sample is even better balanced with only 1-2 unbalanced variables across the two splits. Crucially, willingness to start a business, the baseline dimension most directly related to our outcomes, is balanced both by social closeness and peer centrality.

The few imbalances that emerge include demographic characteristics that could relate to network position. Individuals assigned to closer peers have higher baseline income ($p = 0.010$) and are marginally younger ($p = 0.092$), and those assigned to more central peers have lower rates of *higher* education ($p = 0.002$) while other education metrics (secondary/ elementary/ university) are balanced. To address the possibility that these differences confound our results, we present robustness checks throughout the paper that isolate the effects of network attributes from correlated baseline characteristics.

3 Estimation Strategy

3.1 Endline Outcomes

We measure the effect of the experiment on five endline outcomes collected after the training, consistent with our pre-analysis plan. These include the Knowledge Index, Business Aspirations Index, Business Index, Take-up Index, and Additional Steps Index.

The indices are constructed as follows:

- (1) **Knowledge Index:** We ask five short questions that measure knowledge gained during

the course of the training, and take the proportion of correct responses to construct the knowledge index. We collect this measure for both treatment and control groups.²⁰

(2) **Business Aspirations Index:** We compute a weighted average of business-related aspirations comprising yearly non-agricultural investment, monthly income, and savings, elicited following the procedure in [Bernard & Seyoum Taffesse \(2014\)](#).^{21,22}

(3) **Business Index:** This is constructed as a weighted average of responses to whether the individual is ready to invest in a business, whether they would submit a plan for a business competition, and how likely they are to start a business, capturing forward-looking intention and self-assessed likelihood of taking entrepreneurial action.

(4) **Take-up Index:** This is a weighted average of willingness to seek additional support. First, whether the respondent would be willing to be put in touch with the Federation of Woman Entrepreneurs' Associations of Nepal (FWEAN), a national business-mentoring organization, for further business-training mentoring, and how much she would be hypothetically willing to pay to take part. Second, whether she would be willing to join a commitment-based business-plan training in which a trainer would meet to review and monitor her plans, where she would be asked to pay up front and the payment would be returned only if her plans showed substantial improvement, and how much she would be willing to pay for it. Third, whether she would be keen to take advice about setting up a business from other women in her village. The index aggregates the willingness to take up each option together with stated willingness to pay.

(5) **Additional Steps Index:** This index captures concrete steps toward setting up a business, including whether the individual intends to open a savings account or take a loan for the business. In each case, we construct a weighted average of the sub-components ([Anderson 2008](#)) (to account for multiple hypothesis testing) and normalize indices using the pure control group.²³

Validation of Self-Reported Intentions: With the exception of the Knowledge Index, which tests factual knowledge, these are self-reported measures capturing aspirations, in-

²⁰The questions are: (a) What do you understand by a business? (b) What characteristics are required to be a successful entrepreneur? (c) What do you mean by fixed assets? (d) What sector does a beauty parlor come under? (e) Above what break-even percentage does the business become risky?

²¹We ask individuals the minimum and maximum of the relevant variables in their neighborhood, the current value, and what they aspire to in the corresponding time frame, and construct the weighted average of the individual's aspirations across these dimensions to create the index.

²²We depart from the pre-analysis plan by excluding agricultural investment aspirations from the composite index, as business training could plausibly increase or decrease agricultural aspirations, which would confound the interpretation of the index. Effects on individual aspiration components, including agricultural aspirations, are reported in Tables [SA.20](#) and [SA.29](#) in the Supplementary Appendix.

²³Variables in specifications where the within village control group is the base category are normalized with respect to that group. Variables entering the aspirations index and payment variables in the take-up index were winsorized at the 99th percentile.

tentions, and willingness rather than actual behaviors. This allows us to assess immediate changes in mindset and motivation following the training. One concern with these measures is that stated intentions may not translate into actual behavior. However, higher endline aspirations, take-up, and additional steps indices significantly predict concrete follow-up behaviors one year later, including the outcomes, steps, and mindset indices described below (Table SA.9), suggesting these indices capture genuine intentions. Endline aspirations are the most consistently predictive of follow-up outcomes.

3.2 Follow-up Outcomes

Treatment effects on follow-up outcomes are measured one year later with a subset of the endline sample using phone surveys.

We construct three key indices to capture follow-up outcomes.²⁴ First, the **Outcomes Index** includes monthly income and agricultural profits. Second, the Steps Index captures concrete actions taken toward setting up or expanding a business. It aggregates two binary indicators — whether the respondent opened a new savings account and whether she joined a new cooperative in the past year — together with three monetary measures: agricultural investment, total savings, and the loan amount taken. Third, the **Mindset Index** captures forward-looking motivation and includes income aspirations, willingness to start a business, perceived self-efficacy in running a business, and willingness to open a (hypothetical) commitment savings account (both participation decision and amount committed), where the amount can only be withdrawn for business-related expenses. Finally, as a separate outcome, we also measure whether respondents keep written records of their agricultural profits, reflecting the training’s emphasis on record keeping. As before, all indices are normalized relative to the pure control group.²⁵

3.3 Empirical Specification

3.3.1 Intent-to-treat Effects of the Training

We first measure the immediate impact of the treatment using the specification described below, where we compare individuals in treated villages with women in pure control villages.

$$Y_i = \alpha + \beta_1 \text{Trained Alone}_i + \beta_2 \text{Trained With Peer}_i + \beta_3 (\text{Trained With Peer} + \text{Connections Module})_i + \epsilon_{iv}$$

²⁴We rely on an intuitive categorization since the follow-up survey was not pre-registered.

²⁵All monetary variables were winsorized at the 1st and 99th percentiles to exclude outliers. This includes monthly income, total savings, agricultural profit, agricultural investment, and income aspirations. In contrast, business investment and loan amount were winsorized at the 99th percentile only, since the majority of respondents reported zero values for these variables.

Y_i is an outcome measure for individual i , *Trained Alone* is a binary variable that takes the value 1 if the individual was intended to be treated alone and 0 otherwise. Similarly, *Trained With Peer* is a binary variable that takes value 1 if the individual was intended to be treated with a peer and *Trained With Peer+Connections Module* is a binary variable that takes value 1 if the individual was intended to be treated with a peer and attend the additional connections module. The omitted category is the pure control group. Standard errors are clustered at the village level as we include both pure control and treated villages in this specification. We also discuss robustness to the use of heteroskedasticity-robust standard errors, specifications with within-village variation, and models estimated using post-double-selection Lasso (Belloni et al. 2014).

3.3.2 Impact of Different Types of Peers

Our intention to leverage the random variation in the network identity of the peer and use it to study heterogeneity by social distance and network centrality was pre-registered in our pre-analysis plan. The main tables in the paper estimate specifications on the treated-only sample and take individuals trained alone as the omitted reference category. The coefficients on the peer indicators capture the incremental effect of being paired with a particular kind of peer relative to being trained alone.

We implement specifications using both continuous and discrete measures of social distance, as well as absolute and relative measures of peer centrality (i.e., comparisons with the matched peer’s degree). This is to ensure that our results are not sensitive to the choice of network measure. Let d_{ij} be the network distance between i and j , i.e., the number of links needed by i to reach j , and let ϕ_i be the network centrality (e.g., number of connections) of agent i . In line with our pre-analysis plan, we first implement the following specification:

$$Y_i = \alpha + \beta_1 \text{Trained With Peer}_i + \beta_2 (\text{Trained With Peer}_i \times d_{ij}) + \beta_3 (\text{Trained With Peer}_i \times \phi_j) + \epsilon_i$$

We report heteroskedasticity-robust standard errors in all peer type specifications, as they compare individuals within treated villages and are identified using the individual-level assignment of peers rather than village-level randomization.

Following this, we split individuals into whether or not they are socially close. We define a pair ij as socially close if the network distance between i and j is strictly less than the median social distance in the sample (median $d_{ij} = 3$), i.e., $d_{ij} \leq 2$. Similarly, we classify each peer by her centrality relative to the respondent: a peer is “More Central” if she has strictly more network connections than the respondent, and “Less Central” otherwise.

We then implement the following specification:

$$\begin{aligned}
Y_i = & \alpha + \beta_1 \text{Trained with Peer}_i \\
& + \beta_2 \text{Trained with Close Peer}_i + \beta_3 \text{Trained with a More Central Peer}_i \\
& + \beta_4 (\text{Close}_i \times \text{More Central}_i) + \epsilon_i
\end{aligned}$$

The coefficient on the interaction term, β_4 , captures the additional effect of being matched with a peer who is both socially close and more central than the individual (i.e., $\phi_i - \phi_j < 0$ and $d_{ij} \leq 2$), over and above the effect of each attribute on its own. The effects of different types of peers are obtained by taking linear combinations of the various coefficients.

Additional Controls and Robustness Checks: Because an individual who is more central could be mechanically more likely to be paired with a relatively less central peer even under random assignment, all discrete specifications that use a relative measure of peer centrality (e.g., whether the peer is “More Central” than the individual) control for the individual’s own centrality. Finally, in all regressions that compare peer types or analyze heterogeneity within the paired arms, we control for individuals who were intended to be paired but were not actually matched due to non-compliance.²⁶

We also show robustness of our main results to incomplete network sampling, and additional controls including individual and peer characteristics.

3.3.3 Peer Effects

Finally, we identify peer effects by exploiting random pairing and the panel structure of our data, estimating a linear-in-means specification (Manski 1993) with heterogeneity based on network position. This will be discussed in more detail in Section 6.

4 Short Term Effects

We first present the average effects of the training on pre-registered outcomes and then turn to how these effects depend on the identity of the assigned peer.

Table 1 reports ITT estimates of training effects on endline outcomes. All three training arms significantly improve the knowledge index, by between 0.70 and 0.80 standard deviations ($p < 0.01$). Training alone additionally improves business intentions ($+0.25\sigma$, $p < 0.05$). The paired treatment arm with the connections module has a marginally positive effect on the same index ($+0.21\sigma$, $p < 0.10$). The effect of training alone on take-up is positive but not statistically significant ($+0.11\sigma$). As discussed earlier, these endline intention measures are

²⁶Peer-specific regressors (e.g., social distance, peer centrality) are set to zero for all individuals not originally assigned to Treatment 2 or Treatment 3 in specifications that include all treatment arms.

predictive of follow-up behaviors one year later (Table SA.9).

After FDR correction using q-values, we find that the knowledge effects remain significant in all three arms and the trained-alone effect on business intentions remains marginally significant ($q < 0.10$).²⁷ Pairing does not improve outcomes over training alone on average, and the difference between training alone and training with a peer on the additional steps index is marginally significant ($+0.10\sigma$ vs -0.12σ , $p = 0.06$), with training alone leading to higher effects. These findings do not necessarily imply that peers are unimportant. Instead, it could suggest that the value of a peer depends on the identity of that peer and the average effect combines both productive and unproductive matches. This motivates the analysis of peer identity that follows.

Table 1: Impact of training on immediate outcomes

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.789*** (0.247)	-0.139 (0.127)	0.253** (0.116)	0.101 (0.0878)	0.105 (0.0884)
Trained with Peer	0.704*** (0.253)	-0.145 (0.105)	0.0991 (0.132)	-0.120 (0.110)	-0.0277 (0.0953)
Trained with Peer + Connections Module	0.799*** (0.244)	-0.177 (0.117)	0.210* (0.111)	0.0511 (0.0982)	-0.000710 (0.0909)
Constant	0 (0.234)	-0 (0.0932)	0 (0.0793)	0 (0.0616)	0 (0.0620)
Observations	768	768	765	746	751
R-squared	0.123	0.007	0.009	0.007	0.004
p: Alone vs Peer	0.183	0.951	0.255	0.0596	0.161
p: Alone vs Peer+CM	0.906	0.617	0.702	0.575	0.205
p: Peer vs Peer+CM	0.331	0.646	0.305	0.147	0.748
q: Alone	0.0190	0.205	0.0840	0.205	0.205
q: Peer	0.0530	0.566	0.823	0.616	0.867
q: Peer+CM	0.0160	0.207	0.159	0.436	0.660

Cluster-robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This regression treats individuals in pure control villages as the base category. Standard errors are clustered at the village level. Results are robust to using heteroskedasticity-robust standard errors; the clustered SEs we report are weakly larger for the main effects, so clustering is conservative. Q-values are FDR-adjusted using the Benjamini et al. (2006) sharpened procedure, computed separately for each treatment coefficient across the five outcome indices.

Decomposing each index into its subcomponents (Online Appendix Tables SA.28–SA.32), we find that training alone significantly improves responses to three of the five knowledge questions, two of the three business intention sub-components, and three of the five take-up sub-components, namely willingness to take up mentoring from an organization supporting women entrepreneurs ($p < 0.10$), willingness to seek advice from community members ($p < 0.05$), and willingness to attend additional trainings ($p < 0.10$). Training with a peer significantly improves two knowledge subcomponents but is associated with a marginal reduction in non-agricultural investment aspirations. The average effect on the aspirations

²⁷Table SA.10 shows the types of businesses individuals report that they want to open after being trained, conditional on being ready to invest. We discuss these results further when we explore mechanisms.

index itself is small and slightly negative across all three arms, but this null masks heterogeneity by peer identity that we examine in the next subsection.

We also implement IV specifications capturing local average treatment effects among compliers and find that LATE coefficients are larger than the ITT estimates but follow the same qualitative pattern: training alone significantly increases knowledge and business intentions, while pairing does not deliver additional gains on average (Table SA.57).

Robustness: Even though balance checks show that treated and pure control groups are similar along a wide range of baseline characteristics, we verify that selection into the endline survey is not driving the observed results. We do this by (a) reweighting outcomes of the pure control group using predicted attendance probabilities from the treatment group and (b) trimming observations with low probability of attendance. Both approaches yield estimates similar in magnitude and significance to our main findings and are discussed in detail in the appendix (Tables SA.59 and SA.60).

For completeness, Online Appendix Table SA.55 also reports the analogous within-village ITT specification. As discussed in Section 2.3, within-village comparisons are affected by two-sided non-compliance and would mechanically attenuate the ITT effects. Training alone has a significant effect on the take-up index ($+0.24\sigma$, $p < 0.05$, $q < 0.10$), while the knowledge and intention coefficients are smaller and less precisely estimated. Importantly, our main results on heterogeneous peer effects are not affected by the choice of the comparison group as they compare trained individuals across treatment arms.

4.1 Effect of the Connections Module

Before proceeding to examine the effects of peer identity, we first focus on the effect of the connections module. We find that women assigned to the connections module share 6.5 contacts on average with each other and discuss how these connections can help them set up a business. Within this treatment arm, peers who share more connections are also significantly more likely to report wanting to start a business together, even after controlling for social closeness and peer centrality (Table E.5). Each additional contact exchanged by the pair is associated with an approximately 6 percentage point increase in the probability of wanting to start a business together. However, this does not lead to statistically significant gains in outcomes relative to the other arms. Notably, the interaction of the connections module with the number of contacts pooled is negative across all five indices (Table E.7), suggesting that sharing more contacts does not translate into improved short-term outcomes for this arm.

To unpack this, we first measure the pre-existing network overlap between pairs in this treatment arm by computing the share of common neighbors relative to the total combined unique neighbors of each pair. As shown in Table E.8, lower network overlap is associ-

ated with higher treatment effects of the connections module on the knowledge index and additional steps ($p < 0.10$).

Table E.4 further shows that, as one would expect, significantly more connections were pooled among dyads with a higher social distance (i.e. not socially close) and higher gap between network degrees, as these pairs are less likely to have overlapping social ties. This suggests that the contact-sharing channel operated as designed, even though it did not move outcomes. However, significantly more connections were shared between same-caste pairs (about 1.5 additional contacts). In magnitude, this effect dominates other predictors of contact sharing. A Lasso regression (Figure SA.5) also confirms that caste similarity is the strongest predictor of the number of contacts pooled, followed by own degree centrality and education.

Consistent with these findings, we find that individuals in castes with high homophily (i.e., social groups within the village that have an ingroup bias in terms of network links) share significantly fewer contacts (Table E.6).²⁸ Even among individuals in the same caste, those belonging to high-homophily castes share even fewer contacts with each other ($p = 0.01$). These results are discussed in more detail in Section 5.1.

We do not detect any effects of this module on respondents’ willingness to take advice from others in the village about setting up a business after the training, nor any effect on whether they actually talked to anyone about opening a business one year later. Combined, these patterns suggest that it may be difficult to encourage network interactions in settings with high homophily due to a reluctance to share and approach contacts across caste boundaries.

4.2 Effects of Peer Identity

The main results raise a natural question: why does pairing not improve outcomes on average? We now exploit variation in the identity of the matched peer to understand the conditions under which pairing can be helpful. We restrict the sample to treated individuals and take those trained alone as the omitted category.

First, we find that social proximity improves aspirations. As shown in Table 2 which reports our pre-registered continuous specification, a one unit increase in social distance reduces aspirations by 0.083 standard deviations relative to training alone ($p < 0.01$, $q = 0.025$), and this effect is attenuated but remains significant when we control flexibly for baseline covariates using post-double selection Lasso (Belloni et al. 2014) (Table SA.65). Consistent with this, the discrete specification (Appendix Table C.1) shows that the improvement in aspirations is concentrated among trainees paired with a socially close *and* more central peer: the Close $\times$ More Central interaction is $+0.50\sigma$ ($p < 0.10$) and these pairs have significantly

²⁸Homophily is measured using the inbreeding homophily index (Currarini et al. 2009), as defined in Section 2.

higher aspirations than those trained alone. Among trainees matched with a more central peer, the effect of social closeness on the aspirations index is positive and highly significant ($p=0.008$, $q=0.040$) as shown in Table C.1. This is in line with the literature on “aspirations windows” (Ray 2006, Genicot & Ray 2017) in which a peer who is close and modestly more central inspires, while a distant high-status peer does not.²⁹

Decomposing the aspirations index (Online Appendix Section SA8), we find that the effect of social distance is concentrated in *income aspirations*. Among paired trainees, each additional unit of social distance to the peer lowers stated monthly income aspirations by roughly NPR 11,700 ($p<0.01$, $q<0.01$; Table SA.41). The discrete specification points in the same direction: being matched with a socially close peer increases income aspirations by roughly NPR 30,650 for those matched with less central individuals ($p<0.05$; Table SA.42). Among individuals matched with a more central peer, social closeness significantly improves income and non-agricultural aspirations ($q\text{-value} = 0.027$) relative to those who are socially distant. The largest gains in income aspirations (relative to being trained alone) come from peers who are both socially close and more central ($p\approx 0.05$) although this does not survive FDR correction.

Second, peer centrality improves take-up of additional resources, with the clearest evidence at the subcomponent level. The response to peer degree is strongest for *advice-seeking*: each additional network connection of the peer raises the reported likelihood of seeking advice from community members about opening a business ($p<0.01$, $q<0.05$; Table SA.47), an effect that survives FDR correction across the take-up family. Willingness to take up mentoring (from FWEAN) and additional commitment based training also increase with peer degree, though more marginally (both $p<0.10$) (Table SA.27). Consistent with these component-level results, the index-level interaction of peer degree with take-up is positive in the continuous specification ($p\text{-value}<0.1$, Table 2) and strengthens under post-double-selection Lasso ($p<0.05$, Table SA.65); however, the index interaction does not survive FDR correction across the endline outcome family ($q=0.68$). It is important to note that the aspirations effect is driven by social proximity and its interaction with *relative* centrality (i.e., whether the peer has higher degree than the individual), while the take-up effect responds to *absolute* peer degree regardless of social distance.

4.3 Mechanisms

We asked women who attended the training in pairs to describe how pairing was useful compared to training alone. 37% of paired women report having received encouragement from their matched peer. Another 30% report that the training material was easier to grasp because of the pairing. About 18% say that the training helped because they shared network

²⁹In fact, testing whether a more central but socially *distant* peer lowers income aspirations relative to training alone, we find a significant reduction ($p=0.03$).

Table 2: Effects on endline outcomes by peer type

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained with Peer	0.384** (0.185)	0.382** (0.187)	0.083 (0.240)	-0.093 (0.202)	-0.240 (0.165)
Trained with Peer $\times$ Peer distance	-0.029 (0.028)	-0.083*** (0.029)	-0.024 (0.038)	0.005 (0.032)	0.020 (0.021)
Trained with Peer $\times$ Peer degree	-0.025 (0.022)	-0.024 (0.019)	-0.000 (0.028)	0.011 (0.022)	0.030* (0.017)
Observations	519	519	516	499	504
R-squared	0.076	0.014	0.017	0.028	0.036
q: Paired $\times$ Dist	0.789	0.025	1.000	1.000	0.789
q: Paired $\times$ Deg	0.677	0.677	0.677	0.677	0.677

Notes: Treated-only sample with T1 (Trained alone) is the omitted category. We additionally control for the individual’s own degree centrality in regressions that include relative comparisons between own and peer centrality. Standard errors are robust. The complementary discrete specification (Close, More Central, and their interaction) is reported in Appendix Table C.1 and yields consistent patterns.

contacts.

Our short-term results point to the following key channels: learning, planned collaborations, and encouragement.

Learning: Uniquely, we measure learning during the training itself through performance in interactive exercises, including projected profits from a hypothetical business plan that pairs developed together (see Online Appendix Section SA5 for details). Table E.3 shows that close pairs earn significantly more in this exercise than distant pairs ($p=0.05$) and those trained alone ($p<0.1$), while pairs who are far show no positive effect relative to those trained alone. This effect does not extend to individual game profit or the post-training knowledge index, suggesting that the benefit of social closeness may largely operate through better collaboration on joint tasks. Socially closer peers also perform better on a few questions testing their entrepreneurial knowledge than those matched with socially distant peers: each additional unit of social distance to the peer significantly reduces correct responses on two questions composing the index (both $p<0.05$, $q<0.05$; Table SA.39).

Planned Collaboration: Individuals who are socially close to their matched peer are also significantly more likely to want to open a business together and to meet in the future (Panel A of Table E.9). In contrast, being assigned to a peer with high centrality does not differentially affect these outcomes. Notably, when asked what type of business they would like to open, paired trainees are also significantly more likely to report wanting to open an agricultural business together (Table SA.10), and significantly less likely to want to open a shop or parlor. This is consistent with the *parma* or *bola* system in Nepalese villages, where peers support each other with agriculture-related tasks through reciprocity-based labor provision (Messerschmidt 1981, Sherpa 2005, Bhattarai 2006). These traditional forms of collaboration may lower the perceived cost of joint agricultural enterprise for paired individuals.

Encouragement: Being matched with a more central peer is associated with a 13.4 percentage point higher likelihood ($p < 0.10$) of reporting encouragement as the reason why the training was beneficial. Each additional peer network connection raises this likelihood by 3.0 percentage points ($p < 0.05$, Table SA.16). This evidence is best interpreted as central peers being motivators on the take-up margin rather than on aspirations. The effect of peer centrality on aspirations is ambiguous: absolute peer centrality on its own does not raise aspirations, and relative centrality is helpful only in combination with closeness.

Online Appendix Section SA5 provides additional detail on these and other channels, including additional checks using measures of learning, risk-sharing, and sharing connections.

4.4 Robustness

The key patterns documented above are robust to degree-corrected centrality measures that account for incomplete network sampling (Table SA.68).

The effect of training with a close and more central peer on aspirations is also robust to controlling for homophily, i.e., the effect of being paired with peers who are similar in baseline characteristics. To show this, we construct a similarity index combining whether individuals are in the same income group, age group, caste, marital status, or education.³⁰ Once we additionally control for similarity, the Close $\times$ More Central interaction on aspirations remains positive and significant ($+0.51\sigma$, $p < 0.05$; Table E.1). The interaction also persists when we instead control directly for peer characteristics such as caste, age, education, income, and marital status, remaining positive and marginally significant on aspirations ($+0.40\sigma$, $p < 0.10$; Table E.2). In both specifications, the same close-and-central interaction also raises business intentions by a comparable amount ($+0.37$ – 0.38σ , $p < 0.10$).

Importance of Network Variables using Random Forests. We complement the above findings with a predictive exercise using random forests (Breiman 2001). For each pre-specified index, we train a random forest model on the peer treatment sample that includes both network measures (e.g., peer degree, whether the peer is more central, social distance, interaction between centrality and closeness) and peer demographics (education, income, age, marital status, and caste). We assess variable importance by measuring how much the model’s prediction error increases when the values of a given predictor are randomly shuffled across observations. As seen in Panel A of Figure SA.4, network characteristics of the peer are consistently selected as the most important predictors of the knowledge, aspirations, and business indices.

³⁰As before, we use the method proposed in Anderson (2008) to construct the index.

5 Long Term Effects

We now discuss the effects on outcomes measured one year after the program. About 3% of individuals in the follow-up sample had opened new businesses one year after the training.

As shown in Table 3, training alone significantly improves the business steps index by 0.49 standard deviations, a result that remains statistically significant after adjusting for multiple hypothesis testing ($q < 0.05$). While the effects are positive on all sub-components of the steps index (see Section SA7 and Table SA.35 in the appendix), the effect is primarily driven by significant increases in whether respondents have opened a new savings account or whether they have joined a new cooperative. Training with a peer shows no effects relative to the control group. Focusing on compliers, Table SA.58 shows that LATE estimates show a significant effect on the steps index for those trained alone ($+0.68\sigma$, $p < 0.01$). This effect remains robust to multiple hypothesis correction ($q < 0.05$). We do not detect any changes in the outcomes or mindset index, nor do we find improvements in record-keeping practices: treated respondents are no more likely to keep written records of their agricultural profits.

Table 3: Effects on long-term outcomes

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.194 (0.137)	0.488*** (0.156)	0.173 (0.188)
Trained with Peer	0.103 (0.115)	0.209 (0.127)	0.0110 (0.171)
Constant	0 (0.0699)	-0 (0.0675)	-0 (0.157)
Observations	472	475	475
R-squared	0.004	0.021	0.004
p: Alone vs Paired	0.428	0.0764	0.123
q: Alone	0.201	0.0140	0.323
q: Paired	0.603	0.503	1

Cluster-robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This regression treats the pure control group as the base category. Standard errors are clustered at the village level. Results are robust to using heteroskedasticity-robust standard errors; the clustered SEs we report are weakly larger for the main effects, so clustering is conservative. Q-values are FDR-adjusted using the Benjamini et al. (2006) sharpened procedure, computed separately for each treatment coefficient across the three outcome indices. Treatment effects on individual index components are discussed in the Supplementary Appendix (Section SA7).

5.1 Effects of Peer Identity

As before, the weak average effects in the peer arm likely reflect a combination of productive and unproductive matches. To test for this, we decompose the effects by pair types using continuous and discrete measures of social distance and peer centrality. As in the endline

analysis, the sample is restricted to treated individuals with those trained alone as the omitted base category. The full component estimates that unpack which individual items drive these peer effects at the follow-up are reported in Online Appendix Section SA8.

Two key findings emerge. First, the benefits of being matched with a close peer (relative to a socially distant peer) extend to the longer term. In Table 4, individuals trained with a peer with a high social distance have significantly lower treatment effects on the outcomes and mindset index ($p\text{-value}<0.1$, $q\text{-value}=0.12$). Decomposing these indices (Online Appendix Section SA8), we find that the effect of distance on the mindset index is driven by stated plans to start a business and by business self-efficacy: each additional unit of social distance to the peer significantly reduces both ($p<0.01$ and $p<0.05$, respectively, and $q<0.05$ in both cases; Table SA.53). The discrete specification (Appendix Table D.1) points in the same direction. Closer peers have higher outcomes and mindset indices, though these closeness coefficients are not individually significant. These results are aligned with an additional finding that those matched with a same-caste peer or with someone with low social distance are significantly more likely to have interacted with each other in the one year following the training (Panel B of Table E.9).

Second, the effects of centrality reverse sign relative to the endline. In the continuous specification (Table 4), each additional peer connection significantly lowers the steps index ($p<0.01$, $q<0.05$): trainees matched with a more connected peer take fewer steps toward starting a business. The discrete specification points the same way — those matched with a *more central* peer have a significantly lower steps index than those trained with a less central peer ($q<0.10$; Appendix Table D.1), conditional on being socially distant. It is also important to note that the effect of being paired with a more central peer, relative to a less central peer, on the steps index exceeds the minimum detectable effect at 80% power.³¹ When compared to those trained alone, the effect of peer centrality is even starker: matching with a more central peer significantly lowers the steps index whether or not the peer is socially close i.e., by 0.65σ for a more central, socially distant peer ($q=0.001$) and by 0.64σ for a close *and* more central peer ($q=0.003$; Appendix Table D.1).

The component decomposition also shows that being matched with a more central peer significantly reduces total savings, the loan amount taken, and investments in agriculture, relative to a less central peer ($p<0.05$, $q<0.10$; Table SA.52), again among socially distant pairs. Compared to those trained alone, these level effects are sizable and survive FDR correction: a more central, socially distant peer reduces the loan amount taken ($q=0.004$),

³¹**Power Calculations:** Since these peer-type comparisons are estimated on a smaller subsample, we assess our statistical power to detect differences. We find that the minimum detectable effect at 80% power (two-sided, 5% significance level) is approximately 0.43σ for the more-central versus less-central peer comparison and approximately 0.47σ for the socially close versus socially distant peer comparison. The estimated long-term effect of peer centrality on the Steps Index exceeds this threshold. It is approximately 0.51σ in an OLS regression comparing trainees matched with a more-central rather than a less-central peer.

total savings ($q=0.03$), and agricultural investment ($q=0.03$; Table SA.52). Consistent with this, in the continuous specification each additional peer connection significantly lowers the loan amount taken ($p<0.01$, $q<0.10$; Table SA.51) and savings ($p<0.1$).

5.2 Mechanisms

Disentangling mechanisms, we find that individuals who are trained with a peer are significantly more likely than the control group to report discussing business-related concerns with anyone in the village one year later, though the overall rate of such discussions remains very low (Table E.11). When asked whether they spoke to their matched peer, 31% report having done so. When asked what they spoke about, 83% sought advice, while the remainder discussed borrowing or lending money and setting up businesses. However, these network interactions were significantly more likely to take place (compared to the control group) when the matched peer was socially close but less central. Within the paired arm, such interactions are significantly lower if the matched peer was socially distant (Table E.11). This is likely a combination of two effects: socially distant or central peers are less available for sustained interaction themselves and such pairings also encourage individuals to have fewer interactions with others.

To account for changes in financial assistance, one year following the training, we measured the number of savings groups that the respondent was a part of and whether they had joined a new cooperative in the village. We find that those treated alone are significantly more likely to have joined a new cooperative, but the effect is not significant for those trained with a peer, suggesting that the paired treatments did not increase engagement with the wider network for saving or risk-sharing (Table E.10). However, the lower rate of joining cooperatives among paired individuals could potentially be consistent with bilateral risk-sharing between training partners substituting for formal cooperative membership, but this is suggestive and we do not have direct evidence for this. Consistent with the centrality effects discussed above, trainees matched with a more central and socially distant peer are significantly less likely than those trained alone to join a cooperative ($q=0.04$; Table SA.52). Online Appendix Section SA5 provides further detail on longer-term mechanisms (Table E.10).

5.3 Robustness

These long-term peer heterogeneity results are robust to post-double selection Lasso (Table SA.66): social distance continues to reduce outcomes, while more central peers continue to reduce the steps index. Results are also robust to degree correction accounting for network sampling (Table SA.69). Using inverse probability weighting to account for follow-up attrition (Tables SA.62–SA.63), the negative effects of social distance on outcomes and mindset remain significant ($p<0.10$), while more central peers continue to significantly reduce steps ($p<0.05$).

Table 4: Effects on follow up outcomes by peer type

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained with Peer	0.346 (0.328)	-0.042 (0.387)	0.316 (0.371)
Trained with Peer $\times$ Peer distance	-0.098* (0.054)	0.018 (0.075)	-0.101* (0.053)
Trained with Peer $\times$ Peer degree	-0.017 (0.033)	-0.073*** (0.028)	-0.023 (0.036)
Observations	320	323	323
R-squared	0.010	0.024	0.017
q: Paired $\times$ Dist	0.119	0.368	0.119
q: Paired $\times$ Deg	0.664	0.029	0.664

Notes: Treated-only sample with T1 (Trained alone) is the omitted category. We additionally control for the individual’s own degree centrality in regressions that include relative comparisons between own and peer centrality. Standard errors are robust.

Our key endline and follow-up results are robust: lower social distance improves aspirations in the short term and business mindset outcomes in the longer run, while higher peer centrality reduces the number of steps taken toward opening a business. The effect of peer centrality reverses sign over time. Any marginal positive short-run effects of central peers do not translate into sustained improvements.

Importance of Network Variables using Random Forests. As before, we implement random forests for the paired treatment arms, and rank peer characteristics by how much the model prediction error increases when each variable is randomly permuted. As shown in Panel B of Figure SA.4, peer network characteristics consistently emerge as one of the top 3 most important predictors of follow-up outcomes, often surpassing peer demographic characteristics.

6 Estimation of Peer Effects

We present a simple theoretical framework in the appendix (Section F) in which a peer’s network position affects entrepreneurial effort through two competing channels: motivational benefits that increase with centrality and social closeness, and collaboration costs that rise with centrality but decline with closeness. The model generates three key predictions: (i) closeness increases entrepreneurial effort (Proposition 1), (ii) peer centrality increases entrepreneurial effort only when its motivational benefit exceeds the associated collaboration cost (Proposition 2), and (iii) the effect of peer centrality can reverse over time as interaction costs become more pressing after the training ends, for instance, due to the lack of an experimenter-led platform for interaction (Proposition 3). We now test these predictions by estimating a peer-effects specification that exploits the random assignment of peers.

6.1 Peer Effects by Network Position

We use data from treated individuals in the paired treatment arms to test whether peer effects differ by the peer’s network position.³² To do this, we first construct an outcome index in each wave. We capture short-term effects by constructing a standardized Anderson Index (Anderson 2008) combining all the endline variables that enter the knowledge, business aspirations, business, steps, and take-up indices. Similarly, we combine all follow-up variables into a single outcome index. We interpret these outcome indices as reduced-form proxies for entrepreneurial effort. Using these measures, we estimate the following equation:

$$e_i^* = \theta + \theta_o \mathbf{X}_i + \theta_p \mathbf{X}_j + \theta_e e_j^* + \theta_d c_{ij} e_j^* + \theta_\phi \phi_j e_j^* + \varepsilon_i$$

where e_i^* is the outcomes index of individual i , c_{ij} is the social distance between i and j , and ϕ_j is the peer’s degree centrality.³³ The vector $\mathbf{X}_i$ includes individual characteristics such as age, education, caste, income, and number of social connections, and $\mathbf{X}_j$ includes the analogous peer characteristics. We estimate the equation for the endline and follow-up waves separately because the empirical analysis has previously shown that centrality plays a different role depending on the time horizon over which effects are measured.

While random pairing addresses endogenous selection into pairs, the lack of exogenous variation in peer outcomes does not, on its own, separate the *endogenous* effect of the peer’s contemporaneous outcome from own effort—the “reflection problem” arising from the simultaneous effect of peer effort on own effort (Manski 1993).

The results are shown in Table 5.³⁴ We detect heterogeneity in the magnitude of peer effects depending on how central the peers are – the association between peer outcomes and own outcomes is stronger for respondents matched with more connected peers. This endline effect is robust to controlling for additional own and peer characteristics, including their age, caste, income, and education level. However, these effects vanish in the follow-up wave in this specification (Table 5). This pattern is consistent with our proposed framework: centrality can improve outcomes in the short run when the motivational benefit dominates, but its effects disappear in the long run as collaboration costs become more salient (Propositions 2 and 3 of the model presented in Section F).

³²Throughout this section, we only include treated individuals. We include non-compliers or non-randomized participants who were treated with a peer, as the estimation of peer effects only relies on random pairing.

³³We denote outcomes index as e_i^* as it is a proxy for entrepreneurial effort and is an equilibrium object derived in Section F in the appendix.

³⁴The sample for the paired treatments in the follow-up is smaller, as we only include cases where we observe both the individual and their peer after 1 year.

Table 5: Effect of closeness and centrality in the short and long run

VARIABLES	(1) Outcome (Endline)	(2) Outcome (Endline)	(3) Outcome (Follow-up)	(4) Outcome (Follow-up)
Peer Outcome	-0.0287 (0.104)	-0.0915 (0.0951)	0.180 (0.202)	0.216 (0.193)
Close $\times$ Peer Outcome	-0.116 (0.0872)	-0.0905 (0.0760)	0.171 (0.161)	0.117 (0.169)
Peer Degree $\times$ Peer Outcome	0.0426* (0.0220)	0.0474** (0.0187)	-0.0115 (0.0289)	-0.0105 (0.0281)
Constant	0.199*** (0.0424)	1.047*** (0.272)	-0.0120 (0.0680)	0.919 (0.756)
Observations	454	452	186	186
R-squared	0.027	0.212	0.047	0.160

Notes: This table shows the effect of the matched peer’s outcome on the individual’s own outcome. The first two columns use data from the endline wave, while the last two columns use data from the follow-up wave. Columns (2) and (4) additionally control for individual and peer characteristics.

6.2 Addressing the Reflection Problem

We exploit the panel structure by regressing each individual’s follow-up outcome on their peer’s *endline* outcome, which addresses the reflection problem (Manski 1993) that random matching alone does not. Table 6 reports the results. All columns regress own follow-up outcome on the peer’s endline outcome, the peer’s centrality and closeness, and their interactions with the peer outcome. Columns 3 and 4 add the peer’s follow-up outcome, allowing for the possibility that the peer’s endline outcome is endogenous, being correlated with their own follow-up outcome and therefore with the individual’s. Columns 2 and 4 include village fixed effects.

We find that peer centrality significantly reduces the association between peer endline outcomes and own follow-up outcomes. Each additional peer connection significantly reduces this effect by 0.07–0.09 SD across specifications. Figure 2 plots the marginal peer effect at each level of peer degree, showing how being paired with a more connected peer can adversely affect longer-term outcomes. The marginal effect is positive (≈ 0.3 SD) for the least connected peers and turns negative for the most connected, while the average peer effect across peers with various degrees is close to zero and statistically insignificant.

Some key features of this specification are worth emphasizing. First, we regress an individual’s *follow-up* outcome on the peer’s *endline* outcome to remove the simultaneity that generates the reflection problem, because the peer’s endline outcome is realized before the individual’s follow-up outcome. Second, columns (2) and (4) include village fixed effects to account for village-specific shocks or characteristics and the interaction term is stable in sign and magnitude across these specifications. The remaining concern is that an individual and their peer may have correlated outcomes for reasons unrelated to peer influence, such as a

Table 6: Estimates of Peer Effects

VARIABLES	(1) Outcome (Follow-up)	(2) Outcome (Follow-up)	(3) Outcome (Follow-up)	(4) Outcome (Follow-up)
Peer Outcome (Endline)	0.365** (0.153)	0.351** (0.171)	0.322** (0.150)	0.424** (0.189)
Peer Degree $\times$ Peer Outcome	-0.0726** (0.0321)	-0.0683** (0.0320)	-0.0777*** (0.0283)	-0.0913*** (0.0299)
Close $\times$ Peer Outcome	-0.137 (0.199)	-0.141 (0.196)	-0.0344 (0.182)	-0.0926 (0.209)
Observations	288	288	186	186
R-squared	0.055	0.197	0.099	0.241
Controls	Yes	Yes	Yes	Yes
Peer follow-up outcome	No	No	Yes	Yes
Village fixed effects	No	Yes	No	Yes

Notes: The table reports the effect of the matched peer's endline outcome on the individual's own follow-up outcome. Close $\times$ Peer Outcome interacts the peer outcome with an indicator for whether the peer was socially close. Peer Degree $\times$ Peer Outcome interacts the peer outcome with the peer's degree centrality. All columns control for closeness and peer's degree. Columns (3) and (4) additionally control for the peers' follow up outcome. Columns (2) and (4) include village fixed effects.

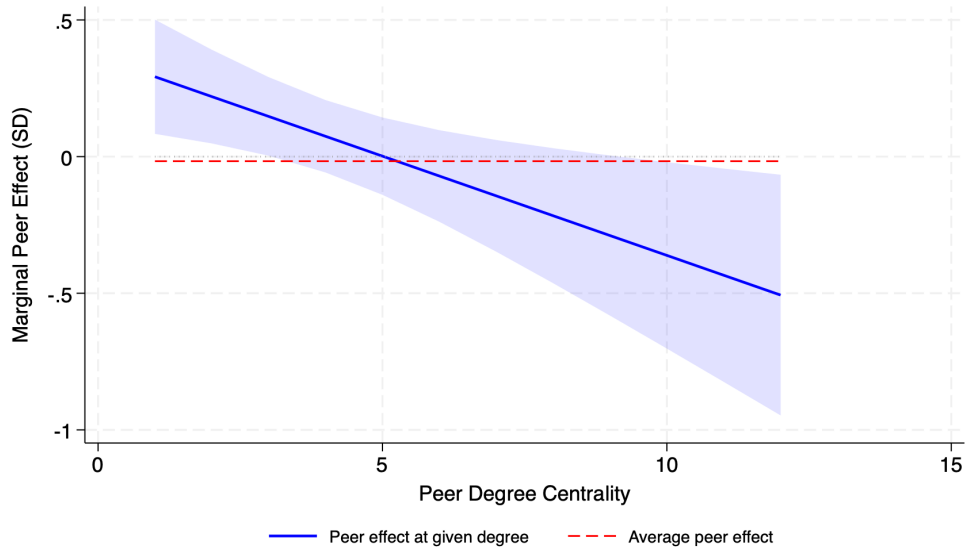


Figure 2: Marginal peer effect by peer degree centrality

Notes: The figure plots the estimated marginal effect of a 1 SD increase in peer endline outcome on own follow-up outcome, at each level of peer degree centrality, based on specification (1) of Table 6. The shaded area represents 90% confidence intervals. The dashed red line shows the sample-average peer effect, computed by averaging the marginal effect across the actual distribution of peer degrees.

shock they both experienced during the training. While village fixed effects absorb shocks that differ across villages, we argue that any residual variation across sessions within a village is likely to be small, because the training content was homogeneous and was delivered by the same six trainers, each prepared by a single master trainer. Moreover, our key result is that

the sign of the peer effect changes as peer degree rises. Thus, a shared shock would affect it only if the shock were itself correlated with the peer’s degree.

6.3 Discussion

While pairings based on social distance can be easy to implement, the feasibility of pairing based on network centralities and the potential returns from such pairings remain unclear. First, it is not clear if it is feasible to collect network data and engineer such pairs in the first place. We find that 63% of individuals in our setting guess the number of connections of their randomly matched peer within a range of ± 2 connections (≈ 1 standard deviation), and about 41% guess it correctly within a range of ± 1 connection. This aligns with findings in the literature showing that individuals can possess accurate knowledge about who is central in their network (Banerjee et al. 2019), and it may not require additional data collection.

Second, it is not clear how many such pairs we can form within a village, given the limited number of ‘central’ individuals in a fixed village network. To check this, we simulate 10,000 counterfactual reassignments within each village: we group individuals into random dyads, compute each dyad’s degree-centrality gap, average across dyads, and repeat. The distribution of the average centrality gaps (Figure SA.8) shows that while some villages have tightly clustered gaps, the majority display much wider distributions, implying that pairs with large gaps in centralities are indeed feasible to implement. Strategic matching on network position is therefore feasible and, depending on the goal, could increase short-term effects or decrease the negative longer-term ones.

Finally, even where strategic pairing is feasible, it is not clear if it generates aggregate gains or redistributes outcomes across individuals. The two network dimensions differ in this respect. Social distance is a property of the *pair* and is symmetric, so pairing two individuals who are socially close improves the outcomes of both. Pairing on closeness can raise aggregate outcomes and is Pareto-improving. Centrality, on the other hand, is an attribute of the *individual*. Pairing a less central person with a more central peer may benefit one while harming the other, so it is not obvious whether reassignment would change overall outcomes or only have distributive effects.

To be precise, consider regressing the steps index on an individual’s own degree d_i , their peer’s degree $d_{p(i)}$, and their interaction:

$$y_i = \alpha + \beta_1 d_i + \beta_2 d_{p(i)} + \beta_3 d_i d_{p(i)} + \varepsilon_i.$$

Summing across all individuals in a village, the aggregate outcome depends on $\sum_i d_i$, $\sum_i d_{p(i)}$, and $\sum_i d_i d_{p(i)}$. The first two are invariant to how individuals are paired, since every individual contributes their own degree once and appears as some other individual’s peer exactly once. Only the interaction term $\sum_i d_i d_{p(i)}$ varies with the assignment, so aggregate out-

comes can increase or decrease depending on peer assignment only if $\beta_3 \neq 0$. Estimating this specification, we find that a more central peer significantly lowers an individual’s own steps ($\hat{\beta}_2 < 0$, $p < 0.05$), while the interaction is small and statistically indistinguishable from zero. Centrality-based pairing is therefore largely redistributive, reallocating which individuals take steps toward a business rather than affecting aggregate outcomes.

7 Conclusion

Development programs frequently incorporate peer interactions, yet the peer’s identity is rarely treated as a design choice. In this paper, we show that who individuals are paired with matters substantially for outcomes, and that the optimal peer depends on the time horizon. Socially close peers generate benefits in both the short and long term: they raise aspirations at endline, collaborate better during training, are more likely to remain in contact, and are more likely to discuss business ideas a year later.

However, one year later, more central peers are associated with significantly worse outcomes. Our estimates of peer effects suggest that this heterogeneity is large enough to reverse the sign of the average effect, implying that programs treating peers as interchangeable may not maximize treatment effects. For entrepreneurship programs in developing countries, where starting a business requires both initial motivation and ongoing support, this distinction is consequential as the peer who raises immediate engagement may not be able to provide support over time. These findings also highlight the limits of the role-model view of peers. A more central peer can potentially act as an effective motivator in the short run, raising aspirations and take-up in a manner consistent with existing evidence on role models (Bernard et al. 2026), but this effect may not carry over into longer-term outcomes when those outcomes require sustained interaction.

We also test whether peers can serve as bridges to the broader social network. Despite structural conditions that strongly favor bridging (73% of pairs share no friends in common, and contact sharing increases with social distance), we find no significant impact on outcomes. Contact sharing is concentrated among individuals of the same caste, and even within same-caste pairs, those in castes with higher homophily share fewer contacts. This suggests that in settings where social networks are organized along group boundaries, facilitating introductions through peers may not be sufficient to generate new productive ties. Future interventions may need to address the underlying reluctance to engage across social boundaries, rather than simply reducing the cost of reaching new contacts.

Finally, our findings suggest that strategic peer assignment, i.e., pairing individuals based on network position, is feasible. The gains from such pairing, however, depend on which network dimension is targeted: pairing on social closeness raises outcomes for both members of a pair and can increase aggregate outcomes (Pareto-improving), whereas pairing on cen-

trality is largely redistributive. The majority of individuals in our sample correctly estimate their peer’s number of connections within one standard deviation, implying that network data collection may not be necessary. Simulations of counterfactual reassignments indicate substantial scope for design improvements. While our results demonstrate that such strategic pairings are feasible and may not require additional network data, they do not capture the private costs of networking. These costs may be higher when interacting with central peers than with less central but socially close ones. Costs can range from informational (e.g., knowing who is central, which we show is not binding in our setting) to interactional (e.g., confidence, time). While we cannot measure such costs, reducing them through appropriate platforms may help leverage the benefits of strategic networking.

Appendix

A Summary Statistics

Table A.1: Summary statistics

Age	37.98	(10.85)
Divorced	0.00141	(0.0375)
Married	0.918	(0.274)
Unmarried	0.0669	(0.250)
Widow	0.0134	(0.115)
Higher Education (Class 11, 12)	0.104	(0.305)
Informal Education	0.126	(0.332)
No Education	0.326	(0.469)
Primary (Class 1-5)	0.155	(0.362)
Secondary (Class 6-10)	0.249	(0.432)
University	0.0402	(0.196)
Belongs to Upper Caste	0.341	(0.474)
Degree Centrality	4.585	(2.140)
Eigen Vector Centrality	0.00978	(0.0133)
Own Non Agr. Business	0.220	(0.415)
Feel not Capable	0.277	(0.448)
Willing to Open a Business	0.419	(0.494)
Risk Aversion (1-6)	4.610	(1.406)
Have no skills	0.496	(0.500)
Financial Reasons	0.239	(0.427)
No support from family	0.0129	(0.113)
Aspirations (Agricultural Expenditure)	245802.6	(331604.3)
Aspirations (Non Agri. Expenditure)	443675.3	(861565.3)
Aspirations (Income)	141669.0	(658774.2)
Aspires to Higher Income	0.890	(0.313)
Aspires to Higher Non Agri. Exp	0.229	(0.420)
Observations	2840	

Notes: This table reports summary statistics (i.e. mean and standard deviation) for baseline characteristics. The sample includes all baseline respondents with non-missing data.

B Implementation

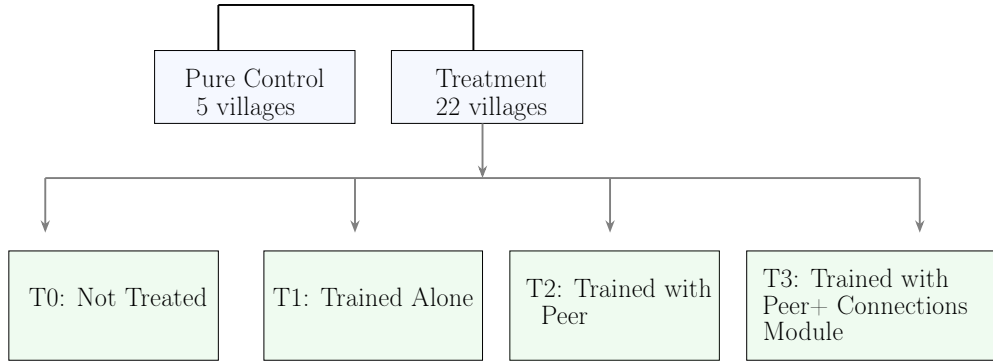


Figure B.1: The figure illustrates the random assignment of 27 villages into pure control and treatment groups. Within treatment villages, individuals were further randomized into one of three arms: trained alone (T1), trained with a peer (T2), or trained with a peer plus a connections module (T3).

Figure B.2: Contact-Sharing List for women in the Connections Module

मूल (नेपाली)	English translation
चिनजान बारे विवरण	Getting-to-know-each-other: details
१. सहभागीको नाम: व्यक्तिगत id	1. Participant's name: Personal ID:
२. सहभागीको नाम: व्यक्तिगत id	2. Participant's name: Personal ID:
१. तपाईं कसरी एक अर्कालाई व्यवसाय खोल्न / खोलेपछि मद्दत गर्न सक्नुहुन्छ?	1. How can you help each other to open a business, or after opening one?
२. यस गाउँमा तपाईंले चिनेका साथीहरूको बारेमा सोच्नुहोस्। तिनीहरूले तपाईंलाई कसरी व्यापार खोल्न मद्दत गर्न सक्छन्?	2. Think about the friends you know in this village. How can they help you open a business?

तपाईंलाई व्यापार खोल्न मद्दत गर्न सक्ने साथीहरू / Friends who could help you start a business

#	साथीहरूको नाम Name of friend	कसरी मद्दत गर्न सक्छन् How they can help
१		
२		
३		
४		
५		
६		
७		
८		
९		
१०		

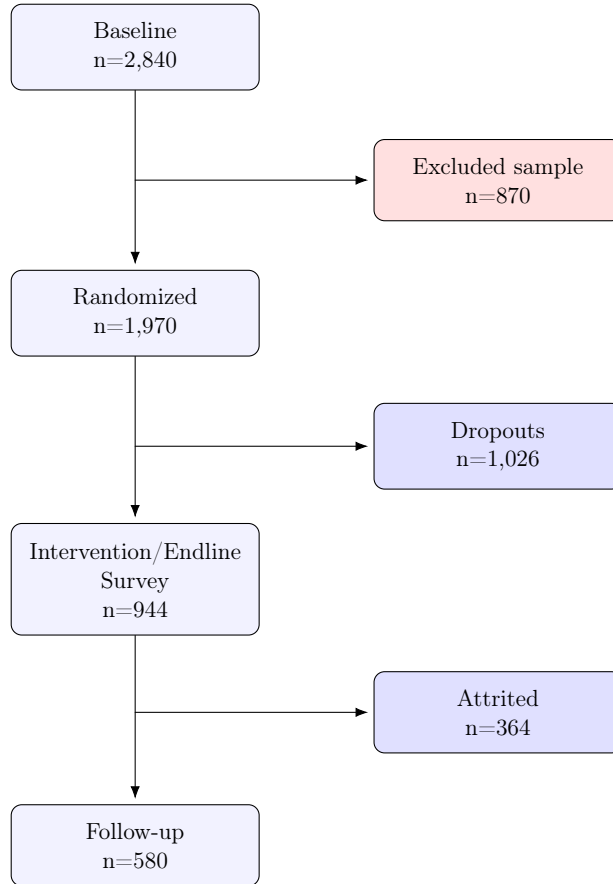


Figure B.3: The figure shows participant numbers at each stage of data collection. From the baseline sample ($n = 2,840$), 1,970 women across 27 of the 31 villages (after excluding 4) were randomized into treatment arms, while 870 were excluded, including those already operating businesses and the four villages where training could not be implemented ex-ante. Of those randomized, 944 participated in the intervention and endline survey, and 580 were successfully followed up one year later. The main intent-to-treat estimation sample excludes the within-village controls. Any individuals not in the randomized sample who nevertheless completed the training or appeared in the endline survey (e.g., excluded baseline respondents or new participants) are excluded from all intent-to-treat analyses. If such individuals were matched with participants in the randomized sample in the peer arms, the randomized participants are retained in the analysis while their non-randomized partners are excluded.

Table B.1: Implementation summary by experimental group

Group	Arrival & assignment revelation	Peer assignment	Training	Endline survey	Follow-up survey	Role in estimation
Pure control	Not invited to training	—	No training	In respondents' homes; in person	Phone survey, one year after training	Comparison group in endline and follow-up regressions
Within-village control	Not invited for training; assignment revealed at the training center prior to survey completion, informed that slots are random.	—	No training; some individuals were reassigned to fill vacant training slots	Training center, survey before training commenced; conducted in respondents' homes in a small number of cases	Phone survey, one year after training	Not used in main analysis due to two-sided non-compliance (see Section 2.3); comparison group in robustness specifications only
T1: Individual training	Invited to training center; assigned after arrival to individual training, informed that slots are random.	Not paired; seated individually	Three-day curriculum completed individually; seated in a different room compared to T2 and T3; activities completed alone	Training center, end of day 3	Phone survey, one year after training	Treatment arm; comparison group in peer specifications
T2: Peer training	Invited to training center; assigned after arrival to paired training, informed that slots are random.	Random pairing after arrival ^a	Three-day curriculum completed in assigned pairs; seated in a different room compared to T1; collaborative activities completed with assigned peer	Training center, end of day 3	Phone survey, one year after training	Treatment arm; pooled with T3 in some specifications
T3: Peer + connections module	Invited to training center; assigned after arrival to paired training with an additional connections module, informed that slots are random.	Same as T2 ^a	Three-day curriculum completed in assigned pairs with connections module; seated in a different room compared to T1	Training center, end of day 3	Phone survey, one year after training	Treatment arm; pooled with T2 in some specifications

Notes: ^a Some women assigned to paired training could not be matched and are flagged as “unmatched” in regressions. Randomization inference confirms that realized pairings are statistically indistinguishable from random assignment.

B.1 Compliance

Compliance is defined as a binary indicator equal to one if the individual's realized treatment status matched their assignment. Overall compliance in the endline sample is 68%, or 58% when excluding pure control villages. This includes those whom we could not find and who do not appear in the endline survey (i.e., dropouts), as well as those who did not adhere to their assigned treatment, e.g. did not get trained despite being assigned to training.

Compliance is not correlated with treatment status in the endline sample (Column 1 in Table B.2) but this pattern disappears if we exclude the pure control villages (Column 2). In that case, compliance is higher for those assigned to treatment, i.e., those invited for training were more likely to attend training compared to compliance among those asked not to attend. In treated villages, 82% of those who arrived at the venue and were assigned to training (under any of the treatments) were actually trained, while the remainder were in the actual control group. Compliance is much lower for the control group as some within-village controls were included in the training to fill spots left by non-attending treatment-assigned individuals. We do not find any key differences in the characteristics of compliers and non-compliers in this subsample, as shown in Table B.3, except that compliers are more likely to be married and more risk-averse.

Table B.2: Correlation of compliance with treatment status (endline sample)

VARIABLES	(1) Compliance	(2) Compliance
Trained Alone	-0.0562 (0.0929)	0.301*** (0.0501)
Trained with Peer	-0.105 (0.0932)	0.252*** (0.0511)
Trained with Peer + CM	-0.0982 (0.0955)	0.259*** (0.0528)
Constant	0.732*** (0.0849)	0.375*** (0.0366)
Observations	944	710
R-squared	0.010	0.058
Trained Alone vs Paired	0.276	0.327
Paired vs Paired+CM	0.868	0.901
Trained Alone vs Paired+CM	0.508	0.413
Robust standard errors in parentheses		
*** p<0.01, ** p<0.05, * p<0.1		

Notes: Compliance is defined as a binary measure indicating adherence to the assigned treatment in the endline sample. Column 2 excludes the pure control villages.

Table B.3: Baseline characteristics of compliers in treated villages (endline sample)

	(1) 0	(2) 1	(3) (1) vs. (2), p-value
Age	38.649	38.849	0.798
Income	24459.091	24311.509	0.940
No Education	0.352	0.366	0.712
Informal Education	0.154	0.156	0.950
Primary Education	0.141	0.176	0.209
Secondary Education	0.265	0.239	0.432
Higher Education	0.081	0.051	0.127
University Education	0.007	0.012	0.447
Upper Caste	0.331	0.324	0.834
Married	0.920	0.954	0.071
Divorced	0.000	0.000	
Widow	0.023	0.012	0.275
Degree Centrality	5.144	5.359	0.233
Willing to Open a Business	0.452	0.404	0.212
Risk Aversion (1-6)	4.525	4.781	0.016

Notes: This table compares baseline characteristics of individuals who complied with the assigned treatment with those who did not, in treated villages, excluding dropouts (i.e. who are not present in the endline survey).

Tables B.4 and B.5 look at these characteristics more closely. Table B.4 compares compliers and non-compliers within each arm. The few differences that arise are concentrated among within-village controls, some of whom filled empty training slots. Table B.5 restricts attention to the paired arm and compares those who were paired with those who were assigned a peer but could not be matched. The unmatched are less connected and less likely to be upper caste, and are otherwise similar in income, age, willingness to open a business, and education. Since we estimate peer effects among those who were actually paired, these differences affect the external validity of the peer effects rather than their internal validity, informing us for whom the estimates are most informative.

To address any issues arising due to compliance, we present intention-to-treat (ITT) estimates in the main results. The main ITT specification compares treated individuals with women in pure control villages and excludes within-village controls, since the within-village control group is partially contaminated by non-compliance. We also present robustness checks using the within-village control group as the comparison group, as well as LATE specifications that instrument actual treatment status with treatment assignment.

Table B.4: Baseline characteristics of compliers and non-compliers, by treatment arm

	Compliers	Non-compliers	Difference	p-value
Panel A: Control (within-village)				
Age	42.455	37.545	-4.909	0.003***
Income	18914.062	24882.353	5968.290	0.015**
No Education	0.500	0.227	-0.273	0.000***
Informal Education	0.182	0.164	-0.018	0.760
Primary Education	0.106	0.127	0.021	0.670
Secondary Education	0.167	0.382	0.215	0.001***
Higher Education	0.030	0.082	0.052	0.129
University Education	0.015	0.018	0.003	0.879
Upper Caste	0.288	0.400	0.112	0.127
Married	0.939	0.927	-0.012	0.754
Divorced	0.000	0.000	0.000	.
Widow	0.030	0.018	-0.012	0.625
Degree Centrality	4.258	5.245	0.988	0.002***
Willing to Open a Business	0.277	0.574	0.297	0.000***
Risk Aversion (1-6)	4.984	4.188	-0.796	0.000***
Panel B: Trained Alone				
Age	37.756	40.492	2.736	0.086*
Income	25447.899	22800.000	-2647.899	0.542
No Education	0.310	0.492	0.182	0.018**
Informal Education	0.151	0.197	0.046	0.448
Primary Education	0.230	0.082	-0.148	0.005***
Secondary Education	0.246	0.131	-0.115	0.050**
Higher Education	0.048	0.098	0.051	0.238
University Education	0.016	0.000	-0.016	0.158
Upper Caste	0.346	0.344	-0.002	0.977
Married	0.953	0.934	-0.018	0.621
Divorced	0.000	0.000	0.000	.
Widow	0.008	0.016	0.009	0.639
Degree Centrality	5.587	5.016	-0.571	0.107
Willing to Open a Business	0.413	0.344	-0.069	0.365
Risk Aversion (1-6)	4.940	4.733	-0.207	0.271
Panel C: Trained with Peer				
Age	38.394	38.719	0.324	0.779
Income	25322.115	24913.710	-408.406	0.896
No Education	0.358	0.394	0.036	0.509
Informal Education	0.151	0.126	-0.025	0.507
Primary Education	0.165	0.181	0.016	0.708
Secondary Education	0.257	0.228	-0.029	0.550
Higher Education	0.060	0.071	0.011	0.688
University Education	0.009	0.000	-0.009	0.158
Upper Caste	0.321	0.266	-0.055	0.272
Married	0.959	0.906	-0.052	0.073*
Divorced	0.000	0.000	0.000	.
Widow	0.009	0.031	0.022	0.188
Degree Centrality	5.560	5.117	-0.442	0.103
Willing to Open a Business	0.437	0.397	-0.041	0.470
Risk Aversion (1-6)	4.629	4.706	0.077	0.619

Notes: Baseline characteristics by compliance status within each assigned arm. Difference and p-value are reported from a regression of the baseline characteristic on a non-complier indicator with robust standard errors. Compliance=1 if realized treatment is the same as assignment. Sample only includes endline respondents, pure-control villages are dropped. *** p<0.01, ** p<0.05, * p<0.1.

Table B.5: Baseline characteristics of matched versus unmatched individuals in the paired arm

	Matched	Unmatched	Difference	p-value
Age	38.539	38.443	-0.096	0.942
Income	25447.368	24362.353	-1085.015	0.738
No Education	0.346	0.443	0.097	0.112
Informal Education	0.160	0.091	-0.069	0.074*
Primary Education	0.167	0.182	0.015	0.760
Secondary Education	0.257	0.216	-0.041	0.430
Higher Education	0.062	0.068	0.006	0.848
University Education	0.008	0.000	-0.008	0.158
Upper Caste	0.341	0.182	-0.159	0.002***
Married	0.934	0.955	0.020	0.452
Divorced	0.000	0.000	0.000	.
Widow	0.019	0.011	-0.008	0.574
Degree Centrality	5.531	5.000	-0.531	0.060*
Willing to Open a Business	0.438	0.376	-0.062	0.315
Risk Aversion (1-6)	4.603	4.812	0.209	0.186
Observations	258	88		

Notes: Among individuals in the paired arms, matched refers to those who were successfully paired and unmatched to those who were not. Difference and p-value are from a regression of the baseline characteristic on an unmatched indicator, with robust standard errors. The sample includes endline respondents in treated villages. *** p<0.01, ** p<0.05, * p<0.1.

B.2 Initial Dropouts

Table B.6 examines whether dropouts from our randomized sample (N=1970), i.e., those who did not appear in the endline survey (N=944), differ by treatment status. In the full sample (Column 1), dropout rates are higher among those in treated villages compared to the pure control group. This difference is driven by survey logistics rather than participant selection as women in treated villages completed the survey at the training center, whereas those in pure control villages were surveyed at home. Balance checks as well as additional robustness exercises ensure that this selection does not affect our main results.

Once we restrict attention to treated villages (Columns 2–3), this pattern disappears: dropout rates are not systematically different between those assigned to treatment and those in the control group within the same village. This suggests that the higher attrition observed in Column 1 reflects survey modality rather than selection related to treatment assignment i.e., surveys were largely conducted at training centers in treatment villages but at households in pure control villages. Importantly, survey attrition does not differ between those trained alone and those trained with a peer, which is the main comparison of interest. While dropout is somewhat higher in the Peer + Connections Module arm relative to the Alone and Peer arms, the differences are only marginally significant, and participants were not informed ex-ante about the module.

Table B.6: Initial dropouts by treatment status

VARIABLES	(1) Initial Dropout	(2) Initial Dropout	(3) Initial Dropout
Within-village control	0.204*** (0.0581)		
Trained Alone	0.176*** (0.0552)	-0.0276 (0.0351)	-0.0276 (0.0351)
Trained with Peer	0.183*** (0.0555)	-0.0213 (0.0351)	
Trained with Peer + CM	0.240*** (0.0559)	0.0364 (0.0349)	
Paired			0.00748 (0.0304)
Constant	0.357*** (0.0476)	0.561*** (0.0248)	0.561*** (0.0248)
Observations	1,970	1,606	1,606
R-squared	0.026	0.003	0.001
p-value: Alone vs. Peer	0.830	0.858	
p-value: Peer vs. Peer+CM	0.0291	0.0991	
p-value: Alone vs. Peer+CM	0.0489	0.0673	
p-value: Alone vs. Paired			0.249

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: This table compares the proportion of dropouts by intended treatment status within the randomized sample. Columns 2 and 3 exclude the pure control villages and Column 3 combines the paired treatment arms into one variable labeled “Paired”.

Comparing those who dropped out before the endline with those who remained, the two groups are balanced on the dimensions most relevant for selection into our outcomes — willingness to open a business, income, age, caste, and risk aversion. They differ only on education and connectedness, with dropouts being somewhat less connected. This connectedness gap does not unbalance our main comparison: treated and pure-control villages remain balanced on a vast majority of baseline variables as discussed in the main text (Table B.8). Imbalance only arises in the within-village contrast (Table SA.8), which is contaminated by two-sided non-compliance and which we therefore use only for robustness.

B.3 Follow-up Attrition

Table B.7 tests whether attrition in the follow-up survey differs by treatment assignment. Overall attrition between the endline and follow-up surveys is approximately 364/944 $\approx$ 38.6%. ³⁵ Attrition does not vary significantly across those trained alone or trained with a peer and pairwise tests confirm no differential attrition. This reduces concern that long-run results are driven by selection.

³⁵The 0.402 reported in the table is the attrition rate of the pure-control group; the regression sample ($N = 768$) excludes the within-village controls.

Table B.7: Correlation of follow-up survey attrition with treatment status

VARIABLES	(1) Attrition
Trained alone	-0.0506 (0.0559)
Trained with Peer	-0.0173 (0.0560)
Constant	0.402*** (0.0372)
Observations	768
R-squared	0.002
T1=Paired	0.623
Robust standard errors in parentheses	
*** p<0.01, ** p<0.05, * p<0.1	

Notes: Attrition is a binary variable equal to 1 if the individual is observed in the endline survey but not in the follow up. This regression compares attrition rates among individuals in the trained alone and paired treatment arms with those in the pure control group. Standard errors are clustered at the village level.

B.4 Balance Tables

Table B.8: Balance test for the endline sample

	0	1	2	3	(0) vs. (1), p-value	(0) vs. (2), p-value	(0) vs. (3), p-value	(1) vs. (2), p-value	(1) vs. (3), p-value	(2) vs. (3), p-value
Willing to Open Business	0.43	0.39	0.40	0.45	0.57	0.64	0.72	0.91	0.27	0.35
Income	24094.72	25317.20	25653.33	23907.65	0.74	0.63	0.95	0.92	0.69	0.56
Income Source- Agri.	0.92	0.87	0.84	0.91	0.27	0.12	0.88	0.49	0.18	0.07
Income Source- Business	0.11	0.06	0.10	0.10	0.31	0.88	0.85	0.20	0.18	0.91
Income Source- Job	0.02	0.02	0.01	0.03	0.99	0.23	0.64	0.33	0.66	0.36
Income Source- Remit.	0.01	0.01	0.01	0.02	0.67	0.80	0.37	0.56	0.11	0.54
Income Source- Other	0.04	0.03	0.04	0.01	0.65	0.83	0.16	0.76	0.27	0.07
Age	39.12	39.70	38.45	39.96	0.73	0.67	0.63	0.24	0.75	0.16
Elementary Education	0.18	0.15	0.19	0.16	0.36	0.67	0.68	0.33	0.70	0.49
Higher Education	0.09	0.09	0.05	0.07	0.99	0.37	0.66	0.29	0.43	0.54
Informal Education	0.32	0.31	0.30	0.33	0.89	0.77	0.82	0.81	0.63	0.54
University Education	0.01	0.01	0.01	0.01	0.74	0.75	0.82	0.99	0.92	0.92
Secondary Education	0.28	0.23	0.27	0.21	0.24	0.79	0.15	0.19	0.55	0.13
Degree	4.78	5.28	5.29	5.32	0.07	0.10	0.02	0.96	0.86	0.90
Brahmin	0.06	0.05	0.07	0.03	0.85	0.87	0.63	0.29	0.62	0.10
Chhetri	0.23	0.30	0.23	0.27	0.66	1.00	0.78	0.07	0.48	0.22
Dalit	0.06	0.01	0.04	0.01	0.21	0.72	0.22	0.31	0.92	0.23
Newar	0.46	0.41	0.50	0.45	0.86	0.87	0.99	0.16	0.39	0.42
Janjati	0.19	0.23	0.16	0.24	0.79	0.80	0.74	0.09	0.88	0.11

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The balance tests compare characteristics of pure control villages (0) with those in Treatment 1 (trained alone), Treatment 2, and Treatment 3 (both paired arms) for the endline sample. The p-values in Columns 5-10 indicate if the difference is statistically significant.

Table B.9: Balance test for the follow-up sample

	0	1	2	(0) vs. (1), p-value	(0) vs. (2), p-value	(1) vs. (2), p-value
Income	25759.514	27694.207	24465.089	0.671	0.697	0.498
Income Source- Agri.	0.914	0.869	0.897	0.345	0.655	0.532
Income Source- Business	0.043	0.000	0.009	0.113	0.225	0.176
Income Source- Job	0.036	0.008	0.023	0.027	0.363	0.160
Income Source- Remit.	0.007	0.008	0.019	0.921	0.309	0.266
Income Source- Other	0.043	0.025	0.028	0.421	0.370	0.841
Age	39.550	39.459	38.737	0.964	0.678	0.371
Elementary Education	0.207	0.180	0.183	0.516	0.553	0.948
Higher Education	0.093	0.090	0.070	0.962	0.668	0.479
Informal Education	0.336	0.270	0.329	0.434	0.929	0.335
University Education	0.007	0.008	0.009	0.922	0.817	0.914
Secondary Education	0.271	0.221	0.254	0.344	0.743	0.325
Degree	4.899	5.438	5.296	0.047	0.099	0.573
Brahmin	0.079	0.041	0.038	0.630	0.586	0.907
Chhetri	0.236	0.311	0.258	0.680	0.902	0.265
Newar	0.400	0.402	0.493	0.995	0.709	0.144
Dalit	0.079	0.008	0.019	0.246	0.342	0.620
Janjati	0.207	0.238	0.192	0.802	0.904	0.253

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The balance tests compare characteristics of those in the pure control group (0) with those in Treatment 1 (1), and paired treatment arms (2) for the follow-up sample. The p-values in Columns 4-6 indicate if the difference is statistically significant.

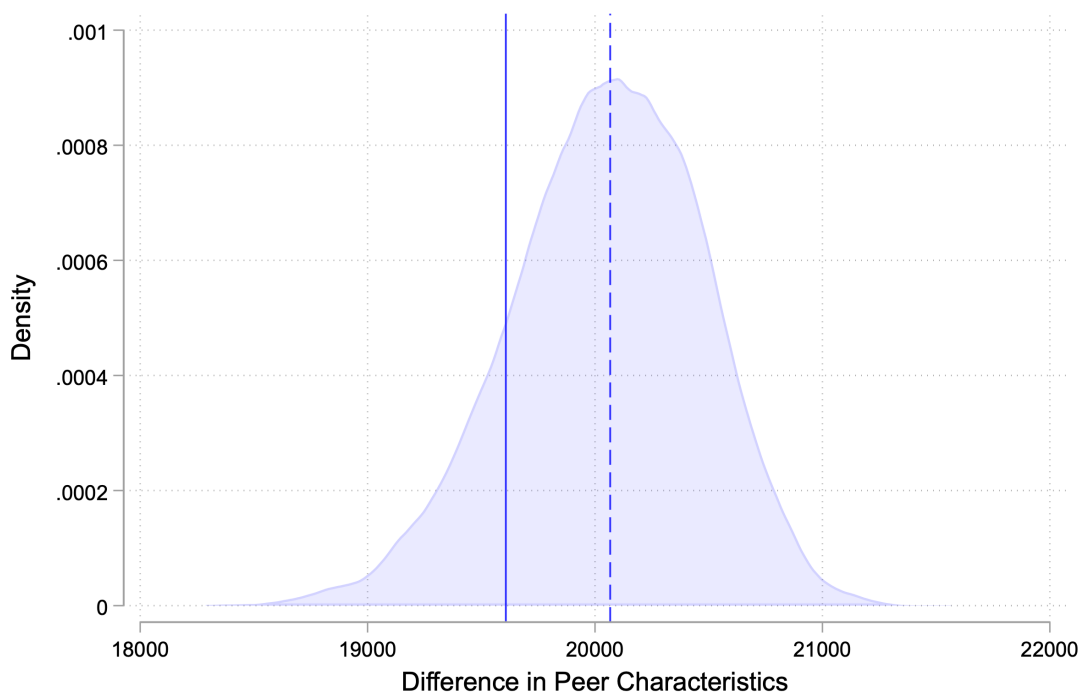
B.5 Randomization Inference for Peer Assignment

We test whether the peer pairs created during the intervention are systematically more or less similar than would be expected under random assignment. To do this, we construct a statistic that captures the overall difference between peers across a set of characteristics including income, source of income, age, education level, caste, marital status, willingness to start a business, income aspirations, agricultural expenditure aspirations, overall degree, outdegree, and risk aversion. In each iteration of the simulation, we match individuals within each village into dyads, compute the sum of the quadratic differences in their baseline characteristics, and add this across the entire sample.

Comparing the observed statistic to the distribution obtained from the simulations allows us to assess whether actual peers are unusually similar or dissimilar. The distribution and the statistic from the actual assignment is plotted in Figure B.4, indicating that the observed pairings are not systematically different from what random assignment would generate.

We find that the p-value of the test exceeds 0.2, so we cannot reject the null hypothesis that pairing is as good as random.

Figure B.4: Randomization inference on peer assignment



Notes: The histogram shows the distribution of peer similarity under 10,000 random re-assignments. The solid vertical line marks the statistic computed from actual peer pairs and the dashed vertical line plots the median difference computed from the re-assignments.

C Endline Results

Table C.1: Effects on endline outcomes by peer type: discrete specification

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained with Peer	0.208** (0.094)	-0.065 (0.096)	0.047 (0.135)	-0.022 (0.121)	-0.035 (0.092)
Trained with Peer $\times$ Close	-0.164 (0.142)	0.103 (0.136)	-0.127 (0.190)	-0.047 (0.169)	-0.060 (0.114)
Trained with Peer $\times$ More Central	-0.131 (0.126)	-0.138 (0.111)	-0.157 (0.182)	0.013 (0.163)	0.077 (0.114)
Trained with Peer $\times$ Close $\times$ More Central	0.323 (0.221)	0.499* (0.263)	0.398 (0.275)	0.030 (0.261)	0.134 (0.151)
Observations	524	524	521	504	509
R-squared	0.086	0.025	0.028	0.029	0.039
p: Close, more-central vs Trained Alone	0.126	0.070	0.342	0.883	0.161
q: Close, more-central vs Trained Alone	0.368	0.368	0.368	0.546	0.368
p: Closeness effect: more-central peers	0.361	0.008	0.185	0.932	0.465
q: Closeness effect: more-central peers	0.860	0.040	0.587	1.000	0.860

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. This is the discrete counterpart to the continuous specification in Table 2 of the main text. It includes Close, More Central, and their interaction Close $\times$ More Central; we additionally control for the individual's own degree centrality. The row “p: Closeness effect: more-central peers” reports the p-value for the marginal effect of social closeness for more central peers (Close + Close $\times$ More Central = 0). Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across the outcome indices.

D Follow-up Results

Table D.1: Effects on follow-up outcomes by peer type: discrete specification

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained with Peer	-0.139 (0.158)	-0.077 (0.223)	-0.177 (0.165)
Trained with Peer $\times$ Close	0.364 (0.241)	-0.033 (0.331)	0.340 (0.278)
Trained with Peer $\times$ More Central	-0.227 (0.215)	-0.576** (0.229)	-0.093 (0.205)
Trained with Peer $\times$ Close $\times$ More Central	0.209 (0.643)	0.050 (0.385)	-0.259 (0.481)
Observations	323	326	326
R-squared	0.017	0.030	0.015
p: Distant, more-central vs Trained Alone	0.072	0.000	0.141
q: Distant, more-central vs Trained Alone	0.078	0.001	0.104
p: Close, more-central vs Trained Alone	0.721	0.001	0.619
q: Close, more-central vs Trained Alone	0.926	0.003	0.926
p: Closeness effect: more-central peers	0.358	0.928	0.843
q: Closeness effect: more-central peers	1.000	1.000	1.000
p: Centrality effect: distant peers	0.292	0.012	0.649
q: Centrality effect: distant peers	0.414	0.039	0.763

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. This is the discrete counterpart to the continuous specification in Table 4 of the main text. It includes Close, More Central, and their interaction Close $\times$ More Central; we additionally control for the individual's own degree centrality. The row “p: Closeness effect: more-central peers” reports the p-value for the marginal effect of social closeness for more central peers (Close + Close $\times$ More Central = 0). Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across the outcome indices.

E Mechanisms

Table E.1: Impact of being trained with a central peer after controlling for similarity

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.774*** (0.248)	-0.150 (0.127)	0.228* (0.114)	0.0753 (0.0854)	0.0616 (0.0918)
Trained with Peer	0.980*** (0.277)	-0.259 (0.194)	0.558** (0.248)	0.196 (0.195)	0.210 (0.181)
Trained with Peer $\times$ Close	-0.134 (0.130)	0.0760 (0.129)	-0.131 (0.145)	-0.0657 (0.100)	-0.0689 (0.108)
Trained with Peer $\times$ More Central	-0.177* (0.0881)	-0.0955 (0.113)	-0.153 (0.146)	0.0350 (0.171)	0.0888 (0.0791)
Trained with Peer $\times$ Close $\times$ More Central	0.315 (0.239)	0.508** (0.230)	0.383* (0.224)	0.0315 (0.302)	0.126 (0.133)
Degree	0.0185 (0.0149)	0.0148 (0.0213)	0.0393** (0.0191)	0.0321* (0.0178)	0.0287* (0.0148)
Constant	-0.0883 (0.244)	-0.0764 (0.107)	-0.187 (0.134)	-0.146 (0.116)	-0.132 (0.0850)
Observations	756	756	753	734	740
R-squared	0.169	0.022	0.031	0.024	0.025
q: Close	1	1	1	1	1
q: More Central	0.383	0.680	0.680	1	0.680
q: Close $\times$ More Central	0.330	0.220	0.247	0.581	0.361

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. We additionally control for the “Similarity Index” which is a weighted index of similarity along characteristics including income, age, caste, marital status, and education and also control for degree-centrality of the individual. Cluster-robust standard errors in parentheses.

Table E.2: Effect of being matched with a central peer after controlling for other characteristics

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.773*** (0.249)	-0.150 (0.127)	0.228* (0.114)	0.0753 (0.0858)	0.0620 (0.0923)
Trained with Peer	1.198*** (0.366)	-0.190 (0.420)	0.232 (0.464)	-0.219 (0.432)	-0.0906 (0.297)
Trained with Peer $\times$ Close	-0.146 (0.140)	0.127 (0.142)	-0.151 (0.162)	-0.0213 (0.0908)	-0.0750 (0.100)
Trained with Peer $\times$ More Central	-0.159* (0.0919)	-0.0259 (0.120)	-0.125 (0.134)	0.0356 (0.173)	0.0806 (0.0854)
Trained with Peer $\times$ Close $\times$ More Central	0.282 (0.265)	0.396* (0.209)	0.372* (0.211)	0.0221 (0.288)	0.121 (0.137)
Degree	0.0197 (0.0150)	0.0158 (0.0217)	0.0403* (0.0197)	0.0321* (0.0183)	0.0279* (0.0154)
Constant	-0.0937 (0.246)	-0.0812 (0.109)	-0.191 (0.137)	-0.145 (0.118)	-0.128 (0.0871)
Observations	755	755	752	733	739
R-squared	0.177	0.035	0.038	0.035	0.030
q: Close	1	1	1	1	1
q: More Central	0.915	1	0.932	1	0.932
q: Close $\times$ More Central	0.404	0.290	0.290	0.920	0.404

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. We additionally control for peer characteristics such as peer age, caste, whether they are married, baseline income, and education. Cluster-robust standard errors in parentheses.

Table E.3: Learning by whether peers are socially close or not

VARIABLES	(1) Knowledge Index	(2) Profit (Game)	(3) Profit (Business Plan)
Trained alone	0.789*** (0.248)		
Paired (Close)	0.946*** (0.249)	55.71 (58.02)	1.011e+06* (566,051)
Paired (Far)	0.934*** (0.246)	48.64 (44.11)	-111,855 (254,422)
Constant	-0 (0.235)	273.3*** (34.14)	534,159*** (168,425)
Observations	763	432	432
R-squared	0.165	0.005	0.023
p: Close vs Far	0.894	0.897	0.0508
p: Close vs T1	0.187		

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages in Column 1 and Trained Alone (T1) in Columns 2 and 3. Cluster-robust standard errors in Column 1 and robust standard errors in Columns 2 and 3 are in parentheses.

Table E.4: Dyadic Regressions: Correlation of number of contacts pooled and peer characteristics

VARIABLES	(1) Number of links pooled	(2) Number of links pooled
Same age group		-0.603 (0.673)
Same caste		1.552*** (0.435)
Same education		0.689 (0.493)
Same income group		-0.267 (0.633)
Quadratic diff. in connections	0.0149* (0.00861)	0.0151* (0.00885)
Network Distance	0.176* (0.0906)	-0.00701 (0.116)
Network Overlap	-0.312 (0.467)	-0.549 (0.418)
Constant	5.570*** (0.368)	5.235*** (0.346)
Dyads	90	90
R-squared	0.072	0.205

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: This regression compares dyads in Treatment Arm 3, including all pairs who attended the connections module, regardless of assignment. Robust standard errors in parentheses.

Table E.5: Perceived future interactions by number of contacts shared

VARIABLES	(1) Pairs will start Business Together	(2) Pairs will meet in Future
Number of links pooled	0.0603*** (0.0186)	-0.00580 (0.0126)
Paired (Socially Close)	0.228** (0.0893)	0.0726 (0.0456)
Paired (High Centrality)	-0.0473 (0.0905)	-0.0862 (0.0685)
Constant	-0.00934 (0.115)	0.938*** (0.0724)
Observations	157	157
R-squared	0.091	0.024

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: Regressions include all individuals who attended Treatment 3, regardless of assignment. Robust standard errors in parentheses.

Table E.6: Relationship between caste homophily and number of contacts pooled

VARIABLES	(1) Number of contacts pooled	(2) Number of contacts pooled
High Homophily	-1.108** (0.461)	-0.476 (0.478)
Same caste	1.059*** (0.255)	2.569** (0.983)
High Homophily x Same Caste		-1.822* (1.006)
Constant	6.020*** (0.429)	5.556*** (0.460)
Observations	134	134
R-squared	0.163	0.211
p: High vs. Low Homophily (same-caste pairs)		0.0106

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: Regressions include all individuals who attended Treatment 3, regardless of assignment. Homophily is measured using the inbreeding homophily index (Currarini et al. 2009): $\frac{\text{share of same-caste links} - \text{share of own caste in population}}{1 - \text{share of own caste in population}}$, which ranges from -1 (maximum heterophily) through 0 (no excess same-type preference) to 1 (maximum homophily) and is comparable across groups of different sizes. Robust standard errors in parentheses.

Table E.7: Heterogeneous treatment effects by number of contacts shared

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.789*** (0.248)	-0.139 (0.127)	0.253** (0.116)	0.101 (0.0879)	0.105 (0.0886)
Trained with Peer	0.886*** (0.251)	-0.121 (0.105)	0.208 (0.123)	0.0146 (0.105)	0.0811 (0.100)
Trained with Peer + Connections Module	0.994*** (0.289)	0.0119 (0.167)	0.437*** (0.155)	0.367* (0.193)	0.265 (0.156)
T3 X Number of contacts pooled	-0.0746 (0.297)	-0.400 (0.300)	-0.287 (0.296)	-0.436* (0.250)	-0.319 (0.272)
Constant	-0 (0.235)	0 (0.0934)	0 (0.0794)	0 (0.0617)	-0 (0.0621)
Observations	743	743	740	723	727
R-squared	0.158	0.009	0.020	0.024	0.020

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Cluster-robust standard errors in parentheses.

Table E.8: Heterogeneous treatment effects by pre-existing network overlap (T3)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.774*** (0.248)	-0.150 (0.127)	0.228* (0.113)	0.0766 (0.0846)	0.0638 (0.0913)
Trained with a Peer	1.181*** (0.291)	0.0727 (0.221)	0.251 (0.282)	-0.00132 (0.166)	-0.178 (0.221)
Trained with a Peer + CM	1.319*** (0.333)	-0.0303 (0.247)	0.366 (0.324)	0.196 (0.191)	-0.173 (0.240)
T3 × Network Overlap	-3.581* (1.909)	4.268 (2.511)	-1.928 (1.966)	-3.704* (2.014)	-0.662 (1.030)
Degree	0.0177 (0.0140)	0.0150 (0.0201)	0.0400** (0.0193)	0.0295 (0.0180)	0.0242* (0.0141)
Peer Degree	-0.0239 (0.0211)	-0.0210 (0.0200)	-0.00406 (0.0311)	0.00709 (0.0166)	0.0313 (0.0214)
Social Distance	-0.0434 (0.0354)	-0.0420 (0.0249)	-0.0131 (0.0506)	-0.0112 (0.0311)	0.0256 (0.0260)
Constant	-0.0846 (0.243)	-0.0772 (0.104)	-0.190 (0.134)	-0.133 (0.116)	-0.110 (0.0819)
Observations	731	731	728	711	716
R-squared	0.169	0.025	0.028	0.031	0.024

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: Network overlap is equal to the share of common friends of i and j divided by the total number of unique friends. The base category is pure control villages, and the regression controls for own degree centrality, peer degree centrality, and social distance, since higher own or peer degree may mechanically reduce overlap, while closer social proximity may be associated with greater overlap. Cluster-robust standard errors in parentheses.

Table E.9: Effect of pair type on collaboration and communication

(a) Willingness to Start a Business Together

VARIABLES	(1) Pairs will meet in Future	(2) Pairs will start Business Together	(3) Pairs will meet in Future	(4) Pairs will start Business Together
Peer Centrality	-0.00227 (0.00720)	0.00349 (0.0137)		
Social Distance	-0.0295** (0.0132)	-0.0278 (0.0198)		
Paired (High Centrality)			-0.0402 (0.0475)	0.00969 (0.0688)
Paired (Close)			0.0822** (0.0415)	0.146** (0.0677)
Constant	0.977*** (0.0619)	0.478*** (0.113)	0.836*** (0.0465)	0.285*** (0.0804)
Observations	325	324	331	330
R-squared	0.642	0.164	0.644	0.177

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

(b) Talking to Matched Peer over 1 Year

VARIABLES	(1) Talk to matched peer	(2) Talk to matched peer	(3) Talk to matched peer	(4) Talk to matched peer
Peer Centrality	0.00472 (0.0197)	0.00365 (0.0194)		
Social Distance	-0.0465** (0.0235)	-0.0393* (0.0233)		
Same caste		0.247*** (0.0835)		0.282*** (0.0801)
Paired (High Centrality)			-0.00122 (0.0943)	-0.00736 (0.0909)
Paired (Close)			0.0853 (0.0961)	0.0660 (0.0938)
Constant	0.472*** (0.166)	0.246 (0.180)	0.283** (0.110)	0.0869 (0.117)
Observations	177	177	181	181
R-squared	0.160	0.193	0.136	0.179

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Panel (a) restricts the endline sample to participants assigned to the paired treatment arms. Panel (b) restricts the follow-up sample to the paired treatment arms. Robust standard errors in parentheses.

Table E.10: Treatment effects on network-based measures of saving

VARIABLES	(1) Number of Savings Groups	(2) Joined Cooperative in last year
Trained Alone	0.144 (0.308)	0.112** (0.0477)
Trained with Peer	0.136 (0.297)	0.00104 (0.0385)
Constant	1.864*** (0.283)	0.0929*** (0.0329)
Observations	474	475
R-squared	0.002	0.022

Cluster-robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Standard errors are clustered at the village level.

Table E.11: Effects on network interactions during the year after training

VARIABLES	(1) Talk to anyone about business	(2) Talk to anyone about business	(3) Talk to anyone about business
Trained Alone	0.0513 (0.0328)	0.0517 (0.0341)	
Close x More Central		0.0909 (0.0684)	
Close x Less Central		0.164** (0.0616)	
Far x Central		0.0588 (0.0427)	
Far x Less Central		0.0649 (0.0408)	
Degree		5.18e-05 (0.00603)	
Trained with Peer	0.0655** (0.0277)		
Peer Centrality			-0.00752 (0.00873)
Social Distance			-0.0353*** (0.0132)
Constant	0.0143 (0.0123)	0.0141 (0.0302)	0.261*** (0.0917)
Observations	475	465	201
R-squared	0.015	0.031	0.045
Paired=Nonpaired	0.689		

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: Column 1 uses pure control villages as the base category with pooled treatment indicators. Column 2 additionally decomposes the paired treatment arms into four peer-type categories. Column 3 restricts the sample to the paired treatment arms and regresses on continuous measures of peer centrality and social distance. Standard errors are clustered at the village level.

F Theoretical Framework

We now present a simple model to understand the patterns in the data and estimate peer effects, exploiting the variation in the identity of the peer. The model has two ingredients: a motivation channel through which peers provide encouragement, and a collaboration cost channel that captures the difficulty of sustained interactions.

Consider two matched women i and j who choose entrepreneurial effort $e_{it} \geq 0$ at time periods $t \in \{0, 1\}$, where $t = 0$ denotes the training/endline period and $t = 1$ denotes the one-year follow-up. Utility in period t can be written as follows:

$$U_{it} = \theta_t e_{it} - \frac{a}{2} e_{it}^2 + \beta_t M(c_{ij}, \phi_j) e_{it} e_{jt} - \kappa_t C(c_{ij}, \phi_j) e_{it} e_{jt}, \quad (1)$$

where $\theta > 0$, $a > 0$, c_{ij} is a measure of closeness between i and j and ϕ_j is peer j 's centrality. The parameters β_t and κ_t measure the relative importance of motivation and collaboration frictions at horizon t .

Motivation. The term $\beta_t M(c_{ij}, \phi_j) e_{it} e_{jt}$ captures the motivational benefit of working with a peer. A central peer (high ϕ_j) has more connections in the village and can serve as

a role model (Bernard et al. 2026, Riley 2024), signaling that entrepreneurship is feasible. Social closeness (high c_{ij}) can amplify this motivational effect, as individuals who are socially proximate can learn better together and support each other.

Collaboration cost. The term $\kappa_t C(c_{ij}, \phi_j) e_{it} e_{jt}$ captures the cost of coordinating with the peer. This cost has a natural microfoundation: a peer with ϕ_j connections allocates their available time and attention across these relationships, so the time they can devote to i could be proportional to $1/\phi_j$, which is decreasing in centrality. The training program subsidizes the cost of interaction, keeping κ_0 low. After the training ($t = 1$), maintaining contact requires active effort and the peer's time constraint can become binding, so $\kappa_1 > \kappa_0$. Social closeness, on the other hand, lowers coordination costs as closer individuals are more likely to share social contexts and face lower barriers to meeting.

Throughout, subscripts denote partial derivatives, so that $M_c \equiv \frac{\partial M(c, \phi)}{\partial c}$, $M_\phi \equiv \frac{\partial M(c, \phi)}{\partial \phi}$, $C_c \equiv \frac{\partial C(c, \phi)}{\partial c}$, and $C_\phi \equiv \frac{\partial C(c, \phi)}{\partial \phi}$. As per our empirical findings, $M_\phi(c, \phi) \geq 0$ (more central peers are more motivating), $M_c(c, \phi) \geq 0$ (closer peers are more motivating), $C_c(c, \phi) < 0$ (closeness reduces collaboration costs), and $C_\phi(c, \phi) > 0$ (collaborating with more central peers is costlier).

F.1 Best Responses and Equilibrium.

Differentiating (1) with respect to e_{it} yields

$$\frac{\partial U_{it}}{\partial e_{it}} = \theta_t - a e_{it} + [\beta_t M(c_{ij}, \phi_j) - \kappa_t C(c_{ij}, \phi_j)] e_{jt} = 0, \quad (2)$$

so the linear best response is

$$e_{it} = A_t + b_t(\phi_j, c_{ij}) e_{jt}, \quad A_t = \frac{\theta_t}{a}, \quad b_t(\phi_j, c_{ij}) = \frac{\beta_t M(c_{ij}, \phi_j) - \kappa_t C(c_{ij}, \phi_j)}{a}. \quad (3)$$

Lemma: Equilibrium existence and uniqueness If $a > 0$ and $b_t(\phi_i, c_{ij}) b_t(\phi_j, c_{ij}) < 1$, then a unique interior Nash equilibrium exists and is given by:

$$e_{it} = \frac{A_t + b_t(\phi_j, c_{ij}) A_t}{1 - b_t(\phi_i, c_{ij}) b_t(\phi_j, c_{ij})}. \quad (4)$$

This follows from solving the pair of best-response equations simultaneously.

Proposition 1: Closeness increases effort. Given $M_c(c_{ij}, \phi) \geq 0$, $C_c(c_{ij}, \phi) < 0$, and $b_t(\phi_i, c_{ij}) b_t(\phi_j, c_{ij}) < 1$, if peer efforts are strategic complements, that is $b_t(\phi_m, c_{ij}) > 0$ for

both $m \in \{i, j\}$ (equivalently $\beta_t M(c_{ij}, \phi) > \kappa_t C(c_{ij}, \phi)$), then holding (ϕ_i, ϕ_j) fixed,

$$\frac{\partial e_{it}}{\partial c_{ij}} > 0.$$

Intuitively, when the motivational effect of peer effort exceeds the cost of collaboration (i.e. $\beta_t M(c_{ij}, \phi) > \kappa_t C(c_{ij}, \phi)$), peer efforts are complements. In this case, greater closeness both raises motivation and lowers collaboration costs, and increases equilibrium effort.

Proposition 2: Peer centrality increases effort if and only if motivation is higher than collaboration cost. Given $M_\phi(c_{ij}, \phi) \geq 0$, $C_\phi(c_{ij}, \phi) > 0$, and $b_t(\phi_i, c_{ij}) b_t(\phi_j, c_{ij}) < 1$, if peer efforts are strategic complements, that is $b_t(\phi_m, c_{ij}) > 0$ for both $m \in \{i, j\}$, then the effect of peer centrality on effort is positive if and only if the motivational effect of centrality exceeds the associated collaboration cost, i.e.

$$\frac{\partial b_t(\phi_j, c_{ij})}{\partial \phi_j} = \frac{1}{a} \left(\beta_t M_\phi(c_{ij}, \phi_j) - \kappa_t C_\phi(c_{ij}, \phi_j) \right) > 0.$$

Intuitively, when effort choices are complements, higher peer centrality raises equilibrium effort only when it increases motivation more than it raises collaboration costs.

The next proposition highlights how the effect of peer centrality can vary over time due to its differential effects on motivation and collaboration.

Proposition 3: The effect of peer centrality can reverse over time If peer efforts are strategic complements, that is $b_t(\phi_m, c_{ij}) > 0$ for both $m \in \{i, j\}$, then higher peer centrality can increase effort in the short run but reduce it in the long run.

This is because at $t = 0$, when κ_0 is small enough, the motivational effect of centrality dominates and $\frac{\partial e_{i0}}{\partial \phi_j} > 0$. At $t = 1$, a larger κ_1 increases the relative cost of collaboration, and for sufficiently high κ_1 the effect can turn negative, so that $\frac{\partial e_{i1}}{\partial \phi_j} < 0$. Intuitively, while the motivational benefits of working with a central peer are immediate, the costs of coordination with a central (but likely busy) peer are larger in the longer term.

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Supplementary (Online) Appendix

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SA1 Baseline Descriptives

Table SA.1: Correlations between number of network connections and baseline characteristics

	Degree Centrality
Age	-0.00957
Divorced	0.0469*
Married	0.0800***
Unmarried	-0.0830***
Widow	-0.0257
Higher Education (Class 11, 12)	-0.0402*
Informal Education	0.0404*
No Education	-0.0744***
Primary (Class 1-5)	0.0216
Secondary (Class 6-10)	0.0847***
University	-0.0552**
Belongs to Upper Caste	0.0719***
Own Non Agr. Business	-0.0437*
Willing to Open a Business	0.0171
Risk Aversion (1-6)	-0.0681***
Aspirations (Agricultural Expenditure)	0.0236
Aspirations (Non Agri. Expenditure)	-0.0374
Aspirations (Income)	-0.0856***
Aspires to Higher Income	-0.0382
Aspires to Higher Non Agri. Exp	-0.0218

Notes: This table reports variable-wise correlations between degree-centrality (i.e. number of connections in the social network) and other baseline characteristics.

Table SA.2: Correlations between whether or not an individual has opened a business and their baseline characteristics

	Own Non Agr. Business
Age	-0.0944***
Divorced	0.00270
Married	-0.0354
Unmarried	0.0499**
Widow	-0.0250
Higher Education (Class 11, 12)	0.129***
Informal Education	-0.00791
No Education	-0.181***
Primary (Class 1-5)	0.00828
Secondary (Class 6-10)	0.00354
University	0.221***
Belongs to Upper Caste	-0.00378
Degree Centrality	-0.0437*
Eigen Vector Centrality	-0.0491**
Risk Aversion (1-6)	-0.0910***
Aspirations (Agricultural Expenditure)	-0.0364
Aspirations (Non Agri. Expenditure)	0.273***
Aspirations (Income)	0.155***
Aspires to Higher Income	-0.0429*
Aspires to Higher Non Agri. Exp	0.382***

Notes: This table reports variable-wise correlations between whether an individual has already opened a non-agricultural business and other baseline characteristics.

Table SA.3: Correlations between whether or not an individual is willing to open a business and their baseline characteristics

	Willing to open a business
Age	-0.391***
Divorced	0.0187
Married	-0.0508*
Unmarried	0.0965***
Widow	-0.0826***
Higher Education (Class 11, 12)	0.167***
Informal Education	-0.104***
No Education	-0.300***
Primary (Class 1-5)	0.0367
Secondary (Class 6-10)	0.251***
University	0.100***
Belongs to Upper Caste	-0.0541*
Degree Centrality	0.0171
Eigen Vector Centrality	0.00140
Own Non Agr. Business	0
Feel not Capable	-0.00718
Willing to Open a Business	1
Risk Aversion (1-6)	-0.103***
Aspirations (Agricultural Expenditure)	0.0455*
Aspirations (Non Agri. Expenditure)	0.167***
Aspirations (Income)	-0.0250
Aspires to Higher Income	0.0372
Aspires to Higher Non Agri. Exp	0.239***

Notes: This table reports variable-wise correlations between baseline willingness to open a non-agricultural business and other baseline characteristics.

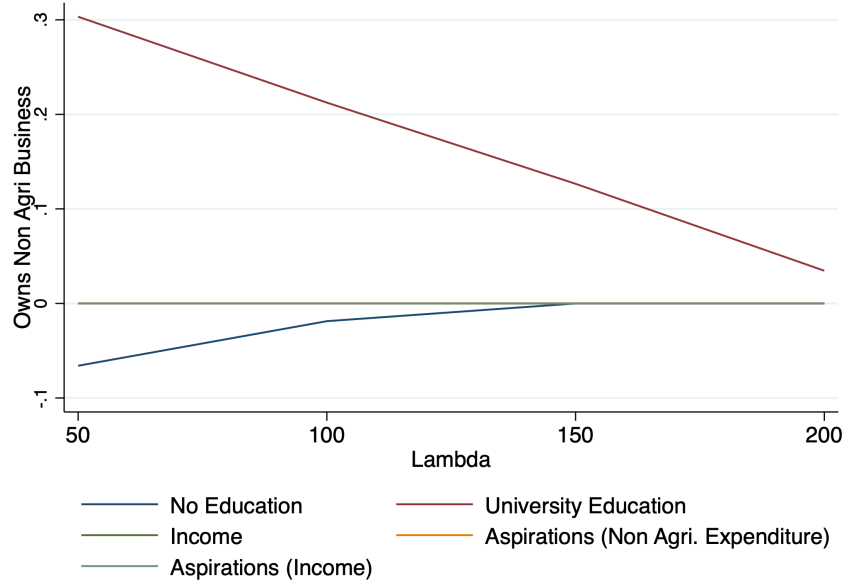


Figure SA.1: This figure plots correlations between selected variables and owning a business for different values of penalty parameters in a Lasso regression. Only variables selected out of a large list of demographic and network variables at $\lambda = 100$ are displayed. No Education and University Education are binary variables equal to 1 if the condition is true, and High Non. Agricultural Expenditure Aspirations is a binary variable equal to 1 if aspirations are higher than the baseline level.

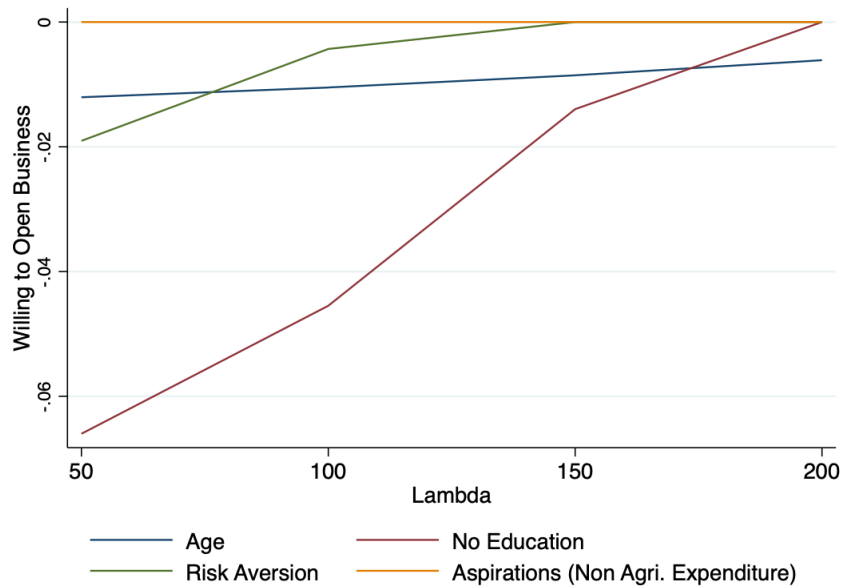


Figure SA.2: This figure plots correlations between selected variables and willingness to open businesses for different values of penalty parameters in a Lasso regression. Only variables selected out of a large list of demographic and network variables at $\lambda = 100$ are displayed. No Education is a binary variable equal to 1 if true. Risk Aversion is measured on a scale of 1-6 using a series of choices between lotteries and fixed outcomes presented to participants in the baseline survey.

SA2 Balance and Design

Table SA.4: Balance by social closeness within paired arms (endline sample)

	(1)	(2)	(3)
	0	1	(1) vs. (2), p-value
Willing to Open Business	0.420	0.472	0.437
Income (Winsorized)	20988.616	37308.989	0.010
Age	39.665	37.562	0.092
Elementary Education	0.134	0.225	0.082
Secondary Education	0.244	0.270	0.657
Higher Education	0.061	0.067	0.844
Informal Education	0.354	0.258	0.112
University Education	0.006	0.011	0.688
Brahmin	0.061	0.079	0.606
Chhetri	0.262	0.292	0.615
Dalit	0.012	0.022	0.568
Newar	0.470	0.461	0.893
Janjati/Tamang	0.195	0.146	0.315
Income: Agriculture	0.854	0.899	0.287
Income: Business	0.006	0.034	0.172
Income: Job	0.024	0.022	0.923
Income: Remittance	0.000	0.000	
Income: Other	0.024	0.000	0.045

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Sample restricted to individuals in paired treatment arms (T2 and T3) who were matched with a peer. Close peers are defined as those with network distance ≤ 2 . P-values from t-tests with standard errors clustered at the village level.

Table SA.5: Balance by relative centrality within paired arms (endline sample)

	(1)	(2)	(3)
	0	1	(1) vs. (2), p-value
Willing to Open Business	0.439	0.442	0.965
Income (Winsorized)	23439.691	31802.062	0.152
Age	38.454	39.887	0.263
Elementary Education	0.151	0.186	0.487
Secondary Education	0.263	0.247	0.782
Higher Education	0.092	0.010	0.002
Informal Education	0.296	0.361	0.293
University Education	0.007	0.010	0.760
Brahmin	0.072	0.062	0.746
Chhetri	0.270	0.289	0.747
Dalit	0.026	0.000	0.045
Newar	0.454	0.485	0.639
Janjati/Tamang	0.178	0.165	0.796
Income: Agriculture	0.888	0.845	0.342
Income: Business	0.013	0.021	0.665
Income: Job	0.033	0.010	0.206
Income: Remittance	0.000	0.000	
Income: Other	0.000	0.031	0.081

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Sample restricted to individuals in paired treatment arms (T2 and T3) who were matched with a peer. More central peers are those with strictly higher degree centrality than the respondent. P-values from t-tests with standard errors clustered at the village level.

Table SA.6: Balance by social closeness within paired arms (follow-up sample)

	(1)	(2)	(3)
	0	1	(1) vs. (2), p-value
Willing to Open Business	0.447	0.426	0.811
Income (Winsorized)	21241.972	31946.809	0.077
Age	39.583	37.234	0.152
Elementary Education	0.167	0.191	0.716
Secondary Education	0.250	0.298	0.546
Higher Education	0.056	0.106	0.315
Informal Education	0.361	0.277	0.295
University Education	0.009	0.021	0.604
Brahmin	0.037	0.064	0.507
Chhetri	0.278	0.277	0.988
Dalit	0.000	0.000	
Newar	0.500	0.489	0.904
Janjati/Tamang	0.185	0.170	0.823
Income: Agriculture	0.898	0.894	0.933
Income: Business	0.000	0.021	0.317
Income: Job	0.019	0.043	0.459
Income: Remittance	0.000	0.000	
Income: Other	0.028	0.000	0.083

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Sample restricted to individuals in paired treatment arms (T2 and T3) who were matched with a peer and appear in the follow-up survey. Close peers are defined as those with network distance ≤ 2 . P-values from t-tests with standard errors clustered at the village level.

Table SA.7: Balance by relative centrality within paired arms (follow-up sample)

	(1)	(2)	(3)
	0	1	(1) vs. (2), p-value
Willing to Open Business	0.455	0.424	0.714
Income (Winsorized)	24536.626	24130.000	0.933
Age	38.044	40.483	0.126
Elementary Education	0.209	0.117	0.126
Secondary Education	0.275	0.267	0.914
Higher Education	0.099	0.017	0.022
Informal Education	0.297	0.400	0.198
University Education	0.011	0.017	0.776
Brahmin	0.066	0.017	0.114
Chhetri	0.275	0.300	0.740
Dalit	0.000	0.000	
Newar	0.495	0.500	0.948
Janjati/Tamang	0.165	0.183	0.772
Income: Agriculture	0.923	0.867	0.283
Income: Business	0.000	0.017	0.318
Income: Job	0.033	0.017	0.518
Income: Remittance	0.000	0.000	
Income: Other	0.000	0.033	0.155

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Sample restricted to individuals in paired treatment arms (T2 and T3) who were matched with a peer and appear in the follow-up survey. More central peers are those with strictly higher degree centrality than the respondent. P-values from t-tests with standard errors clustered at the village level.

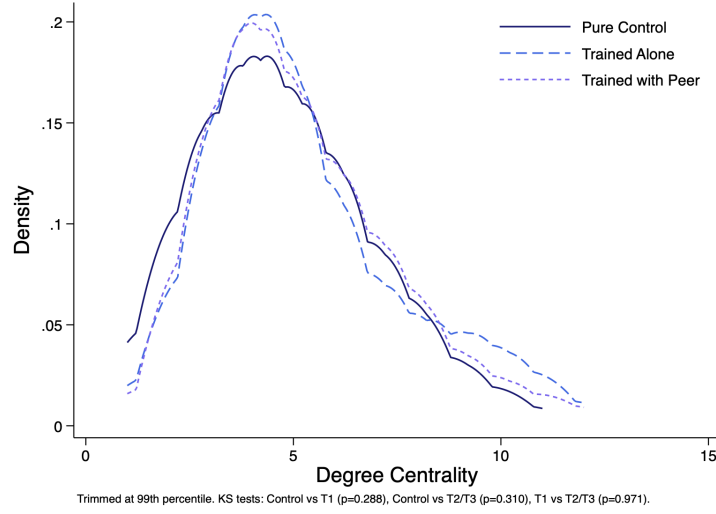
Table SA.8: Balance test for within-village comparisons (endline sample)

	0	1	2	3	(0) vs. (1), p-value	(0) vs. (2), p-value	(0) vs. (3), p-value	(1) vs. (2), p-value	(1) vs. (3), p-value	(2) vs. (3), p-value
Willing to Open Business	0.46	0.39	0.40	0.45	0.17	0.21	0.85	0.90	0.25	0.31
Income	24666.47	25317.20	25653.33	23907.65	0.81	0.72	0.79	0.91	0.65	0.58
Income Source- Agri.	0.85	0.87	0.84	0.91	0.48	0.82	0.06	0.35	0.22	0.03
Income Source- Business	0.06	0.06	0.10	0.10	0.87	0.17	0.22	0.12	0.16	0.92
Income Source- Job	0.03	0.02	0.01	0.03	0.66	0.23	0.89	0.42	0.57	0.20
Income Source- Remit.	0.01	0.01	0.01	0.02	0.53	0.96	0.59	0.56	0.27	0.55
Income Source- Other	0.07	0.03	0.04	0.01	0.08	0.14	0.00	0.76	0.21	0.13
Age	40.37	39.70	38.45	39.96	0.53	0.08	0.71	0.23	0.81	0.17
Elementary Education	0.10	0.15	0.19	0.16	0.18	0.02	0.11	0.30	0.75	0.50
Higher Education	0.06	0.09	0.05	0.07	0.29	0.91	0.66	0.24	0.56	0.58
Informal Education	0.32	0.31	0.30	0.33	0.84	0.67	0.83	0.81	0.68	0.53
University Education	0.02	0.01	0.01	0.01	0.29	0.30	0.35	0.99	0.91	0.92
Secondary Education	0.32	0.23	0.27	0.21	0.06	0.32	0.03	0.36	0.69	0.20
Degree	4.79	5.28	5.29	5.32	0.04	0.04	0.03	0.96	0.88	0.93
Brahmin	0.05	0.05	0.07	0.03	0.89	0.45	0.35	0.36	0.42	0.09
Chhetri	0.31	0.30	0.23	0.27	0.94	0.09	0.50	0.10	0.54	0.32
Dalit	0.02	0.01	0.04	0.01	0.29	0.14	0.35	0.02	0.91	0.02
Newar	0.39	0.41	0.50	0.45	0.66	0.04	0.26	0.11	0.47	0.42
Janjati	0.23	0.23	0.16	0.24	0.92	0.09	0.95	0.11	0.87	0.09

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The balance tests compare characteristics of the control group within treated villages (0) with those in Treatment 1 (trained alone), Treatment 2, and Treatment 3 (both paired arms) for the endline sample. The p-values in Columns 5-10 indicate if the difference is statistically significant.

Figure SA.3: Distribution of degree centrality by treatment group



Notes: Kernel density estimates of degree centrality for controls within treated villages, Trained Alone (T1), and Trained with Peer (T2 and T3 pooled). Kolmogorov-Smirnov tests confirm no statistically significant differences in distributions across groups. Sample trimmed at 99th percentile.

Table SA.9: Correlation between endline and follow-up outcomes

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Knowledge Index	0.0544 (0.0586)	-0.00575 (0.0611)	0.0972* (0.0530)
Aspirations Index	0.210** (0.0862)	0.123** (0.0524)	0.131** (0.0564)
Business Index	-0.0466 (0.0845)	0.121 (0.0736)	0.0317 (0.0719)
Additional Steps	0.151** (0.0699)	0.00350 (0.0635)	0.145** (0.0622)
Take-up Index	-0.0842 (0.0563)	0.156** (0.0633)	0.0814 (0.0576)
Observations	561	565	565
R-squared	0.053	0.054	0.081

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: Each column reports coefficients from regressing the follow up outcome (column header) on endline outcomes. Endline outcomes are self-reported measures collected immediately after training. Follow-up outcomes are measured one year later and capture actual behaviors. Sample restricted to individuals appearing in both surveys. Robust standard errors in parentheses.

SA3 Additional Endline Tables

Table SA.10: Types of businesses that individuals are willing to open

VARIABLES	(1) Agricultural Business	(2) Sewing	(3) Shop/Parlor	(4) Other Business
Trained Alone	0.0940 (0.0639)	0.0166 (0.0528)	-0.146 (0.0878)	0.0356 (0.0243)
Trained with Peer	0.174*** (0.0509)	-0.0441 (0.0472)	-0.148* (0.0823)	0.0188 (0.0151)
Constant	0.468*** (0.0414)	0.144*** (0.0410)	0.369*** (0.0791)	0.0180* (0.00882)
Observations	413	413	413	413
R-squared	0.021	0.006	0.022	0.005

Cluster-robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Business type is equal to 1 if the individual reports wanting to open such a business and reports being ready to invest. Standard errors are clustered at the village level.

Table SA.11: Effects on short-term outcomes by social closeness and peer centrality (pure-control base)

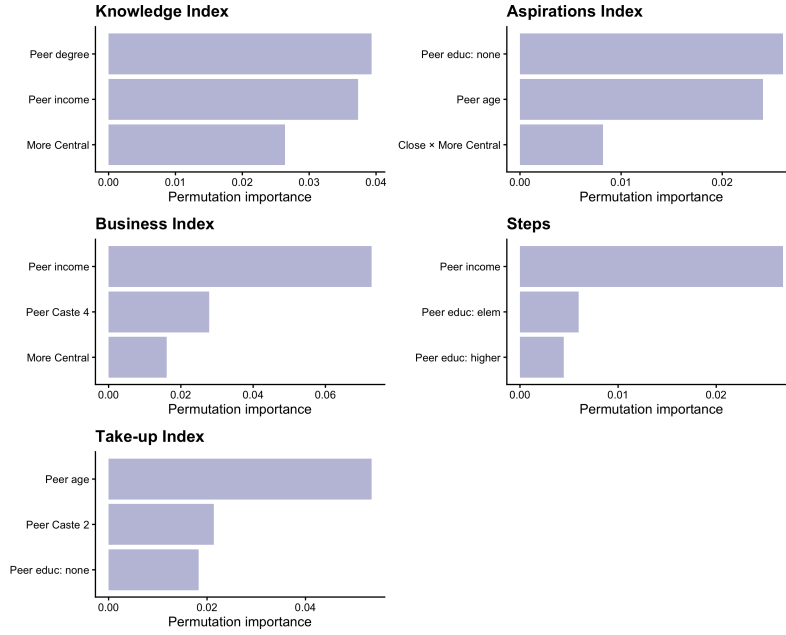
VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.774*** (0.248)	-0.150 (0.127)	0.229* (0.114)	0.0754 (0.0854)	0.0617 (0.0918)
Trained with Peer	0.995*** (0.246)	-0.226* (0.119)	0.277** (0.119)	0.0483 (0.0905)	0.0254 (0.0970)
Trained with Peer × Close	-0.134 (0.134)	0.0747 (0.127)	-0.122 (0.147)	-0.0590 (0.100)	-0.0636 (0.105)
Trained with Peer × More Central	-0.176* (0.0877)	-0.0944 (0.113)	-0.165 (0.149)	0.0319 (0.172)	0.0819 (0.0820)
Trained with Peer × Close × More Central	0.315 (0.237)	0.507** (0.229)	0.397* (0.225)	0.0332 (0.304)	0.134 (0.136)
Degree	0.0185 (0.0149)	0.0149 (0.0213)	0.0389* (0.0192)	0.0319* (0.0178)	0.0285* (0.0149)
Constant	-0.0883 (0.244)	-0.0765 (0.107)	-0.185 (0.135)	-0.145 (0.116)	-0.131 (0.0851)
Observations	756	756	753	734	740
R-squared	0.169	0.022	0.028	0.023	0.023
q: Close	1	1	1	1	1
q: More Central	0.381	0.702	0.702	1	0.702
q: Close x More Central	0.289	0.221	0.221	0.577	0.334

Cluster-robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

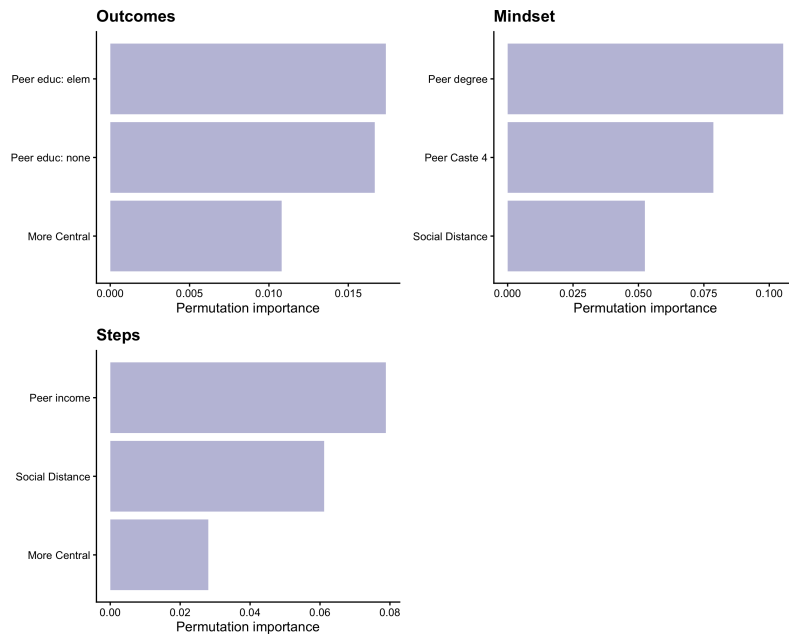
Notes: The base category is pure control villages. Standard errors are clustered at the village level.

Figure SA.4: Outcome predictors using random forests

(A) Endline

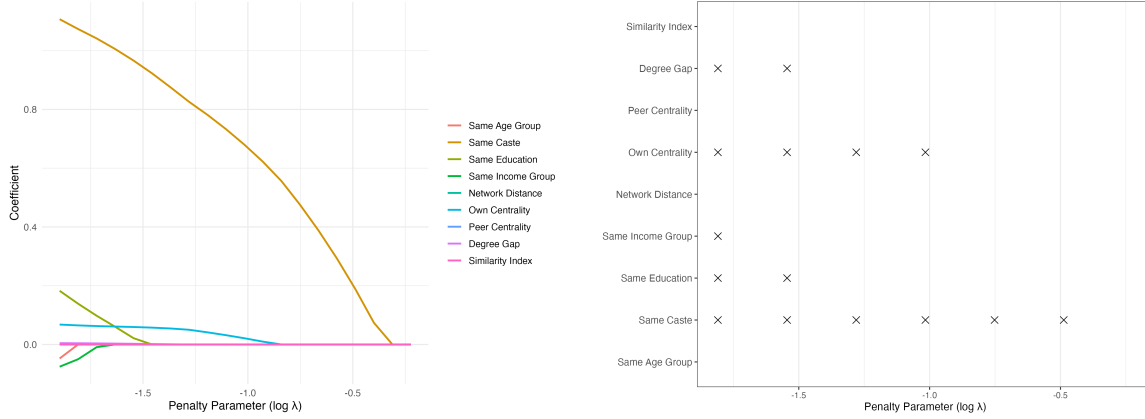


(B) Follow-up



Notes: The bars show the top three predictors of each outcome at endline (top) and follow-up (bottom), estimated using random forest models. Variable importance is measured by the increase in root mean squared error (RMSE) when the predictor is permuted. Outcomes are indices pre-specified in the analysis plan.

Figure SA.5: Lasso predictors of number of pooled network links



Notes: The left panel shows LASSO coefficient paths for predicting the number of links pooled within each pair who attended Treatment 3, regardless of assignment; the right panel shows when each variable enters the model as the penalty parameter (λ) decreases.

SA4 Additional Follow-up Tables

Table SA.12: Effects on long-term outcomes by social closeness and peer centrality (pure-control base)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.205 (0.134)	0.448*** (0.145)	0.174 (0.180)
Trained with Peer	0.0647 (0.148)	0.362* (0.204)	-0.0118 (0.200)
Trained with Peer × Close	0.360* (0.195)	-0.0731 (0.285)	0.302 (0.246)
Trained with Peer × More Central	-0.221 (0.184)	-0.518** (0.208)	-0.0367 (0.196)
Trained with Peer × Close × More Central	0.210 (0.663)	0.0584 (0.341)	-0.251 (0.436)
Degree	-0.0186 (0.0280)	0.0141 (0.0320)	-0.0108 (0.0239)
Constant	0.0836 (0.163)	-0.0745 (0.173)	0.0466 (0.201)
Observations	462	465	465
R-squared	0.016	0.036	0.009
q: Close	0.298	0.530	0.300
q: More Central	0.320	0.0620	0.571
q: Close x More Central	1	1	1

Cluster-robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. The regression includes an indicator for those who were intended to be paired but remained unmatched. We additionally control for the individual's own degree centrality in regressions that include relative comparisons between own and peer centrality. Standard errors are clustered at the village level.

SA5 Mechanisms

This section provides additional details on the mechanisms underlying the peer effects documented in Section 4. We examine short-term mechanisms including joint learning during training, encouragement, risk-sharing, and pooling of contacts, and then turn to longer-term mechanisms, asking whether peers facilitate risk-sharing, sustain collaboration, and remain in contact one year later.

We asked women who attended the training in pairs to describe how pairing was useful compared to training alone. 37% of women paired report having received encouragement from their matched peer. Another 30% report that the training material was easier to grasp because of the pairing. About 18% say that the training helped because they shared network contacts. We now examine each mechanism in detail.

Do pairs learn better together? To evaluate this mechanism, we test how the intervention affects knowledge for different types of peers. Column 1 of Table SA.13 shows that while the intervention improved the knowledge index, this does not differ by whether the individual was trained alone or with a peer.

Uniquely, we also measure learning during the training itself through performance in interactive exercises. We use two additional variables measured in exercises conducted during the training: game profit and yearly profit. Game profit records performance on the first day of the training, where women played an investment and savings game. Yearly profit was measured on the last day of the training as the projected profit from a hypothetical business plan that pairs collaboratively developed. Columns 2 and 3 of Table SA.13 show that these within-training measures of performance do not differ by whether the individual was trained alone or with a peer.

Table SA.13: Effect on learning during and after the training

VARIABLES	(1) Knowledge Index	(2) Profit (Game)	(3) Profit (Business Plan)
Trained alone	0.789*** (0.247)		
Trained with Peer	0.748*** (0.244)	52.51 (41.13)	203,402 (263,911)
Constant	0 (0.234)	273.3*** (34.06)	534,159*** (168,030)
Observations	768	437	436
R-squared	0.122	0.004	0.001
Treatment 1=Pair	0.466		

Robust standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages in Column 1 and Trained Alone (T1) in Columns 2 and 3. Cluster-robust standard errors in Column 1 and robust standard errors in Columns 2 and 3 are in parentheses.

Women who attended the training with a socially close peer may have been better able to discuss the material being taught. Table E.3 shows that close pairs earn significantly more in the joint business plan exercise than far pairs ($p=0.05$), while far pairs show no positive effect relative to those trained alone. This suggests that socially close peers may be learning better together. The effect on projected profits does not extend to the individual game profit or the post-training knowledge index, suggesting that the benefit of social closeness may largely operate through better collaboration on joint tasks rather than individual learning.

Do pairs help each other share risk? Pairing can also improve outcomes if individuals are risk-averse and the matched peer is less risk-averse or can provide financial assistance in case business endeavors do not proceed as planned. Table SA.14 shows heterogeneous effects by the individual’s own risk aversion, taking pure control villages as the base. We find no robust evidence that pairing differentially benefits risk-averse or risk-loving individuals.

Table SA.14: Heterogeneous effects by own risk aversion

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.596*** (0.135)	-0.0324 (0.170)	0.207 (0.194)	0.00662 (0.101)	0.105 (0.145)
Trained with Peer	0.658*** (0.133)	-0.0263 (0.147)	0.107 (0.169)	-0.118 (0.113)	0.111 (0.118)
Risk Averse (Binary)	-0.0899 (0.162)	0.0280 (0.184)	-0.177 (0.219)	-0.109 (0.140)	-0.127 (0.169)
T1 x Risk Averse	-0.0476 (0.190)	-0.119 (0.234)	-0.0522 (0.284)	-0.00635 (0.197)	0.0428 (0.221)
T2/T3 x Risk Averse	0.105 (0.209)	-0.189 (0.219)	0.103 (0.263)	0.190 (0.183)	-0.0251 (0.191)
Constant	0.256*** (0.0812)	-0.0441 (0.118)	0.168 (0.126)	0.139** (0.0544)	0.0456 (0.0788)
Observations	673	673	670	651	656
R-squared	0.110	0.008	0.019	0.019	0.027

Cluster-robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This regression treats individuals in pure control villages as the base category. Risk Aversion is measured on a scale of 1–6 using a series of choices between lotteries and fixed outcomes presented to participants in the baseline survey. Risk Averse (Binary) equals 1 if the individual’s own risk aversion score is 5 or 6 (most risk-averse choices), and 0 otherwise. Cluster robust standard errors in parentheses.

Table SA.15 shows effects by the peer’s risk aversion relative to the individual. Being paired with a less risk-averse peer does not improve outcomes and, if anything, significantly reduces aspirations in T3 (-0.36σ , $p < 0.01$) and T2 (-0.27σ , $p < 0.05$). The interaction also raises take-up significantly in T3 ($+0.14\sigma$, $p < 0.05$), suggesting that less risk-averse peers may encourage seeking additional resources while reducing aspirations.

Table SA.15: Heterogeneous effects by peer risk aversion

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.574** (0.220)	-0.0943 (0.108)	0.206* (0.103)	0.0882 (0.0882)	0.109 (0.0802)
Trained with Peer x Peer Less Risk Averse	0.569** (0.226)	-0.273** (0.121)	0.0396 (0.265)	0.212 (0.182)	0.0516 (0.114)
Trained with Peer x Peer Same/More Risk Averse	0.705*** (0.236)	-0.0453 (0.103)	0.184 (0.145)	-0.0829 (0.108)	0.120 (0.130)
Trained with Peer + Connections Module x Peer Less Risk Averse	0.627** (0.293)	-0.362*** (0.113)	0.325 (0.275)	0.143 (0.212)	0.139** (0.0676)
Trained with Peer + Connections Module x Peer Same/More Risk Averse	0.748*** (0.215)	-0.0410 (0.113)	0.246** (0.119)	0.109 (0.115)	0.0910 (0.0977)
Constant	0.215 (0.212)	-0.0449 (0.0762)	0.0472 (0.0715)	0.0133 (0.0562)	-0.00426 (0.0497)
Observations	768	768	765	746	751
R-squared	0.101	0.007	0.016	0.021	0.018

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: This regression treats individuals in pure control villages as the base category. Peer Less Risk Averse equals 1 if the matched peer's risk aversion score is lower than the individual's own score. Peer Same/More Risk Averse equals 1 if the peer's score is the same or higher. Cluster robust standard errors in parentheses.

To account for financial assistance, one year following the training, we measured the number of savings groups that the respondent was a part of and whether they had joined a new cooperative in the village. We find that those treated alone are significantly more likely to have joined a new cooperative, but the effect is not significant for those trained with a peer, suggesting that the paired treatments did not increase engagement with the wider network for saving or risk-sharing (Table E.10). However, the lower rate of joining cooperatives among paired individuals could also be consistent with bilateral risk-sharing between training partners substituting for formal cooperative membership. If pairs provide mutual support to each other, they may have less need to seek risk-sharing through external groups. We provide additional information on network interactions later in this section.

Do pairs provide access to network connections? As documented in Section 4, the connections module did not produce significant gains relative to other arms, and contact sharing was concentrated within caste groups.

Considering the pair-level dyadic data, we find that among all those who attended the connections module, dyads with a higher social distance and higher gap in network degrees pool significantly more contacts, while higher network overlap has a negative albeit insignificant impact (Table E.4). While the treatment increases contact sharing along these dimensions, contact sharing remains concentrated within caste groups. Same-caste pairs share significantly more contacts than other pairs, with the effect dominating other predictors in magnitude. Inward-looking social networks therefore limit the scope for contact sharing.

Do pairs encourage each other? A key channel driving short-term peer effects is encouragement. The most common benefit of pairing reported by respondents was encouragement. Regressing an indicator for whether the respondent reported encouragement as a benefit of pairing on peer characteristics in Table SA.16, we find that being matched with a more central peer increases the likelihood of reporting encouragement by 13.4 percentage points ($p < 0.10$), and each additional peer network connection raises this likelihood by 3.0 percentage points ($p < 0.05$).

Table SA.16: Determinants of reporting encouragement as a benefit of pairing

VARIABLES	(1) Encouragement	(2) Encouragement
Peer caste = 2	-0.161 (0.128)	-0.151 (0.123)
Peer caste = 3	-0.00793 (0.125)	-5.77e-05 (0.120)
Peer caste = 4	-0.300** (0.135)	-0.274** (0.130)
Peer education = 1	-0.239*** (0.0872)	-0.240*** (0.0878)
Peer education = 2	-0.109 (0.0850)	-0.112 (0.0854)
Peer education = 3	-0.147 (0.121)	-0.135 (0.121)
Peer income	-8.61e-07 (8.25e-07)	-8.96e-07 (8.04e-07)
Peer marital status = 1	0.0188 (0.131)	0.0305 (0.126)
Peer marital status = 3	0.333 (0.288)	0.285 (0.279)
Peer age	0.00241 (0.00402)	0.00254 (0.00401)
Own network degree	-1.08e-05 (0.0130)	
Peer is more central	0.134* (0.0707)	
Peer network degree		0.0302** (0.0122)
Constant	0.411 (0.256)	0.282 (0.250)
Observations	247	247
R-squared	0.112	0.117

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: The dependent variable is an indicator equal to 1 if the respondent reported encouragement as a reason why pairing was beneficial. The sample is restricted to individuals who were assigned to the paired treatment arm and were actually trained with a peer. Robust standard errors in parentheses.

Are pairs likely to interact in the future? Finally, we find that individuals with lower social distance (i.e., socially close) to their matched peer are significantly more likely to want to open a business together and to meet in the future (Panel A of Table E.9). In contrast, being assigned to a peer with high centrality does not differentially affect these outcomes. Notably, paired trainees are also significantly more likely to report wanting to open an agricultural business together (Table SA.10). This is consistent with the *parma* or *bola* system in Nepalese villages, where peers support each other with agriculture-related tasks through reciprocity-based labor provision (Messerschmidt 1981, Sherpa 2005, Bhattarai 2006). These traditional forms of collaboration may lower the perceived cost of joint agricultural enterprise

for paired individuals.

One year later, we find that individuals who are trained with a peer are significantly more likely than the control group to report discussing business-related concerns with anyone in the village, though the overall rate of such discussions remains low (Table E.11). When asked whether they spoke to their matched peer, 31% report having done so. When asked what they spoke about, 83% sought advice, while the remainder discussed borrowing or lending money and setting up businesses. However, as shown in Table E.11, these network interactions were significantly more likely than in the control group when the matched peer was less central (Column 2), and within the paired arm, when the matched peer was socially close (Column 3).

Importantly, as shown in Panel B of Table E.9, those matched with a same-caste peer or with someone with low social distance are significantly more likely to have interacted with each other. Being matched with someone with high centrality does not affect these outcomes. This highlights the potential role played by the ease of collaboration among individuals with a similar social standing and corroborates the previous empirical findings.

SA5.1 Summary of Mechanisms

Overall, our evidence highlights two key channels through which peer identity affects outcomes. First, socially close peers facilitate collaboration: close pairs perform significantly better on joint training exercises, are more likely to plan future meetings and express willingness to start businesses together, and when they share the same caste, are significantly more likely to maintain contact one year later.

Second, more central peers raise short-term take-up of additional resources: respondents paired with central peers were significantly more likely to report encouragement as the reason pairing was beneficial, and being matched with a more central peer is associated with a marginal increase in the take-up index at endline (about $+0.03\sigma$ per additional peer connection, $p < 0.10$), although this effect is not robust to multiple-hypothesis correction ($q = 0.68$). However, these marginal short-term benefits do not persist. Those matched with less central peers were significantly more likely to talk to others about opening a business one year later and produce larger long-term gains on the steps index, consistent with central peers being less available for sustained interaction.

Taken together, the results suggest that ease of collaboration with close peers and short-term take-up gains from central peers drive effects in the short term, while interaction costs (shaped by social distance and caste boundaries) determine who is effective in the long run.

SA6 Endline Index Components

SA6.0.1 Outcome Descriptions

Knowledge Index. This index measures comprehension of business concepts covered during training. It comprises five quiz questions, each coded as 1 if answered correctly and 0 otherwise:

1. “What do you understand by a business?” (Correct: intended to make a profit)
2. “What characteristics are required to be a successful entrepreneur?” (Correct: all listed characteristics)
3. “What do you mean by fixed assets?” (Correct: lasting more than 1 year)
4. “What sector does a beauty parlor come under?” (Correct: service industry)
5. “Above what break-even percentage does the business become risky?” (Correct: 60%)

Aspirations Index. This index captures business-related aspirations following the elicitation procedure in [Bernard & Seyoum Taffesse \(2014\)](#). Respondents were asked: “What is the level you would like to reach?” for each of the following:

1. Yearly non-agricultural business investment (NPR)
2. Monthly income (NPR)
3. Monthly savings (NPR)

All monetary variables are winsorized at the 99th percentile. Agricultural investment aspirations are excluded from the composite index because business training could plausibly increase or decrease them, conflating the interpretation of the measure. Effects on agricultural aspirations are reported separately in Tables [SA.19](#) and [SA.20](#).

Business Index. This index captures forward-looking entrepreneurial intentions:

1. Ready to invest in a business (ordinal: 0=not ready, 1=maybe, 2=of course)
2. Would submit business plan for a competition (binary)
3. Likelihood of starting a business (scale 1–5)

Additional Steps Index. This index captures concrete preparatory actions for starting a business:

1. Intends to open a savings account for business purposes (binary)
2. Has taken a loan for business purposes (binary)

Take-up Index. This index measures willingness to seek additional support post-training:

1. Would attend additional training (binary)

2. Would take up NGO/advisory mentoring (binary)
3. Would seek advice from community members (binary)
4. Willingness to pay for trainer services (NPR, winsorized at 99th percentile)
5. Willingness to pay for mentoring services (NPR, winsorized at 99th percentile)

Table SA.17: Endline knowledge index components: ITT effects

VARIABLES	(1) Q1	(2) Q2	(3) Q3	(4) Q4	(5) Q5
Trained Alone	0.0228 (0.0310)	-0.00387 (0.0525)	0.0228 (0.0513)	0.0843* (0.0470)	-0.00182 (0.0424)
Trained with Peer	0.0367 (0.0301)	-0.0773 (0.0523)	-0.0425 (0.0526)	0.0393 (0.0487)	0.0215 (0.0416)
Trained with Peer + Connections Module	0.0457 (0.0302)	-0.0301 (0.0545)	0.0505 (0.0532)	0.0705 (0.0493)	0.0325 (0.0426)
Constant	0.897*** (0.0237)	0.477*** (0.0378)	0.640*** (0.0376)	0.704*** (0.0360)	0.807*** (0.0307)
Observations	690	710	679	680	688
R-squared	0.004	0.004	0.005	0.006	0.001
p: T1 vs T2	0.609	0.153	0.199	0.313	0.565
p: T1 vs T3	0.401	0.625	0.589	0.760	0.408
p: T2 vs T3	0.733	0.377	0.0773	0.507	0.787
q: T1	1	1	1	0.577	1
q: T2	1	1	1	1	1
q: T3	0.621	0.621	0.621	0.621	0.621

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.18: Endline knowledge index components: effects by peer type

VARIABLES	(1) Q1	(2) Q2	(3) Q3	(4) Q4	(5) Q5
Trained Alone	0.0182 (0.0253)	-0.0119 (0.0611)	0.0114 (0.0573)	0.0770 (0.0549)	-0.00503 (0.0374)
Trained with Peer	0.0829*** (0.0210)	-0.0630 (0.0602)	0.0822 (0.0587)	0.109** (0.0519)	0.109** (0.0420)
Trained with Peer $\times$ Close	-0.0625* (0.0337)	-0.0289 (0.0589)	-0.0125 (0.0835)	0.0525 (0.0711)	-0.0651 (0.0569)
Trained with Peer $\times$ More Central	-0.0533 (0.0376)	0.0381 (0.0897)	-0.0394 (0.0679)	-0.0150 (0.0541)	-0.0454 (0.0418)
Trained with Peer $\times$ Close $\times$ More Central	0.0583 (0.0703)	0.0511 (0.153)	0.123 (0.106)	0.0734 (0.0882)	0.117 (0.0847)
Constant	0.849*** (0.0288)	0.426*** (0.0586)	0.528*** (0.0594)	0.629*** (0.0515)	0.783*** (0.0356)
Observations	680	700	669	670	678
R-squared	0.020	0.005	0.068	0.074	0.042
q: Close	0.632	1	1	1	1
q: More Central	1	1	1	1	1
q: Close $\times$ More Central	1	1	1	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. “Close $\times$ More Central” denotes pairing with a socially close peer who is more central. Robust standard errors in parentheses.

Table SA.19: Endline aspirations index components: ITT effects

VARIABLES	(1) Agricultural Aspirations	(2) Non-Agricultural Aspirations	(3) Income Aspirations	(4) Savings Aspirations
Trained Alone	20,470 (22,327)	2,054 (31,803)	5,185 (7,378)	1,759 (1,417)
Trained with Peer	-19,451 (20,807)	-11,492 (26,730)	3,646 (7,965)	2,092 (1,453)
Trained with Peer + Connections Module	3,494 (22,719)	62,137 (40,257)	-4,437 (7,394)	993.1 (1,255)
Constant	225,369*** (14,846)	103,226*** (20,879)	57,768*** (4,998)	5,883*** (844.5)
Observations	707	604	709	706
R-squared	0.005	0.009	0.002	0.003
p: T1 vs T2	0.0719	0.643	0.852	0.839
p: T1 vs T3	0.479	0.153	0.211	0.602
p: T2 vs T3	0.309	0.0547	0.328	0.465
q: T1	1	1	1	1
q: T2	1	1	1	1
q: T3	1	0.973	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family. Agricultural aspirations are reported here for completeness but are excluded from the Aspirations Index.

Table SA.20: Endline aspirations index components: effects by peer type

VARIABLES	(1) Agricultural Aspirations	(2) Non-Agricultural Aspirations	(3) Income Aspirations	(4) Savings Aspirations
Trained Alone	16,916 (16,254)	1,655 (25,377)	3,842 (5,455)	1,785 (1,425)
Trained with Peer	5,549 (27,445)	13,035 (39,174)	-13,939** (6,078)	2,817 (1,740)
Trained with Peer $\times$ Close	-16,367 (35,154)	-27,204 (32,205)	30,490** (12,825)	-2,725 (1,887)
Trained with Peer $\times$ More Central	-49,053* (26,419)	-68,932** (28,939)	-435.0 (7,296)	-1,665 (2,829)
Trained with Peer $\times$ Close $\times$ More Central	89,491* (46,549)	227,834** (88,184)	35,932 (21,236)	3,590 (3,926)
Constant	204,661*** (23,683)	138,955*** (29,251)	51,483*** (8,012)	6,280*** (2,102)
Observations	697	597	699	696
R-squared	0.011	0.024	0.044	0.005
q: Close	0.486	0.486	0.122	0.325
q: More Central	0.132	0.120	0.911	0.600
q: Close $\times$ More Central	0.114	0.0750	0.118	0.164

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. “Close $\times$ More Central” denotes pairing with a socially close peer who is more central. Robust standard errors in parentheses. Agricultural aspirations are reported here for completeness but are excluded from the Aspirations Index.

Table SA.21: Endline business index components: ITT effects

VARIABLES	(1) Ready to Invest	(2) Submit Plan	(3) Business Likelihood
Trained Alone	0.0886 (0.0841)	0.0452 (0.0476)	0.0959 (0.134)
Trained with Peer	-0.0335 (0.0876)	0.0131 (0.0485)	-0.111 (0.136)
Trained with Peer + Connections Module	0.00500 (0.0888)	0.0571 (0.0490)	0.246* (0.137)
Constant	1.320*** (0.0607)	0.708*** (0.0349)	3.586*** (0.0957)
Observations	706	681	564
R-squared	0.003	0.003	0.012
p: T1 vs T2	0.156	0.492	0.124
p: T1 vs T3	0.338	0.801	0.270
p: T2 vs T3	0.671	0.360	0.00972
q: T1	0.902	0.902	0.902
q: T2	1	1	1
q: T3	0.578	0.324	0.283

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.22: Endline business index components: effects by peer type

VARIABLES	(1) Ready to Invest	(2) Submit Plan	(3) Business Likelihood
Trained Alone	0.0708 (0.104)	0.0326 (0.0501)	0.0830 (0.138)
Trained with Peer	0.0751 (0.106)	0.0601 (0.0505)	0.112 (0.206)
Trained with Peer $\times$ Close	-0.0628 (0.138)	-0.0882* (0.0480)	-0.102 (0.226)
Trained with Peer $\times$ More Central	-0.190 (0.130)	0.0786 (0.0524)	-0.123 (0.169)
Trained with Peer $\times$ Close $\times$ More Central	0.334* (0.166)	0.0288 (0.0929)	0.371 (0.233)
Constant	1.165*** (0.118)	0.588*** (0.0501)	3.472*** (0.171)
Observations	696	671	555
R-squared	0.024	0.030	0.006
q: Close	0.781	0.318	0.781
q: More Central	0.314	0.314	0.314
q: Close $\times$ More Central	0.209	0.339	0.209

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. “Close $\times$ More Central” denotes pairing with a socially close peer who is more central. Robust standard errors in parentheses.

Table SA.23: Endline additional steps index components: ITT effects

VARIABLES	(1) Savings	(2) Loans
Trained Alone	0.0690* (0.0389)	0.00349 (0.0432)
Trained with Peer	-0.00258 (0.0422)	-0.0758* (0.0456)
Trained with Peer + Connections Module	0.0360 (0.0420)	0.0152 (0.0441)
Constant	0.808*** (0.0301)	0.797*** (0.0308)
Observations	685	680
R-squared	0.006	0.008
p: T1 vs T2	0.0635	0.0803
p: T1 vs T3	0.390	0.790
p: T2 vs T3	0.354	0.0490
q: T1	0.182	0.880
q: T2	0.907	0.240
q: T3	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.24: Endline additional steps index components: effects by peer type

VARIABLES	(1) Savings	(2) Loans
Trained Alone	0.0619 (0.0443)	-0.00258 (0.0480)
Trained with Peer	0.0336 (0.0502)	0.0259 (0.0370)
Trained with Peer $\times$ Close	0.00698 (0.0362)	-0.0672 (0.0498)
Trained with Peer $\times$ More Central	0.00803 (0.0662)	0.00697 (0.0622)
Trained with Peer $\times$ Close $\times$ More Central	0.0330 (0.107)	0.0227 (0.124)
Constant	0.749*** (0.0536)	0.748*** (0.0456)
Observations	675	671
R-squared	0.024	0.021
q: Close	0.738	0.622
q: More Central	1	1
q: Close $\times$ More Central	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. “Close $\times$ More Central” denotes pairing with a socially close peer who is more central. Robust standard errors in parentheses.

Table SA.25: Endline take-up index components: ITT effects

VARIABLES	(1) Advice	(2) Mentoring	(3) Mentoring Payment	(4) Trainer	(5) Trainer Payment
Trained Alone	0.0818*** (0.0314)	0.0475 (0.0322)	169.4 (132.4)	0.0895** (0.0369)	69.78 (175.5)
Trained with Peer	0.0280 (0.0350)	-0.000659 (0.0352)	90.97 (105.6)	0.0405 (0.0396)	169.5 (170.9)
Trained with Peer + Connections Module	0.0610* (0.0338)	0.0103 (0.0359)	43.02 (110.4)	0.0486 (0.0406)	138.7 (186.8)
Constant	0.862*** (0.0262)	0.874*** (0.0251)	388.4*** (68.84)	0.815*** (0.0296)	609.8*** (112.4)
Observations	690	692	688	685	678
R-squared	0.011	0.004	0.003	0.009	0.001
p: T1 vs T2	0.0643	0.131	0.572	0.154	0.593
p: T1 vs T3	0.451	0.255	0.374	0.250	0.732
p: T2 vs T3	0.297	0.758	0.684	0.833	0.876
q: T1	0.0410	0.164	0.178	0.0410	0.336
q: T2	1	1	1	1	1
q: T3	0.561	1	1	0.861	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.26: Endline take-up index components: effects by peer type

VARIABLES	(1) Advice	(2) Mentoring	(3) Mentoring Payment	(4) Trainer	(5) Trainer Payment
Trained Alone	0.0757** (0.0338)	0.0393 (0.0304)	113.0 (142.1)	0.0784* (0.0429)	15.79 (169.2)
Trained with Peer	0.0623* (0.0324)	0.0151 (0.0338)	80.89 (130.7)	0.0579 (0.0386)	306.0* (171.5)
Trained with Peer $\times$ Close	-0.0718 (0.0508)	0.0110 (0.0425)	32.17 (130.7)	-0.00961 (0.0408)	15.39 (261.0)
Trained with Peer $\times$ More Central	0.0541** (0.0243)	0.0415 (0.0437)	-4.197 (192.3)	0.0434 (0.0366)	-180.1 (324.3)
Trained with Peer $\times$ Close $\times$ More Central	0.0945 (0.0577)	0.0334 (0.0683)	5.616 (175.8)	0.0424 (0.0543)	-100.2 (356.2)
Constant	0.804*** (0.0420)	0.795*** (0.0299)	358.0** (158.5)	0.713*** (0.0511)	718.8*** (238.1)
Observations	681	683	679	677	671
R-squared	0.041	0.049	0.002	0.053	0.005
q: Close	1	1	1	1	1
q: More Central	0.228	0.892	1	0.892	1
q: Close $\times$ More Central	1	1	1	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. “Close $\times$ More Central” denotes pairing with a socially close peer who is more central. Robust standard errors in parentheses.

Table SA.27: Endline take-up index components: effects by social distance and peer centrality

VARIABLES	(1) Advice	(2) Mentoring	(3) Mentoring Payment	(4) Trainer	(5) Trainer Payment
Trained Alone	0.0818*** (0.0315)	0.0475 (0.0322)	169.4 (132.6)	0.0895** (0.0370)	69.78 (175.8)
Trained with Peer	-0.0618 (0.0676)	0.0236 (0.0561)	80.45 (285.7)	0.0627 (0.0568)	-96.70 (428.0)
Trained with Peer $\times$ Peer distance	0.0165** (0.00784)	-0.00499 (0.00870)	0.988 (36.88)	-0.00627 (0.0103)	82.35 (71.39)
Trained with Peer $\times$ Peer degree	0.0169*** (0.00607)	0.00895* (0.00523)	4.138 (32.75)	0.0101* (0.00553)	7.562 (50.76)
Constant	0.862*** (0.0263)	0.874*** (0.0252)	388.4*** (68.95)	0.815*** (0.0297)	609.8*** (112.6)
Observations	676	678	674	672	666
R-squared	0.037	0.037	0.003	0.037	0.005
q: Alone	0.0420	0.165	0.179	0.0420	0.338
q: Paired	1	1	1	1	1
q: Paired $\times$ Dist	0.219	1	1	1	0.994
q: Paired $\times$ Deg	0.0290	0.133	0.562	0.133	0.562

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. The regression includes interactions of paired treatment with peer social distance and peer degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

SA6.1 Endline Index Subcomponents

The tables in this subsection report the ITT effect of each treatment arm on the individual subcomponents of the five endline indices, using pure control villages as the omitted base category.

Table SA.28: Endline knowledge index components: ITT effects (pure-control base)

VARIABLES	(1) Q1	(2) Q2	(3) Q3	(4) Q4	(5) Q5
Trained alone	0.0356 (0.0438)	0.196** (0.0889)	0.340*** (0.0851)	0.394*** (0.108)	0.111 (0.0896)
Trained with Peer	0.0495 (0.0439)	0.122 (0.0970)	0.275*** (0.0877)	0.349*** (0.107)	0.134 (0.0887)
Trained with Peer + Connections Module	0.0585 (0.0414)	0.169* (0.0855)	0.368*** (0.0890)	0.381*** (0.110)	0.145 (0.0868)
Constant	0.884*** (0.0390)	0.278*** (0.0752)	0.323*** (0.0775)	0.394*** (0.101)	0.695*** (0.0803)
Observations	715	768	704	706	712
R-squared	0.007	0.026	0.089	0.125	0.020
p: T1 vs T2	0.528	0.297	0.178	0.0826	0.645
p: T1 vs T3	0.317	0.624	0.548	0.802	0.438
p: T2 vs T3	0.674	0.487	0.0398	0.555	0.789
q: T1	0.205	0.0390	0.00300	0.00300	0.129
q: T2	0.194	0.194	0.0110	0.0110	0.168
q: T3	0.108	0.0620	0.00200	0.00500	0.0880

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.29: Endline aspirations index components: ITT effects (pure-control base)

VARIABLES	(1) Agricultural Aspirations	(2) Non-Agricultural Aspirations	(3) Income Aspirations	(4) Savings Aspirations
Trained alone	-32,624 (29,199)	-80,111 (56,817)	178.6 (6,809)	-2,137 (2,014)
Trained with Peer	-72,545*** (24,375)	-93,657* (53,759)	-1,360 (6,755)	-1,803 (1,852)
Trained with Peer + Connections Module	-49,600 (33,405)	-20,028 (59,772)	-9,443 (6,943)	-2,902 (1,856)
Constant	278,463*** (19,597)	185,390*** (50,323)	62,774*** (4,215)	9,778*** (1,380)
Observations	762	654	768	765
R-squared	0.016	0.014	0.002	0.005
p: T1 vs T2	0.132	0.658	0.817	0.871
p: T1 vs T3	0.559	0.0356	0.148	0.593
p: T2 vs T3	0.476	0.0370	0.256	0.441
q: T1	0.661	0.661	0.661	0.661
q: T2	0.0260	0.163	0.728	0.293
q: T3	0.329	0.329	0.329	0.329

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family. Agricultural aspirations are reported here for completeness but are excluded from the Aspirations Index.

Table SA.30: Endline business index components: ITT effects (pure-control base)

VARIABLES	(1) Ready to Invest	(2) Submit plan	(3) Business Likelihood
Trained alone	0.194* (0.102)	0.0760 (0.0662)	0.428** (0.192)
Trained with Peer	0.0719 (0.113)	0.0438 (0.0705)	0.222 (0.183)
Trained with Peer + Connections Module	0.110 (0.0978)	0.0878 (0.0629)	0.578*** (0.178)
Constant	1.215*** (0.0760)	0.677*** (0.0545)	3.254*** (0.151)
Observations	764	739	616
R-squared	0.008	0.006	0.037
p: T1 vs T2	0.211	0.492	0.202
p: T1 vs T3	0.320	0.792	0.259
p: T2 vs T3	0.633	0.334	0.000929
q: T1	0.116	0.117	0.116
q: T2	1	1	1
q: T3	0.219	0.212	0.0100

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.31: Endline additional steps index components: ITT effects (pure-control base)

VARIABLES	(1) Savings	(2) Loans
Trained alone	0.0581 (0.0385)	0.00435 (0.0354)
Trained with Peer	-0.0134 (0.0445)	-0.0750* (0.0422)
Trained with Peer + Connections Module	0.0252 (0.0390)	0.0160 (0.0414)
Constant	0.819*** (0.0298)	0.796*** (0.0200)
Observations	745	738
R-squared	0.005	0.007
p: T1 vs T2	0.0993	0.119
p: T1 vs T3	0.302	0.796
p: T2 vs T3	0.352	0.0562
q: T1	0.402	0.824
q: T2	0.621	0.212
q: T3	1	1
Cluster-robust standard errors in parentheses		
*** p<0.01, ** p<0.05, * p<0.1		

Notes: The base category is pure control villages. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.32: Endline take-up index components: ITT effects (pure-control base)

VARIABLES	(1) Advice	(2) Mentoring	(3) Mentoring- Payment (W)	(4) Trainer	(5) Trainer- Payment (W)
Trained alone	0.0769** (0.0335)	0.0849* (0.0466)	-134.2 (215.7)	0.0719* (0.0379)	-38.09 (211.1)
Trained with Peer	0.0232 (0.0377)	0.0367 (0.0469)	-212.6 (235.9)	0.0229 (0.0409)	61.64 (193.1)
Trained with Peer + Connections Module	0.0561 (0.0371)	0.0477 (0.0480)	-260.6 (208.4)	0.0310 (0.0375)	30.87 (207.2)
Constant	0.867*** (0.0290)	0.837*** (0.0410)	692.1*** (185.8)	0.833*** (0.0299)	717.6*** (140.2)
Observations	749	750	745	745	737
R-squared	0.011	0.009	0.005	0.006	0.000
p: T1 vs T2	0.0901	0.0850	0.676	0.158	0.612
p: T1 vs T3	0.403	0.187	0.386	0.260	0.665
p: T2 vs T3	0.275	0.709	0.777	0.817	0.864
q: T1	0.154	0.154	0.370	0.154	0.523
q: T2	1	1	1	1	1
q: T3	1	1	1	1	1
Cluster-robust standard errors in parentheses					
*** p<0.01, ** p<0.05, * p<0.1					

Notes: The base category is pure control villages. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

SA7 Follow-up Index Components

SA7.1 Outcome Descriptions

The following indices capture outcomes measured one year after the training program via phone surveys.

Outcomes Index. This index captures realized economic outcomes:

1. Monthly income (NPR, winsorized at 1st and 99th percentiles)
2. Agricultural profits (NPR, winsorized at 1st and 99th percentiles)

Steps Index. This index captures concrete actions taken toward starting or expanding a business or agricultural activities:

1. Opened a new savings account with a bank (binary)
2. Joined a new cooperative in the past year (binary)
3. Total savings (NPR, winsorized at 1st and 99th percentiles)
4. Loan amount (NPR, winsorized at 99th percentile)
5. Investment in agriculture (NPR, winsorized at 1st and 99th percentiles)

Mindset Index. This index captures forward-looking motivation and self-beliefs:

1. Income aspirations: “How much would you want to earn per month?” (NPR, winsorized at 1st and 99th percentiles)
2. Likelihood of starting a business (scale 0–5)
3. Business self-efficacy (scale 1–5)
4. Committed to a savings account where funds can only be used for business (binary)
5. Amount committed to business savings account (NPR)

Table SA.33: Follow-up outcomes index components: ITT effects

VARIABLES	(1)	(2)
	Monthly Income	Agriculture Profits
Trained Alone	4,096 (3,625)	16,310 (20,075)
Trained with Peer	2,327 (3,000)	3,650 (19,796)
Constant	23,007*** (2,452)	112,699*** (14,531)
Observations	444	453
R-squared	0.004	0.002
p: T1 vs Paired	0.492	0.287
q: T1	0.736	0.736
q: Paired	1	1

Cluster-robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. T2 and T3 are pooled into a single paired treatment indicator. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.34: Follow-up outcomes index components: effects by peer type

VARIABLES	(1)	(2)
	Monthly Income	Agriculture Profits
Trained Alone	4,035 (3,629)	17,851 (20,841)
Trained with Peer	1,411 (3,154)	5,476 (25,483)
Trained with Peer × Close	7,616 (6,250)	41,334* (21,347)
Trained with Peer × More Central	-2,444 (6,607)	-30,581 (19,819)
Trained with Peer × Close × More Central	-1,287 (13,969)	-17,237 (48,738)
Constant	24,206*** (3,770)	117,497*** (22,587)
Observations	434	444
R-squared	0.010	0.011
q: Close	0.147	0.147
q: More Central	0.556	0.370
q: Close x More Central	1	1

Cluster-robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. “Close × More Central” denotes pairing with a socially close peer who is more central. Standard errors are clustered at the village level.

Table SA.35: Follow-up steps index components: ITT effects

VARIABLES	(1) Investment in Agriculture	(2) New Savings Account	(3) Joined Cooperative	(4) Savings	(5) Loan
Trained Alone	27,196 (21,583)	0.0991** (0.0430)	0.112** (0.0477)	1,282 (813.7)	12,060 (14,907)
Trained with Peer	18,786 (15,493)	0.0304 (0.0358)	0.00104 (0.0385)	952.9 (575.5)	16,182 (14,752)
Constant	98,953*** (9,899)	0.171*** (0.0205)	0.0929*** (0.0329)	2,177*** (361.3)	21,071** (9,747)
Observations	362	475	475	434	475
R-squared	0.006	0.008	0.022	0.009	0.003
p: T1 vs Paired	0.623	0.127	0.00734	0.649	0.797
q: T1	0.197	0.0800	0.0800	0.146	0.270
q: Paired	0.892	0.892	0.892	0.892	0.892

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. T2 and T3 are pooled into a single paired treatment indicator. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family. Note that agricultural investment is missing for 23.8%. Missingness of agricultural investment is significantly higher among treated respondents than in the pure control group. Crucially, this missingness does *not* differ between those trained alone and those trained with a peer, so it does not affect our within-treatment comparisons of peer identity. Our main approach is to keep these observations as missing. We also find that the training-alone effect on the steps index is not driven by this: excluding agricultural investment from the index leaves the effect unchanged ($+0.50\sigma$, $p < 0.01$, versus $+0.49\sigma$ in the full index), consistent with the component estimates above. Setting the missing values to zero, either for all respondents or only for those reporting agriculture as an income source also leaves our follow-up results qualitatively unchanged.

Table SA.36: Follow-up steps index components: effects by peer type

VARIABLES	(1) Investment in Agriculture	(2) New Savings Account	(3) Joined Cooperative	(4) Savings	(5) Loan
Trained Alone	28,903 (21,497)	0.0951** (0.0412)	0.107** (0.0484)	962.8 (699.4)	12,257 (14,531)
Trained with Peer	37,717 (26,246)	0.0347 (0.0534)	0.00531 (0.0504)	999.3 (707.7)	42,229 (30,450)
Trained with Peer $\times$ Close	-21,913 (43,775)	-0.0377 (0.0770)	0.0534 (0.0699)	352.3 (1,123)	-25,643 (40,006)
Trained with Peer $\times$ More Central	-64,817** (25,060)	-0.0211 (0.0872)	-0.00613 (0.0655)	-1,440* (778.7)	-63,612** (30,663)
Trained with Peer $\times$ Close $\times$ More Central	67,807 (58,405)	-0.0357 (0.107)	-0.0937 (0.112)	-234.3 (1,295)	38,953 (41,322)
Constant	104,736*** (18,566)	0.130** (0.0577)	0.114** (0.0478)	1,418* (746.1)	21,909 (17,473)
Observations	355	465	465	424	465
R-squared	0.017	0.017	0.022	0.023	0.015
q: Close	1	1	1	1	1
q: More Central	0.0870	0.589	0.589	0.113	0.107
q: Close $\times$ More Central	1	1	1	1	1

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. “Close $\times$ More Central” denotes pairing with a socially close peer who is more central. Standard errors are clustered at the village level.

Table SA.37: Follow-up mindset index components: ITT effects

VARIABLES	(1) Income Aspirations	(2) Plan to Start Business	(3) Self-Efficacy	(4) Commitment Savings	(5) Commitment Amount
Trained Alone	-1,495 (5,299)	0.357 (0.377)	0.0302 (0.326)	0.0612 (0.0719)	906.2 (685.7)
Trained with Peer	-5,909 (4,459)	0.0724 (0.358)	-0.336 (0.303)	0.0541 (0.0679)	734.4 (619.0)
Constant	52,329*** (4,069)	2.181*** (0.313)	2.871*** (0.261)	0.439*** (0.0524)	900.4* (493.3)
Observations	472	466	475	474	474
R-squared	0.005	0.005	0.008	0.003	0.005
p: T1 vs Paired	0.237	0.159	0.117	0.895	0.658
q: T1	1	1	1	1	1
q: Paired	0.856	0.856	0.856	0.856	0.856

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. T2 and T3 are pooled into a single paired treatment indicator. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes within each index family.

Table SA.38: Follow-up mindset index components: effects by peer type

VARIABLES	(1) Income Aspirations	(2) Plan to Start Business	(3) Self-Efficacy	(4) Commitment Savings	(5) Commitment Amount
Trained Alone	-1,425 (5,204)	0.393 (0.369)	0.00648 (0.327)	0.0553 (0.0691)	857.1 (677.5)
Trained with Peer	-3,837 (5,086)	0.232 (0.399)	-0.312 (0.392)	0.0322 (0.0763)	-158.4 (640.6)
Trained with Peer × Close	2,630 (9,128)	0.362 (0.404)	0.212 (0.400)	0.00875 (0.121)	1,703 (1,818)
Trained with Peer × More Central	-9,712** (3,783)	-0.345 (0.389)	-0.0642 (0.413)	0.107 (0.0891)	1,836* (1,011)
Trained with Peer × Close × More Central	11,683 (17,488)	-0.184 (0.376)	-0.0657 (0.453)	-0.213 (0.140)	-4,023* (2,006)
Constant	56,457*** (4,436)	2.367*** (0.374)	2.656*** (0.330)	0.380*** (0.0890)	334.7 (632.2)
Observations	462	457	465	464	464
R-squared	0.013	0.013	0.014	0.009	0.019
q: Close	1	1	1	1	1
q: More Central	0.0900	0.402	0.674	0.319	0.194
q: Close x More Central	0.893	0.893	1	0.384	0.384

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. “Close × More Central” denotes pairing with a socially close peer who is more central. Standard errors are clustered at the village level.

SA8 Peer Type Regressions for Index Components

This section decomposes each index into its individual components under the same specifications used for the main tables (Tables 2 and 4), with those trained alone (T1) as the omitted base category.

SA8.1 Endline Index Components (Trained-Alone Base)

Table SA.39: Endline knowledge components by social distance and peer degree

VARIABLES	(1) Knowledge Index	(2) Q1	(3) Q2	(4) Q3	(5) Q4	(6) Q5
Trained with Peer	0.384** (0.185)	0.0728 (0.0449)	-0.0493 (0.119)	0.238** (0.0965)	0.148** (0.0752)	0.173** (0.0717)
Trained with Peer $\times$ Peer distance	-0.0294 (0.0280)	0.00518 (0.00641)	-0.00498 (0.0191)	-0.0424** (0.0170)	-0.0337** (0.0140)	-0.00469 (0.0108)
Trained with Peer $\times$ Peer degree	-0.0255 (0.0218)	-0.0113** (0.00555)	0.00471 (0.0136)	-0.00344 (0.0113)	0.00615 (0.00813)	-0.0131 (0.00870)
Constant	0.789*** (0.0686)	0.920*** (0.0200)	0.473*** (0.0366)	0.663*** (0.0350)	0.788*** (0.0303)	0.805*** (0.0293)
Observations	519	510	519	500	503	507
R-squared	0.076	0.015	0.003	0.084	0.096	0.054
q: Paired x Dist		0.721	0.912	0.0430	0.0430	0.912
q: Paired x Deg		0.268	0.842	0.842	0.818	0.361

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.40: Endline knowledge components by social closeness and peer centrality

VARIABLES	(1) Knowledge Index	(2) Q1	(3) Q2	(4) Q3	(5) Q4	(6) Q5
Trained with Peer	0.208** (0.0945)	0.0658*** (0.0216)	-0.0494 (0.0618)	0.0709 (0.0567)	0.0361 (0.0487)	0.114*** (0.0396)
Trained with Peer $\times$ Close	-0.164 (0.142)	-0.0600* (0.0357)	-0.0248 (0.0866)	-0.0124 (0.0756)	0.0610 (0.0585)	-0.0633 (0.0575)
Trained with Peer $\times$ More Central	-0.131 (0.126)	-0.0572 (0.0384)	0.0317 (0.0844)	-0.0395 (0.0786)	-0.0286 (0.0670)	-0.0483 (0.0543)
Trained with Peer $\times$ Close $\times$ More Central	0.323 (0.221)	0.0577 (0.0669)	0.0500 (0.135)	0.123 (0.118)	0.0711 (0.0899)	0.116 (0.0841)
Constant	0.576*** (0.116)	0.877*** (0.0339)	0.430*** (0.0640)	0.540*** (0.0591)	0.739*** (0.0529)	0.785*** (0.0483)
Observations	524	515	524	505	508	512
R-squared	0.086	0.021	0.004	0.087	0.090	0.055
p: Close, more-central vs Trained Alone		0.897	0.934	0.0592	0.00936	0.0261
q: Close, more-central vs Trained Alone		0.597	0.597	0.0700	0.0500	0.0560
p: Closeness effect: more-central peers		0.968	0.811	0.226	0.0588	0.406
q: Closeness effect: more-central peers		1	1	0.823	0.417	1

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.41: Endline aspirations components by social distance and peer degree

VARIABLES	(1) Aspirations Index	(2) Non-Agricultural Aspirations	(3) Income Aspirations	(4) Savings Aspirations
Trained with Peer	0.382** (0.187)	79,126 (64,884)	37,841* (19,581)	3,041 (3,569)
Trained with Peer $\times$ Peer distance	-0.0830*** (0.0293)	-13,546 (12,121)	-11,685*** (3,245)	-89.51 (466.2)
Trained with Peer $\times$ Peer degree	-0.0241 (0.0187)	-5,763 (6,400)	-239.0 (1,717)	-539.0 (372.2)
Constant	-0.139** (0.0589)	105,280*** (24,048)	62,952*** (5,438)	7,642*** (1,140)
Observations	519	437	519	516
R-squared	0.014	0.008	0.036	0.004
q: Paired x Dist		0.360	0.00200	0.658
q: Paired x Deg		0.801	1	0.801

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.42: Endline aspirations components by social closeness and peer centrality

VARIABLES	(1) Aspirations Index	(2) Non-Agricultural Aspirations	(3) Income Aspirations	(4) Savings Aspirations
Trained with Peer	-0.0647 (0.0962)	12,300 (34,992)	-17,714*** (6,607)	1,039 (2,137)
Trained with Peer $\times$ Close	0.103 (0.136)	-24,644 (36,583)	30,650** (15,593)	-2,708 (1,989)
Trained with Peer $\times$ More Central	-0.138 (0.111)	-73,123* (37,756)	-683.4 (7,180)	-1,691 (2,693)
Trained with Peer $\times$ Close $\times$ More Central	0.499* (0.263)	227,329** (93,856)	35,888 (29,730)	3,586 (3,704)
Constant	-0.122 (0.111)	151,296*** (42,961)	55,927*** (10,139)	8,128*** (2,107)
Observations	524	442	524	521
R-squared	0.025	0.029	0.054	0.003
p: Close, more-central vs Trained Alone		0.102	0.0552	0.936
q: Close, more-central vs Trained Alone		0.182	0.182	0.454
p: Closeness effect: more-central peers		0.0190	0.00850	0.784
q: Closeness effect: more-central peers		0.0270	0.0270	0.354

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.43: Endline business index components by social distance and peer degree

VARIABLES	(1) Business Index	(2) Ready to Invest	(3) Submit Plan	(4) Business Likelihood
Trained with Peer	0.0835 (0.240)	0.183 (0.191)	-0.0555 (0.0959)	-0.124 (0.272)
Trained with Peer $\times$ Peer distance	-0.0235 (0.0381)	-0.0424 (0.0306)	0.0192 (0.0148)	-0.0465 (0.0462)
Trained with Peer $\times$ Peer degree	-0.000329 (0.0276)	-0.0151 (0.0223)	0.00434 (0.0107)	0.0593** (0.0283)
Constant	0.253*** (0.0759)	1.409*** (0.0584)	0.753*** (0.0325)	3.682*** (0.0939)
Observations	516	516	495	405
R-squared	0.017	0.017	0.018	0.014
q: Paired x Dist		0.412	0.412	0.412
q: Paired x Deg		0.841	0.841	0.124

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.44: Endline business index components by social closeness and peer centrality

VARIABLES	(1) Business Index	(2) Ready to Invest	(3) Submit Plan	(4) Business Likelihood
Trained with Peer	0.0466 (0.135)	0.0115 (0.0993)	0.0294 (0.0525)	0.0313 (0.158)
Trained with Peer $\times$ Close	-0.127 (0.190)	-0.0469 (0.143)	-0.0839 (0.0748)	-0.0960 (0.219)
Trained with Peer $\times$ More Central	-0.157 (0.182)	-0.215 (0.145)	0.0718 (0.0698)	-0.131 (0.203)
Trained with Peer $\times$ Close $\times$ More Central	0.398 (0.275)	0.330 (0.216)	0.0277 (0.113)	0.369 (0.311)
Constant	0.0249 (0.137)	1.297*** (0.101)	0.637*** (0.0572)	3.574*** (0.164)
Observations	521	521	500	410
R-squared	0.028	0.025	0.030	0.006
p: Close, more-central vs Trained Alone		0.542	0.549	0.364
q: Close, more-central vs Trained Alone		1	1	1
p: Closeness effect: more-central peers		0.0876	0.513	0.225
q: Closeness effect: more-central peers		0.357	0.510	0.357

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.45: Endline additional steps index components by social distance and peer degree

VARIABLES	(1) Additional Steps	(2) Savings	(3) Loans
Trained with Peer	-0.0931 (0.202)	-0.0299 (0.0692)	-0.00972 (0.0923)
Trained with Peer $\times$ Peer distance	0.00483 (0.0320)	-0.00560 (0.0112)	0.00816 (0.0144)
Trained with Peer $\times$ Peer degree	0.0108 (0.0219)	0.00726 (0.00756)	-0.00144 (0.0106)
Constant	0.101 (0.0692)	0.877*** (0.0247)	0.800*** (0.0304)
Observations	499	498	494
R-squared	0.028	0.026	0.023
q: Paired x Dist		1	1
q: Paired x Deg		1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.46: Endline additional steps components by social closeness and peer centrality

VARIABLES	(1) Additional Steps	(2) Savings	(3) Loans
Trained with Peer	-0.0219 (0.121)	-0.0259 (0.0440)	0.0295 (0.0490)
Trained with Peer $\times$ Close	-0.0472 (0.169)	0.0126 (0.0597)	-0.0647 (0.0718)
Trained with Peer $\times$ More Central	0.0133 (0.163)	-0.000750 (0.0632)	0.00303 (0.0673)
Trained with Peer $\times$ Close $\times$ More Central	0.0300 (0.261)	0.0315 (0.0931)	0.0220 (0.111)
Constant	-0.0245 (0.121)	0.832*** (0.0435)	0.755*** (0.0536)
Observations	504	503	499
R-squared	0.029	0.026	0.025
p: Close, more-central vs Trained Alone		0.765	0.891
q: Close, more-central vs Trained Alone		1	1
p: Closeness effect: more-central peers		0.546	0.621
q: Closeness effect: more-central peers		1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.47: Endline take-up components by social distance and peer degree

VARIABLES	(1) Take-up Index	(2) Advice	(3) Mentoring	(4) Mentoring Payment	(5) Trainer	(6) Trainer Payment
Trained with Peer	-0.240 (0.165)	-0.144** (0.0647)	-0.0239 (0.0541)	-88.94 (299.6)	-0.0267 (0.0532)	-166.5 (434.7)
Trained with Peer $\times$ Peer distance	0.0203 (0.0209)	0.0165** (0.00784)	-0.00499 (0.00871)	0.988 (36.90)	-0.00627 (0.0103)	82.35 (71.43)
Trained with Peer $\times$ Peer degree	0.0295* (0.0172)	0.0169*** (0.00608)	0.00895* (0.00524)	4.138 (32.77)	0.0101* (0.00553)	7.562 (50.79)
Constant	0.105* (0.0567)	0.944*** (0.0173)	0.922*** (0.0202)	557.8*** (113.3)	0.904*** (0.0221)	679.5*** (135.1)
Observations	504	502	503	500	499	495
R-squared	0.036	0.045	0.052	0.002	0.047	0.005
q: Paired x Dist		0.221	1	1	1	0.997
q: Paired x Deg		0.0300	0.134	0.563	0.134	0.563

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.48: Endline take-up components by social closeness and peer centrality

VARIABLES	(1) Take-up Index	(2) Advice	(3) Mentoring	(4) Mentoring Payment	(5) Trainer	(6) Trainer Payment
Trained with Peer	-0.0350 (0.0916)	-0.0111 (0.0311)	-0.0240 (0.0363)	-37.21 (163.3)	-0.0172 (0.0386)	283.1 (230.8)
Trained with Peer $\times$ Close	-0.0604 (0.114)	-0.0667 (0.0516)	0.0115 (0.0446)	20.04 (159.3)	-0.000923 (0.0497)	-2.396 (273.5)
Trained with Peer $\times$ More Central	0.0769 (0.114)	0.0459 (0.0375)	0.0406 (0.0491)	14.77 (197.2)	0.0309 (0.0527)	-154.1 (358.0)
Trained with Peer $\times$ Close $\times$ More Central	0.134 (0.151)	0.0932 (0.0572)	0.0333 (0.0630)	8.860 (258.2)	0.0396 (0.0743)	-94.50 (430.4)
Constant	-0.0571 (0.0845)	0.899*** (0.0349)	0.836*** (0.0356)	425.2*** (135.3)	0.822*** (0.0424)	671.7*** (218.7)
Observations	509	507	508	505	504	500
R-squared	0.039	0.044	0.063	0.002	0.055	0.005
p: Close, more-central vs Trained Alone		0.000655	0.0618	0.971	0.217	0.872
q: Close, more-central vs Trained Alone		0.00400	0.141	0.636	0.278	0.636
p: Closeness effect: more-central peers		0.279	0.323	0.887	0.478	0.762
q: Closeness effect: more-central peers		1	1	1	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

SA8.2 Follow-up Index Components (Trained-Alone Base)

Table SA.49: Follow-up outcomes components by social distance and peer degree

VARIABLES	(1) Outcomes Index	(2) Monthly Income	(3) Agriculture Profits
Trained with Peer	0.346 (0.328)	9,315 (8,582)	20,394 (40,120)
Trained with Peer $\times$ Peer distance	-0.0980* (0.0540)	-2,034 (1,274)	-7,883 (6,473)
Trained with Peer $\times$ Peer degree	-0.0173 (0.0329)	-701.8 (718.8)	-1,356 (4,165)
Constant	0.194* (0.102)	27,104*** (2,530)	129,009*** (13,334)
Observations	320	299	306
R-squared	0.010	0.010	0.006
q: Paired x Dist		0.287	0.287
q: Paired x Deg		1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. T2 and T3 are pooled into a single paired indicator. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.50: Follow-up outcomes components by social closeness and peer centrality

VARIABLES	(1) Outcomes Index	(2) Monthly Income	(3) Agriculture Profits
Trained with Peer	-0.139 (0.158)	-2,392 (3,287)	-13,003 (22,613)
Trained with Peer $\times$ Close	0.364 (0.241)	8,326 (6,616)	37,106 (31,822)
Trained with Peer $\times$ More Central	-0.227 (0.215)	-3,424 (5,963)	-24,898 (25,840)
Trained with Peer $\times$ Close $\times$ More Central	0.209 (0.643)	-1,460 (12,723)	-16,352 (55,801)
Constant	0.305 (0.197)	30,865*** (4,506)	119,698*** (23,798)
Observations	323	301	309
R-squared	0.017	0.013	0.014
p: Distant, more-central vs Trained Alone		0.338	0.0913
q: Distant, more-central vs Trained Alone		0.224	0.224
p: Close, more-central vs Trained Alone		0.917	0.701
q: Close, more-central vs Trained Alone		1	1
p: Closeness effect: more-central peers		0.552	0.655
q: Closeness effect: more-central peers		1	1
p: Centrality effect: distant peers		0.566	0.336
q: Centrality effect: distant peers		1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.51: Follow-up steps components by social distance and peer degree

VARIABLES	(1) Steps Index	(2) Investment in Agriculture	(3) New Savings Account	(4) Joined Cooperative	(5) Savings	(6) Loan
Trained with Peer	-0.0419 (0.387)	21,646 (55,686)	-0.156 (0.123)	-0.0437 (0.0915)	611.8 (1,557)	24,979 (43,987)
Trained with Peer $\times$ Peer distance	0.0185 (0.0751)	-471.9 (11,713)	0.0180 (0.0217)	-0.00905 (0.0154)	-44.46 (240.2)	4,559 (10,525)
Trained with Peer $\times$ Peer degree	-0.0731*** (0.0280)	-6,544 (4,315)	-0.00173 (0.0121)	-0.00637 (0.00757)	-230.6* (135.9)	-7,275*** (2,688)
Constant	0.488*** (0.133)	126,149*** (18,515)	0.270*** (0.0405)	0.205*** (0.0368)	3,459*** (576.7)	33,131*** (11,167)
Observations	323	226	323	323	291	323
R-squared	0.024	0.006	0.018	0.026	0.010	0.011
q: Paired $\times$ Dist		1	1	1	1	1
q: Paired $\times$ Deg		0.212	0.550	0.279	0.212	0.0380

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. T2 and T3 are pooled into a single paired indicator. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.52: Follow-up steps components by social closeness and peer centrality

VARIABLES	(1) Steps Index	(2) Investment in Agriculture	(3) New Savings Account	(4) Joined Cooperative	(5) Savings	(6) Loan
Trained with Peer	-0.0767 (0.223)	9,524 (32,893)	-0.0608 (0.0661)	-0.0993* (0.0521)	141.4 (838.3)	30,134 (28,293)
Trained with Peer $\times$ Close	-0.0332 (0.331)	-20,213 (43,709)	-0.0397 (0.0901)	0.0644 (0.0771)	614.8 (1,448)	-24,923 (36,523)
Trained with Peer $\times$ More Central	-0.576** (0.229)	-68,656** (31,703)	-0.0182 (0.0820)	-0.0222 (0.0606)	-1,838** (843.9)	-64,665** (28,922)
Trained with Peer $\times$ Close $\times$ More Central	0.0503 (0.385)	67,852 (52,373)	-0.0353 (0.129)	-0.0959 (0.104)	-286.6 (1,793)	38,807 (39,429)
Constant	0.533*** (0.191)	142,676*** (30,599)	0.217*** (0.0707)	0.265*** (0.0616)	3,408*** (845.1)	37,043** (18,541)
Observations	326	227	326	326	293	326
R-squared	0.030	0.016	0.016	0.029	0.018	0.018
p: Distant, more-central vs Trained Alone		0.0216	0.276	0.0467	0.0166	0.000698
q: Distant, more-central vs Trained Alone		0.0300	0.0620	0.0380	0.0300	0.00400
p: Close, more-central vs Trained Alone		0.727	0.0597	0.0178	0.219	0.163
q: Close, more-central vs Trained Alone		0.377	0.136	0.0980	0.197	0.196
p: Closeness effect: more-central peers		0.124	0.415	0.651	0.776	0.225
q: Closeness effect: more-central peers		1	1	1	1	1
p: Centrality effect: distant peers		0.0314	0.825	0.715	0.0302	0.0261
q: Centrality effect: distant peers		0.0560	0.493	0.493	0.0560	0.0560

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.53: Follow-up mindset components by social distance and peer degree

VARIABLES	(1) Mindset Index	(2) Income Aspirations	(3) Plan to Start Business	(4) Self-Efficacy	(5) Commitment Savings	(6) Commitment Amount
Trained with Peer	0.316 (0.371)	1,313 (10,194)	1.272** (0.590)	0.601 (0.565)	-0.0961 (0.156)	631.7 (2,601)
Trained with Peer $\times$ Peer distance	-0.101* (0.0535)	-494.2 (1,505)	-0.250*** (0.0889)	-0.198** (0.0865)	-0.00118 (0.0243)	-347.0 (354.6)
Trained with Peer $\times$ Peer degree	-0.0226 (0.0361)	-788.8 (1,041)	-0.112 (0.0684)	-0.0349 (0.0670)	0.0158 (0.0173)	57.00 (230.0)
Constant	0.173* (0.103)	50,833*** (3,387)	2.538*** (0.176)	2.902*** (0.173)	0.500*** (0.0456)	1,807*** (508.9)
Observations	323	320	316	323	323	323
R-squared	0.017	0.005	0.033	0.025	0.003	0.007
q: Paired $\times$ Dist		1	0.0270	0.0490	1	0.490
q: Paired $\times$ Deg		1	1	1	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

Table SA.54: Follow-up mindset components by social closeness and peer centrality

VARIABLES	(1) Mindset Index	(2) Income Aspirations	(3) Plan to Start Business	(4) Self-Efficacy	(5) Commitment Savings	(6) Commitment Amount
Trained with Peer	-0.177 (0.165)	-2,145 (4,787)	-0.151 (0.310)	-0.326 (0.306)	-0.0213 (0.0786)	-1,000 (639.2)
Trained with Peer $\times$ Close	0.340 (0.278)	3,568 (7,625)	0.412 (0.456)	0.180 (0.446)	0.0165 (0.116)	1,770 (1,874)
Trained with Peer $\times$ More Central	-0.0934 (0.205)	-11,132** (4,896)	-0.413 (0.394)	-0.0173 (0.373)	0.0954 (0.105)	1,738 (1,130)
Trained with Peer $\times$ Close $\times$ More Central	-0.259 (0.481)	11,509 (16,070)	-0.198 (0.723)	-0.0592 (0.693)	-0.215 (0.178)	-4,036* (2,098)
Constant	0.376** (0.173)	58,828*** (5,730)	2.948*** (0.315)	2.535*** (0.313)	0.467*** (0.0851)	1,459** (688.3)
Observations	326	323	320	326	326	326
R-squared	0.015	0.018	0.020	0.019	0.007	0.014
p: Distant, more-central vs Trained Alone		0.00528	0.0965	0.284	0.430	0.520
q: Distant, more-central vs Trained Alone		0.0280	0.240	0.398	0.454	0.454
p: Close, more-central vs Trained Alone		0.900	0.500	0.657	0.306	0.00266
q: Close, more-central vs Trained Alone		1	1	1	1	0.0140
p: Closeness effect: more-central peers		0.313	0.704	0.821	0.149	0.0304
q: Closeness effect: more-central peers		0.597	0.972	0.972	0.427	0.180
p: Centrality effect: distant peers		0.0237	0.295	0.963	0.362	0.125
q: Centrality effect: distant peers		0.135	0.454	0.826	0.454	0.333

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The sample is restricted to treated individuals, with Trained Alone (T1) as the omitted base category. The discrete specification includes Close, More Central, and their interaction Close $\times$ More Central (pairing with a socially close and more central peer). We additionally control for the individual's own degree centrality. Robust standard errors in parentheses. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across components.

SA9 Within-Village Robustness Specifications

Choice of the Control Group and Robustness Exercises The choice of the control group involves a tradeoff. Our main specification uses pure control villages, which raises a potential selection concern: only women who arrived at the training center enter the endline sample in treatment villages, while pure control women were surveyed at home. However, the two groups are very well balanced on observables (Section 2.3), dropouts within treated villages do not differ by treatment assignment or on willingness to start a business, and results are robust even when we account for selection. While the alternative within-village comparison shares the same selection process (as the within-village controls also arrived at the training center), this comparison is imperfect: two-sided non-compliance mechanically attenuates ITT estimates.

As robustness exercises, we (1) discuss results comparing treated individuals to control individuals within treated villages to ensure completeness and (2) also estimate specifications that instrument actual treatment status with assignment to obtain local average treatment effects. Importantly, the key results of the paper around peer heterogeneity are unaffected by this choice, as those specifications compare within the treated sample only.

The following table shows the ITT effects when we compare those assigned to treatment with those assigned to the within village control group.

Table SA.55: Impact of training on immediate outcomes (within-village comparison)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.114 (0.0945)	0.125 (0.112)	0.108 (0.102)	0.126 (0.0986)	0.243** (0.0969)
Trained with Peer	0.0209 (0.0957)	0.108 (0.115)	-0.0314 (0.105)	-0.0753 (0.106)	0.0928 (0.105)
Trained with Peer + Connections Module	0.122 (0.0963)	0.0483 (0.122)	0.0735 (0.111)	0.0760 (0.103)	0.149 (0.105)
Constant	0 (0.0754)	0 (0.0754)	0 (0.0754)	0 (0.0758)	-0 (0.0756)
Observations	710	710	707	688	693
R-squared	0.004	0.002	0.003	0.007	0.009
p: Alone vs Peer	0.259	0.889	0.167	0.0378	0.113
p: Alone vs Peer+CM	0.921	0.545	0.743	0.595	0.321
p: Peer vs Peer+CM	0.231	0.642	0.338	0.137	0.584
q: Alone	0.301	0.301	0.301	0.301	0.0660
q: Peer	1	1	1	1	1
q: Peer+CM	1	1	1	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. Standard errors are robust. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across the five outcome indices.

The main-text peer-heterogeneity tables (Table 2 for endline and Table 4 for follow-up) are estimated on treated individuals only, with T1 (Trained alone) as the omitted reference category, normalizing outcomes by the pure control group.

As a robustness check, this section reports the within-village specification, which uses within-village controls (untreated women in treatment villages) as the base category and normalizes by that subgroup. The results are similar.

Table SA.56: Effects on endline outcomes by peer type (within-village base)

(a) By Social Distance and Peer Centrality (continuous)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.114 (0.0946)	0.125 (0.113)	0.108 (0.102)	0.126 (0.0987)	0.243** (0.0970)
Trained with Peer	0.420** (0.171)	0.695** (0.274)	0.176 (0.219)	0.0383 (0.188)	-0.158 (0.215)
Trained with Peer × Peer distance	-0.0282 (0.0254)	-0.135*** (0.0438)	-0.0210 (0.0344)	0.00379 (0.0292)	0.0471* (0.0283)
Trained with Peer × Peer degree	-0.0190 (0.0195)	-0.0295 (0.0270)	0.00125 (0.0250)	0.0105 (0.0199)	0.0480** (0.0213)
Constant	-0 (0.0755)	-0 (0.0755)	-0 (0.0755)	-0 (0.0759)	-0 (0.0757)
Observations	695	695	692	673	679
R-squared	0.047	0.016	0.013	0.020	0.037
q: Alone	0.302	0.302	0.302	0.302	0.0670
q: Paired	0.0380	0.0380	0.532	1	0.532
q: Paired x Dist	0.366	0.0110	0.687	0.807	0.238
q: Paired x Deg	0.786	0.786	1	1	0.143

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

(b) By Social Closeness and Peer Centrality (Close, More Central, and their interaction)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.0855 (0.0937)	0.111 (0.114)	0.0794 (0.101)	0.112 (0.0989)	0.204** (0.0943)
Trained with Peer	0.234** (0.0965)	-0.0288 (0.129)	0.113 (0.125)	0.0856 (0.118)	0.204* (0.119)
Trained with Peer × Close	-0.152 (0.124)	0.210 (0.206)	-0.135 (0.172)	-0.0450 (0.153)	-0.151 (0.162)
Trained with Peer × More Central	-0.0761 (0.112)	-0.153 (0.146)	-0.108 (0.163)	0.0214 (0.148)	0.0842 (0.144)
Trained with Peer × Close × More Central	0.274 (0.197)	0.715* (0.397)	0.366 (0.249)	0.0281 (0.238)	0.239 (0.201)
Degree	0.0471*** (0.0129)	0.00277 (0.0198)	0.0528*** (0.0163)	0.0263 (0.0160)	0.0375*** (0.0142)
Constant	-0.225** (0.103)	-0.0133 (0.114)	-0.253** (0.114)	-0.126 (0.112)	-0.180 (0.111)
Observations	700	700	697	678	684
R-squared	0.063	0.026	0.029	0.023	0.038
q: Close	1	1	1	1	1
q: More Central	1	1	1	1	1
q: Close x More Central	0.377	0.377	0.377	0.416	0.377

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is within-village controls. The regressions include an indicator for those who were intended to be paired but remained unmatched. We additionally control for the individual's own degree centrality in regressions that include relative comparisons between own and peer centrality. Standard errors are robust. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes.

SA10 Local Average Treatment Effects

Table SA.57: Impact of training on immediate outcomes (LATE)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.999*** (0.306)	-0.169 (0.169)	0.345** (0.157)	0.167 (0.120)	0.166 (0.119)
Trained with Peer	0.895*** (0.284)	-0.196 (0.126)	0.165 (0.137)	-0.0740 (0.111)	-0.0391 (0.105)
Constant	0 (0.230)	-0 (0.0913)	0 (0.0777)	-0 (0.0604)	0 (0.0607)
Observations	768	768	765	746	751
R-squared	0.232	0.004	0.022	-0.000	0.003
Under-ID p(KP LM)	1.65e-05	1.65e-05	1.28e-05	9.67e-06	9.20e-06
First-stage F: Trained Alone	216.9	216.9	208.5	181.8	183.5
First-stage F: Trained with peer	290.2	290.2	288.4	283.4	292.8
p: Trained Alone = Trained with peer	0.282	0.835	0.315	0.0820	0.0958
q: Trained Alone	0.00600	0.235	0.0610	0.141	0.141
q: Trained with Peer	0.00900	0.315	0.427	0.614	0.616

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages; within-village controls are excluded. Local average treatment effects (LATE) are estimated by instrumenting actual treatment status with assigned treatment. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across the five outcome indices.

Table SA.58: Impact of the training on follow-up outcomes (LATE)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.265 (0.181)	0.675*** (0.216)	0.253 (0.238)
Trained with Peer	0.110 (0.143)	0.208 (0.160)	-0.0124 (0.205)
Constant	0 (0.0685)	0 (0.0661)	-0 (0.154)
Observations	472	475	475
R-squared	-0.003	0.004	0.001
Under-ID p(KP LM)	1.18e-05	1.17e-05	1.17e-05
First-stage F: Trained Alone	112.3	119	119
First-stage F: Trained with peer	223.3	233.4	233.4
p: Trained Alone = Trained with peer	0.386	0.0598	0.102
q: Trained Alone	0.166	0.00600	0.239
q: Trained with Peer	1	1	1

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages; within-village controls are excluded. Standard errors are clustered at the village level. Q-values are FDR-adjusted using the sharpened procedure of [Benjamini et al. \(2006\)](#), computed separately for each treatment coefficient across outcomes.

SA11 Robustness to Selection

We estimate the probability that a woman appears at the training center for the endline survey, conditional on baseline characteristics, and use it to reweight or trim the pure-control sample. The selection model is fit on the women assigned to training in treated villages, with baseline covariates including age, winsorized income, degree centrality, education indicators, caste indicators, marital status, stated willingness to start a business, and income aspirations. The fitted model is then used to generate predicted attendance probability $\hat{p}_i$ for each pure-control woman.

Following this, we implement two robustness exercises. First (Table SA.59), we run the main ITT specification with weights $w_i = 1$ for endline participants assigned to treatment (alone or with a peer) and $w_i = \hat{p}_i$ for endline participants in the pure-control. The pure control is therefore reweighted to be similar to the attender subpopulation.

Second (Table SA.60), we drop pure-control women whose $\hat{p}_i$ falls below the 25th percentile of the $\hat{p}$ distribution for endline participants assigned to treatment and re-run the unweighted ITT on the trimmed sample.

The two exercises produce ITT estimates that are similar in magnitude and significance to the main Table 1, suggesting that the main effects are not driven by selection.

Table SA.59: Endline ITT with would-be-attender IPW weighting

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.821*** (0.248)	-0.142 (0.142)	0.251** (0.107)	0.0707 (0.0781)	0.0747 (0.112)
Trained with Peer	0.736*** (0.253)	-0.148 (0.123)	0.0969 (0.124)	-0.151 (0.102)	-0.0580 (0.117)
Trained with Peer + Connections Module	0.831*** (0.245)	-0.180 (0.134)	0.208** (0.101)	0.0204 (0.0896)	-0.0311 (0.113)
Constant	-0.0316 (0.235)	0.00288 (0.113)	0.00223 (0.0648)	0.0307 (0.0468)	0.0304 (0.0920)
Observations	752	752	749	730	735
R-squared	0.089	0.005	0.007	0.008	0.005
p: T1 vs T2	0.183	0.951	0.255	0.0596	0.161
p: T1 vs T3	0.906	0.617	0.702	0.575	0.205
p: T2 vs T3	0.331	0.646	0.305	0.147	0.748

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Pure-control observations are reweighted by their predicted probability of attending training. Standard errors are clustered at the village level.

Table SA.60: Endline ITT after trimming pure control to common support

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained alone	0.879*** (0.245)	-0.223 (0.163)	0.284** (0.130)	0.183* (0.0997)	0.106 (0.103)
Trained with Peer	0.794*** (0.251)	-0.229 (0.147)	0.130 (0.144)	-0.0384 (0.120)	-0.0267 (0.109)
Trained with Peer + Connections Module	0.889*** (0.242)	-0.261 (0.156)	0.242* (0.125)	0.133 (0.109)	0.000234 (0.105)
Constant	-0.0897 (0.232)	0.0839 (0.139)	-0.0312 (0.0985)	-0.0815 (0.0777)	-0.000944 (0.0815)
Observations	692	692	689	671	676
R-squared	0.127	0.012	0.010	0.008	0.004
p: T1 vs T2	0.183	0.951	0.255	0.0597	0.161
p: T1 vs T3	0.906	0.617	0.702	0.576	0.205
p: T2 vs T3	0.331	0.646	0.306	0.147	0.748

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Pure-control respondents whose predicted attendance probability $\hat{p}_i$ lies below the 25th percentile of the treatment-attender $\hat{p}$ distribution are dropped. Standard errors are clustered at the village level.

SA12 Robustness to Attrition: IPW-Weighted Results

The following tables replicate the main follow-up results using Inverse Probability Weighting (IPW) to account for potential attrition bias between the endline and follow-up surveys. We estimate the probability of being observed at follow-up, conditional on being in the endline sample, using a probit model. The covariates include treatment assignment indicators and all baseline characteristics used in the balance tables. The IPW weight for each individual is the inverse of their predicted probability of being observed, $w_i = 1/\hat{p}_i$. All follow-up regressions are then re-estimated using these weights.

Table SA.61: Follow-up effects on main outcomes (IPW)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.191 (0.139)	0.434*** (0.150)	0.155 (0.184)
Trained with Peer	0.143 (0.116)	0.246* (0.135)	0.0365 (0.169)
Constant	-0.0160 (0.0675)	-0.0283 (0.0690)	-0.0264 (0.156)
Observations	458	461	461
R-squared	0.004	0.017	0.003
p: T1 vs Paired	0.679	0.267	0.276

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Observations are weighted by the inverse probability of survey response. Standard errors are clustered at the village level.

Table SA.62: Follow-up effects by social distance and peer centrality (IPW)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.191 (0.139)	0.434*** (0.151)	0.155 (0.185)
Trained with Peer	0.617* (0.340)	0.457 (0.380)	0.553 (0.364)
Trained with Peer × Peer distance	-0.107* (0.0527)	0.0254 (0.0710)	-0.110* (0.0553)
Trained with Peer × Peer degree	-0.0182 (0.0256)	-0.0756** (0.0295)	-0.0241 (0.0301)
Constant	-0.0160 (0.0677)	-0.0283 (0.0692)	-0.0264 (0.156)
Observations	449	452	452
R-squared	0.013	0.030	0.012

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Observations are weighted by the inverse probability of survey response. Standard errors are clustered at the village level.

Table SA.63: Follow-up effects by social closeness and peer centrality (IPW)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.209 (0.136)	0.422*** (0.147)	0.163 (0.180)
Trained with Peer	0.116 (0.155)	0.401* (0.229)	-0.000951 (0.203)
Trained with Peer $\times$ Close	0.373* (0.206)	-0.0941 (0.289)	0.361 (0.257)
Trained with Peer $\times$ More Central	-0.264 (0.200)	-0.536** (0.215)	-0.0402 (0.216)
Trained with Peer $\times$ Close $\times$ More Central	0.285 (0.714)	0.0851 (0.348)	-0.270 (0.473)
Degree	-0.0280 (0.0310)	0.0195 (0.0336)	-0.0128 (0.0251)
Constant	0.119 (0.167)	-0.122 (0.177)	0.0354 (0.205)
Observations	451	454	454
R-squared	0.020	0.037	0.008
q: Close	0.327	0.347	0.327
q: More Central	0.246	0.0620	0.420
q: Close $\times$ More Central	1	1	1

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. The discrete specification includes Close, More Central, and their interaction (Close $\times$ More Central). Observations are weighted by the inverse probability of survey response. Standard errors are clustered at the village level.

SA13 PDS-Lasso Robustness Checks

SA13.0.1 Short-Term Outcomes

We replicate the main endline specifications using post-double selection Lasso (Belloni et al. 2014) to account for baseline covariates that may be correlated with treatment status. Table SA.64 reports the basic ITT (with pure control villages as the base category), and Table SA.65 reports the peer-heterogeneity specifications on the treatment sample with T1 (Trained alone) as the omitted base.

Table SA.64: Impact of training on immediate outcomes (PDS-Lasso)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained Alone	0.758*** (0.253)	-0.162** (0.0808)	0.299** (0.145)	0.0837 (0.0965)	0.0692 (0.0868)
Trained with Peer	0.695*** (0.252)	-0.160** (0.0765)	0.0994 (0.147)	-0.127 (0.110)	-0.0384 (0.0999)
Trained with Peer + Connections Module	0.779*** (0.248)	-0.202** (0.0792)	0.247* (0.137)	0.0236 (0.109)	-0.0144 (0.0884)
Constant	-0.112 (0.247)	-0.424*** (0.0735)	0.690*** (0.264)	0.335 (0.213)	0.533*** (0.163)
Observations	748	748	745	726	731
Number of groups	0	0	0	0	0
p: T1 vs T2	0.340	0.975	0.159	0.0480	0.156
p: T1 vs T3	0.803	0.467	0.644	0.472	0.304
p: T2 vs T3	0.388	0.417	0.168	0.175	0.758

Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: These regressions treat individuals in pure control villages as the base category. Covariates include income, source of income, age, education, caste, and number of network connections. Standard errors are clustered at the village level.

Table SA.65: Effects on endline outcomes by peer type (PDS-Lasso)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Index	(4) Additional Steps	(5) Take-up Index
Trained with Peer	0.280 (0.176)	0.246* (0.140)	-0.137 (0.227)	-0.162 (0.198)	-0.303* (0.164)
Trained with Peer x Peer distance	-0.0177 (0.0264)	-0.0345* (0.0196)	0.00400 (0.0349)	0.0146 (0.0307)	0.0318 (0.0205)
Trained with Peer x Peer degree	-0.0157 (0.0204)	-0.0316** (0.0144)	0.0155 (0.0243)	0.0153 (0.0215)	0.0378** (0.0171)
Constant	1.213*** (0.177)	-0.596*** (0.0531)	1.148*** (0.235)	0.673*** (0.214)	0.791*** (0.140)
Observations	505	505	502	485	490
Number of groups	0	0	0	0	0

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: These regressions use post double selection Lasso ([Belloni et al. 2014](#)) to account for baseline covariates that may be correlated with treatment status while comparing those trained alone and with a peer. Covariates include income, source of income, age, education, and caste. The regression includes the sample assigned to treatment with T1 (Trained alone) as the omitted base. We additionally control for the individual's own degree centrality in regressions that include relative comparisons between own and peer centrality.

SA13.0.2 Long-Term Outcomes

As at endline, the follow-up specifications are run twice: Table [SA.67](#) reports the basic ITT with pure control villages as the base category, and Table [SA.66](#) reports the peer-heterogeneity specifications on the treated-only sample with T1 (Trained alone) as the omitted base.

Table SA.66: Effects on long-term outcomes by peer type (PDS-Lasso)

(a) By Social Distance and Peer Centrality

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained with Peer	0.347 (0.326)	-0.0238 (0.383)	0.147 (0.354)
Trained with Peer x Peer distance	-0.0993* (0.0537)	0.0194 (0.0747)	-0.0739 (0.0509)
Trained with Peer x Peer degree	-0.0167 (0.0327)	-0.0735*** (0.0277)	-0.0121 (0.0359)
Constant	0.198* (0.102)	0.467*** (0.128)	1.400*** (0.285)
Observations	318	321	321
Number of groups	0	0	0

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

(b) By Social Closeness and Peer Centrality (Close, More Central, and their interaction)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained with Peer	-0.159 (0.157)	-0.0965 (0.221)	-0.224 (0.160)
Trained with Peer × Close	0.356 (0.234)	-0.0330 (0.327)	0.310 (0.262)
Trained with Peer × More Central	-0.202 (0.214)	-0.555** (0.227)	0.0386 (0.190)
Trained with Peer × Close × More Central	0.231 (0.647)	0.00591 (0.383)	-0.421 (0.455)
Degree	-0.0233 (0.0310)	-0.0154 (0.0353)	-0.0441 (0.0293)
Constant	0.326* (0.196)	0.554*** (0.190)	1.708*** (0.357)
Observations	322	325	325
Number of groups	0	0	0
q: Close	0.557	0.557	0.557
q: More Central	0.525	0.0460	1
q: Close x More Central	1	1	1

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: These regressions use post double selection Lasso (Belloni et al. 2014) to account for baseline covariates that may be correlated with treatment status. Covariates include income, source of income, age, education, and caste. The regression includes the sample assigned to treatment with T1 (Trained alone) as the omitted base. We additionally control for the individual's own degree centrality in regressions that include relative comparisons between own and peer centrality.

SA14 Robustness to Network Sampling

Figure SA.6 shows the proportion of village populations for which we elicited network data. On average, this rate is roughly 80%. We correct our measure of degree centrality to account for unobserved links by dividing the observed degree by the village-specific sampling proportion. The intuition is as follows: if individual i has d_i links in an observed network with x nodes, then in a village with population y they are predicted to have $\frac{d_i}{x} \times y$ links. Figure SA.7 shows the distribution of village densities based on this corrected measure. Density is higher than the earlier conservative estimate but remains low overall, indicating sparse networks.

Table SA.67: Effects on long-term outcomes (PDS-Lasso)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained Alone	0.154 (0.135)	0.459*** (0.143)	0.203 (0.170)
Trained with Peer	0.0560 (0.120)	0.199 (0.121)	0.0207 (0.155)
Constant	0.660* (0.378)	0.629*** (0.204)	0.747** (0.317)
Observations	469	472	472
Number of groups	0	0	0
p: Trained Alone v/s Trained with Peer	0.338	0.0749	0.0494

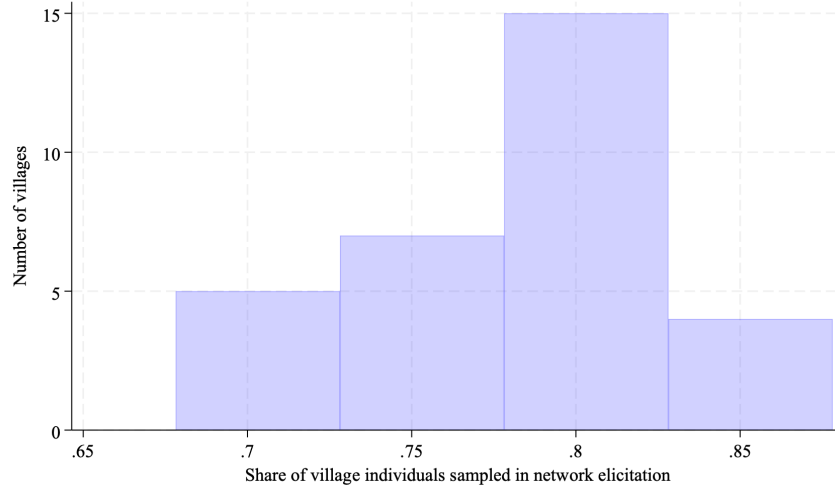
Cluster-robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: The base category is pure control villages. Standard errors are clustered at the village level. We implement PDS-Lasso (Belloni et al. 2014) with baseline controls for income (level and sources), age, education (elementary, secondary, higher, university, informal), and caste.

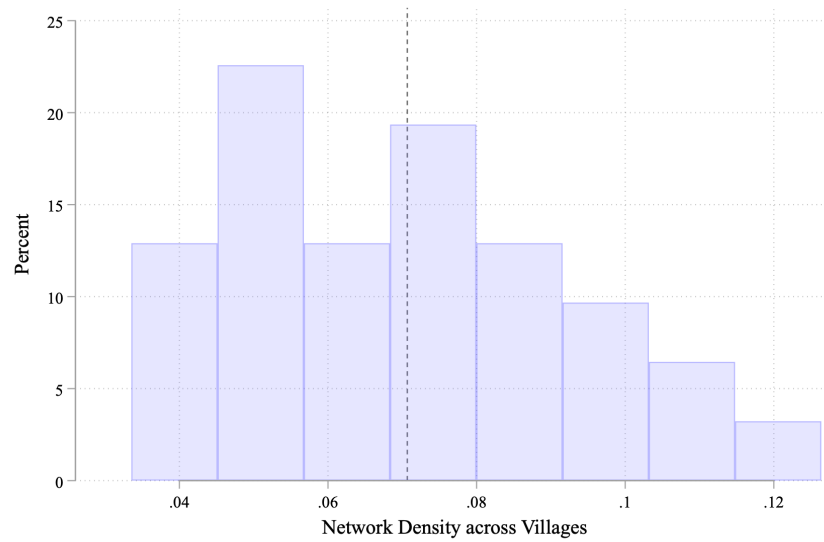
Next, we present endline and follow-up results using this corrected measure of centrality in Table SA.68 and Table SA.69. The results show that the positive effects of close and central peers on aspirations at endline, and of close and less central peers at follow-up, are robust to this correction.

Figure SA.6: Fraction of individuals per village sampled in network elicitation



Notes: The figure plots the distribution of the share of each village's population for which network data were elicited. Village population is defined as the union of all individuals who had network data collected or were listed as contacts by others but do not have network data. The sampling rate for each village is the ratio of individuals with network data to this constructed village population.

Figure SA.7: Network density across villages



Notes: The figure shows the distribution of village-level network density computed using the degree measure corrected for sampling coverage. For each village, we first adjust individual degrees accounting for individuals who are not included in the network elicitation, and then calculate density based on these corrected degrees.

Table SA.68: Effects on endline outcomes by peer type (degree-corrected)

(a) By Social Distance and Peer Centrality

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Intention	(4) Steps Index	(5) Take-up Index
Trained with Peer	0.395** (0.186)	0.351* (0.182)	0.060 (0.238)	-0.071 (0.198)	-0.306* (0.167)
Trained with Peer $\times$ Peer distance	-0.028 (0.028)	-0.083*** (0.029)	-0.021 (0.038)	0.003 (0.032)	0.025 (0.021)
Trained with Peer $\times$ Peer degree	-0.023 (0.017)	-0.015 (0.015)	0.002 (0.021)	0.006 (0.017)	0.031** (0.014)
Observations	519	519	516	499	504
R-squared	0.072	0.013	0.017	0.027	0.044
q: Paired $\times$ Dist	0.747	0.025	1.000	1.000	0.747
q: Paired $\times$ Deg	0.573	0.750	1.000	1.000	0.159

(b) By Social Closeness and Peer Centrality (Close, More Central, and their interaction)

VARIABLES	(1) Knowledge Index	(2) Aspirations Index	(3) Business Intention	(4) Steps Index	(5) Take-up Index
Trained with Peer	0.209** (0.094)	-0.070 (0.096)	0.045 (0.135)	-0.024 (0.121)	-0.043 (0.092)
Trained with Peer $\times$ Close	-0.162 (0.141)	0.102 (0.137)	-0.136 (0.190)	-0.043 (0.168)	-0.084 (0.119)
Trained with Peer $\times$ More Central	-0.144 (0.127)	-0.129 (0.111)	-0.141 (0.181)	0.012 (0.163)	0.098 (0.115)
Trained with Peer $\times$ Close $\times$ More Central	0.317 (0.223)	0.502* (0.265)	0.395 (0.275)	0.030 (0.260)	0.159 (0.154)
Observations	524	524	521	504	509
R-squared	0.080	0.025	0.030	0.029	0.047
p: Close, more-central vs Trained Alone	0.158	0.068	0.334	0.884	0.091
q: Close, more-central vs Trained Alone	0.296	0.296	0.335	0.547	0.296
p: Closeness effect: more-central peers	0.379	0.008	0.203	0.947	0.435
q: Closeness effect: more-central peers	0.769	0.041	0.683	1.000	0.769

Notes: The base category is pure control villages. The regressions include an indicator for those who were intended to be paired but remained unmatched. We additionally control for the individual's own degree centrality in regressions that include relative comparisons between own and peer centrality. Standard errors are clustered at the village level.

Table SA.69: Effects on long-term outcomes (degree-corrected)

(a) By Social Distance and Peer Centrality

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained with Peer	0.370 (0.325)	-0.063 (0.386)	0.338 (0.371)
Trained with Peer $\times$ Peer distance	-0.099* (0.053)	0.010 (0.074)	-0.102* (0.053)
Trained with Peer $\times$ Peer degree	-0.016 (0.029)	-0.049** (0.024)	-0.020 (0.030)
Observations	320	323	323
R-squared	0.011	0.021	0.017
q: Paired $\times$ Dist	0.101	0.426	0.101
q: Paired $\times$ Deg	0.612	0.157	0.612

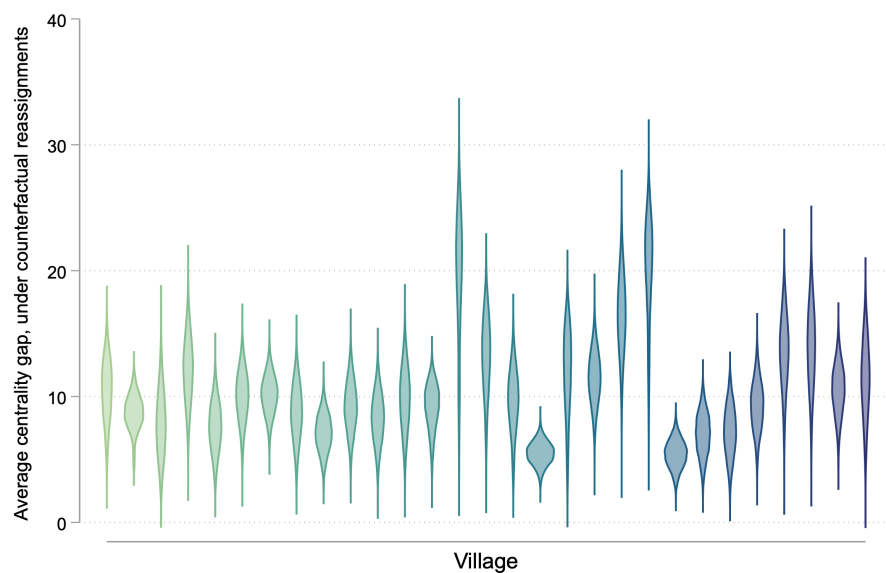
(b) By Social Closeness and Peer Centrality (Close, More Central, and their interaction)

VARIABLES	(1) Outcomes Index	(2) Steps Index	(3) Mindset Index
Trained with Peer	-0.137 (0.158)	-0.098 (0.221)	-0.176 (0.165)
Trained with Peer $\times$ Close	0.381 (0.240)	-0.025 (0.324)	0.354 (0.278)
Trained with Peer $\times$ More Central	-0.246 (0.215)	-0.570** (0.225)	-0.105 (0.205)
Trained with Peer $\times$ Close $\times$ More Central	0.204 (0.641)	0.193 (0.393)	-0.264 (0.479)
Observations	323	326	326
R-squared	0.018	0.027	0.017
p: Distant, more-central vs Trained Alone	0.059	0.000	0.126
q: Distant, more-central vs Trained Alone	0.063	0.001	0.093
p: Close, more-central vs Trained Alone	0.728	0.024	0.613
q: Close, more-central vs Trained Alone	0.943	0.078	0.943
p: Closeness effect: more-central peers	0.348	0.439	0.828
q: Closeness effect: more-central peers	1.000	1.000	1.000
p: Centrality effect: distant peers	0.253	0.012	0.609
q: Centrality effect: distant peers	0.340	0.037	0.614

Notes: The base category is pure control villages. The regression includes an indicator for those who were intended to be paired but remained unmatched. We additionally control for the individual's own degree centrality in regressions that include relative comparisons between own and peer centrality. Standard errors are clustered at the village level.

SA15 Peer Effects and Feasibility of Peer Pairings

Figure SA.8: Feasibility of counterfactual pair reassignments



Notes: The figure plots the distribution of the *average within-village degree-centrality gap* across 10,000 random reassignments of peer dyads (without replacement), holding the network structure fixed.